



Inserm

**Institut national
de la santé et de la recherche médicale**

Inria
INVENTEURS DU MONDE NUMÉRIQUE

DATA SCIENCE IN BIOLOGY

Hédi SOULA

UMR 1269 - NUTRIOMICS

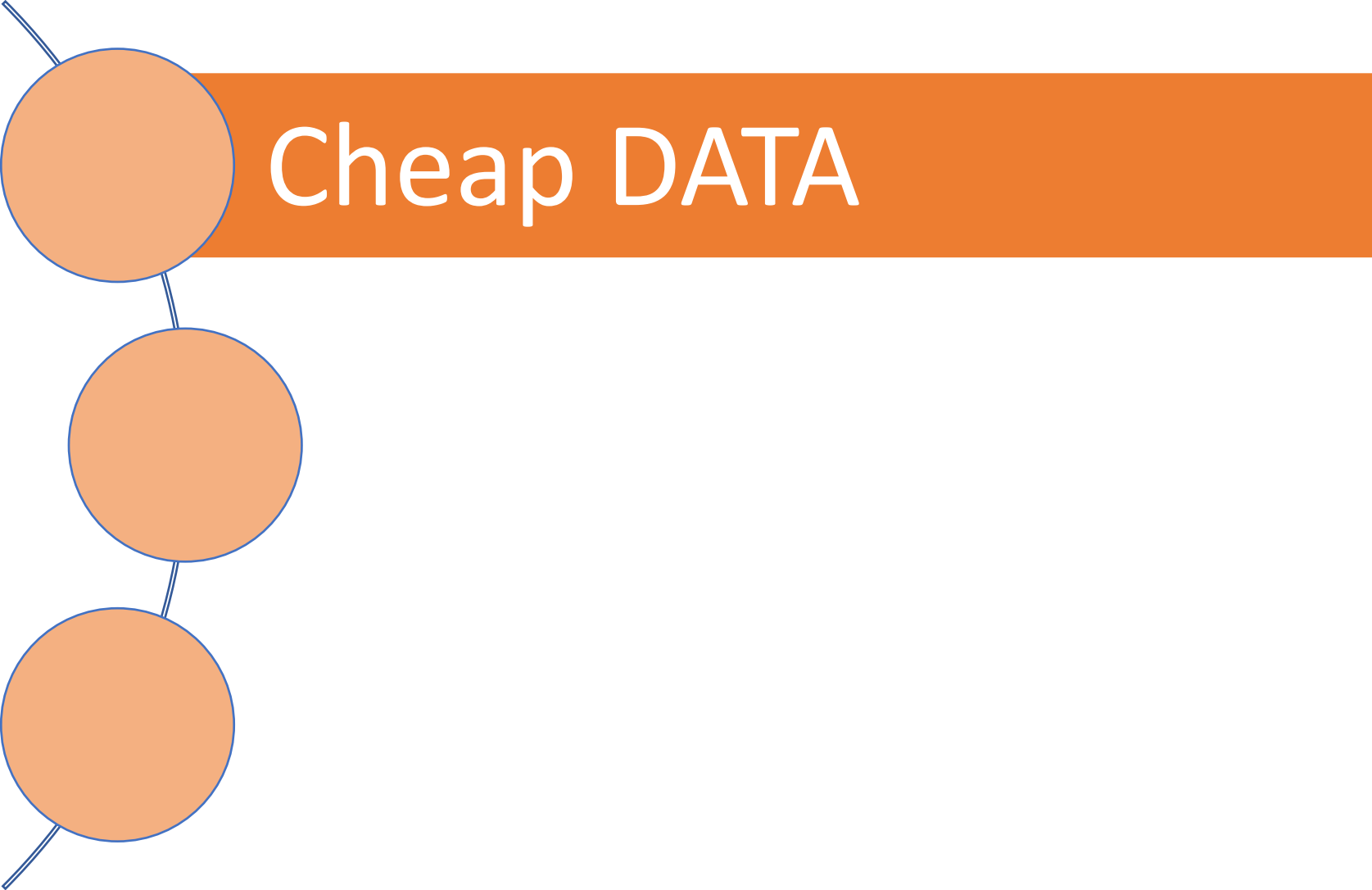
AIDA - DMEP

Comrad **SOULA** is warning you

in **2025**
You **MUST**
do **MATHS**
and **IT**

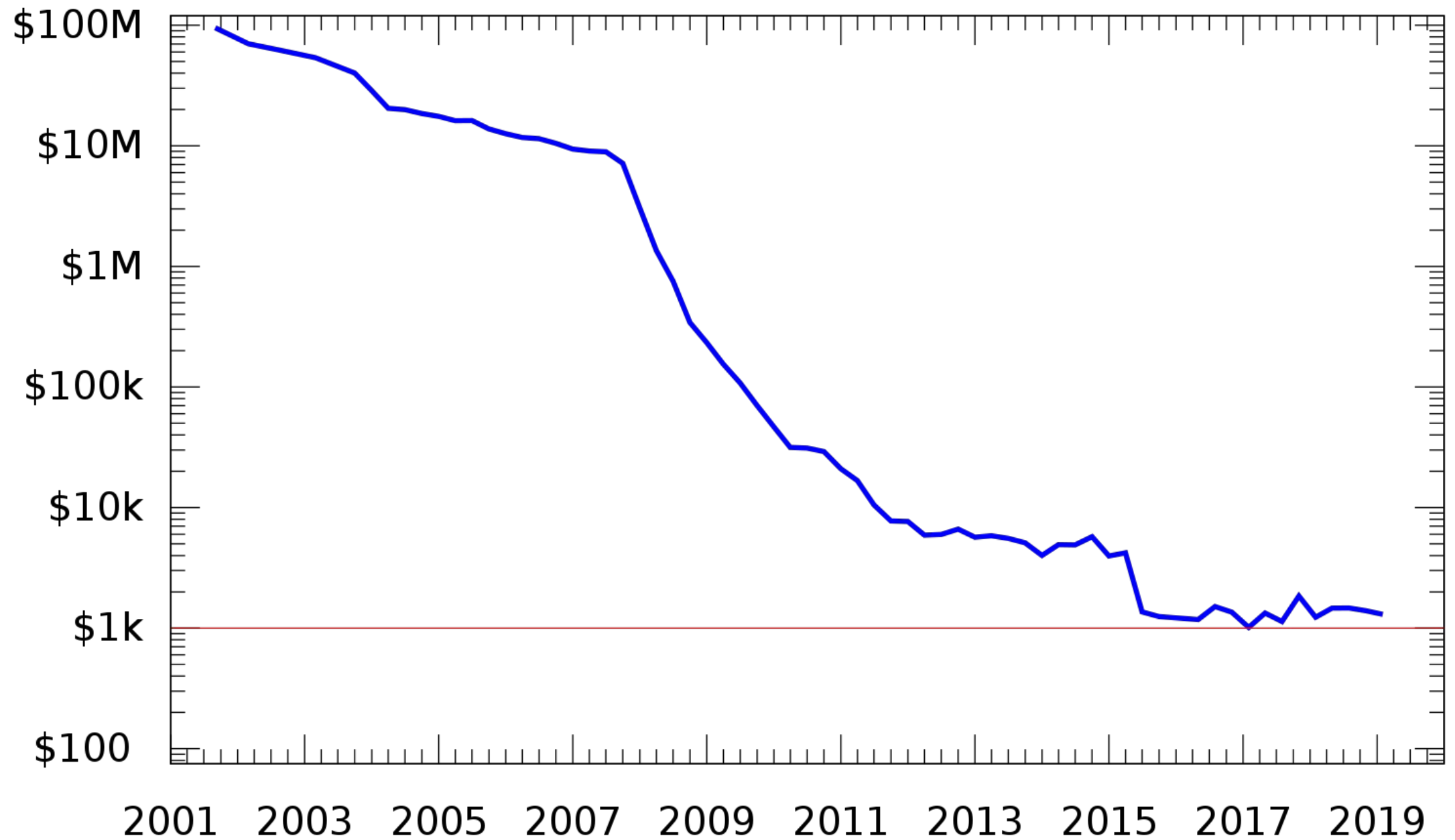


WHY DATA SCIENCE ?



Cheap DATA

Coût de séquençage d'un génome humain (USD)



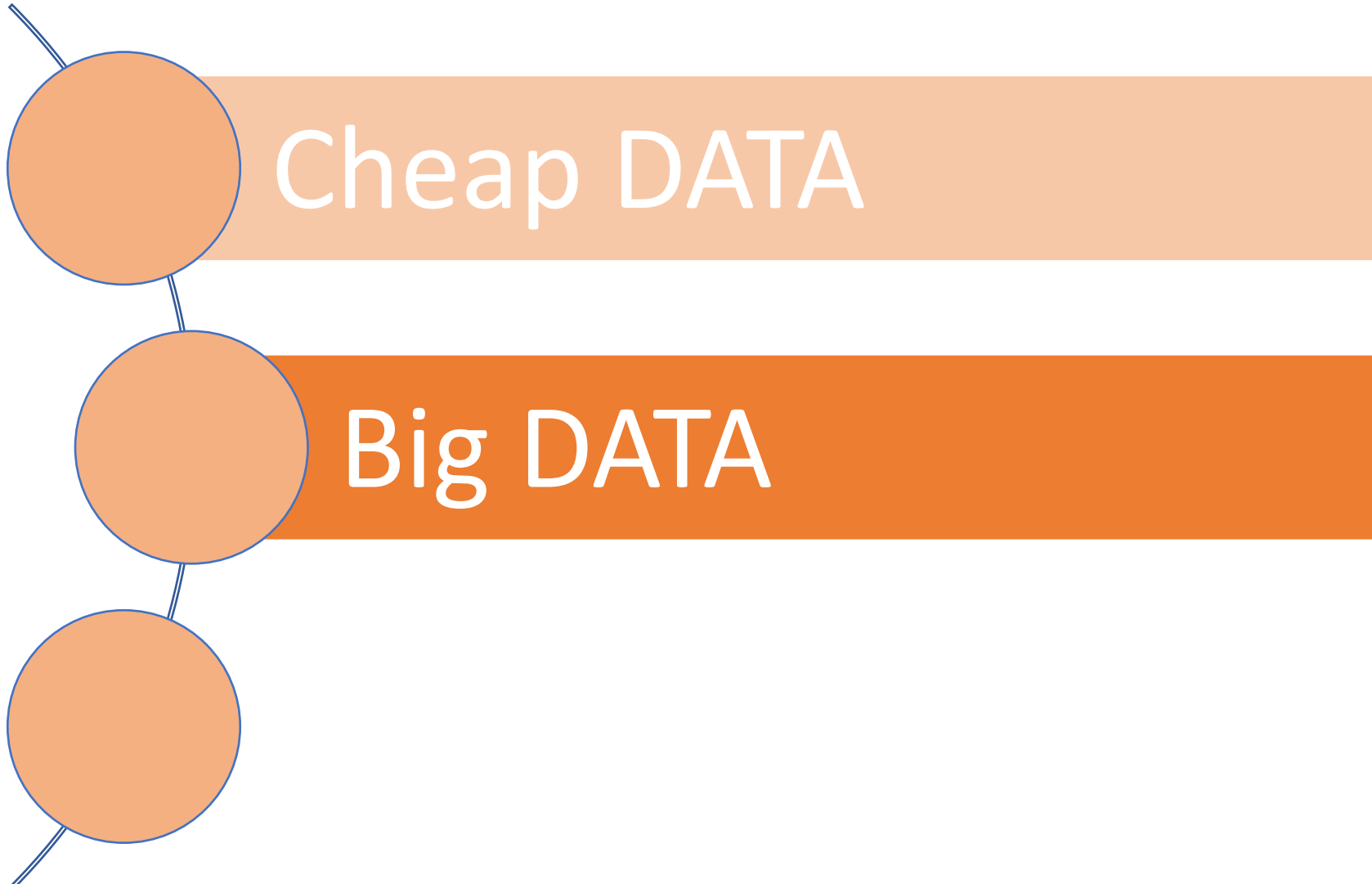


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[order now](#)**\$199**

WHY DATA SCIENCE ?



Cheap DATA

Big DATA



Accession	Title	Reported trait	Ancestry Category	No of Individuals
G22217	Analysis of Genetic Host Response Risk Factors in Severe COVID-19 Patients	Severe response to COVID-19 (Sepsis controls)	European	2083
G22218	Analysis of Genetic Host Response Risk Factors in Severe COVID-19 Patients	Severe response to COVID-19 (Sepsis controls)	European	2083
G22219	Identification and Analysis of Shared Risk Factors in Sepsis and High Mortality Risk COVID-19 Patients	Sepsis	European	12696
G22220	Identification and Analysis of Shared Risk Factors in Sepsis and High Mortality Risk COVID-19 Patients	Sepsis	European	12696
G11993	Clinical laboratory test-wide association scan of polygenic scores identifies biomarkers of complex disease	Alpha 1 antitrypsin [Mass/volume] in Serum or Plasma	European	1756
G26371	GWAS and ExWAS of blood Mitochondrial DNA copy number identifies 71 loci and highlights a potential causal role in dementia	mitochondrial DNA copy number	European South Asian African unspecified	395718
G26372	GWAS and ExWAS of blood Mitochondrial DNA copy number identifies 71 loci and highlights a potential causal role in dementia	mitochondrial DNA copy number	European	383476
G26373	GWAS and ExWAS of blood Mitochondrial DNA copy number identifies 71 loci and highlights a potential causal role in dementia	mitochondrial DNA copy number	South Asian	6172
G26374	GWAS and ExWAS of blood Mitochondrial DNA copy number identifies 71 loci and highlights a potential causal role in dementia	mitochondrial DNA copy number	African unspecified	6133

Growth in healthcare data

1 exabyte = 1 billion gigabytes

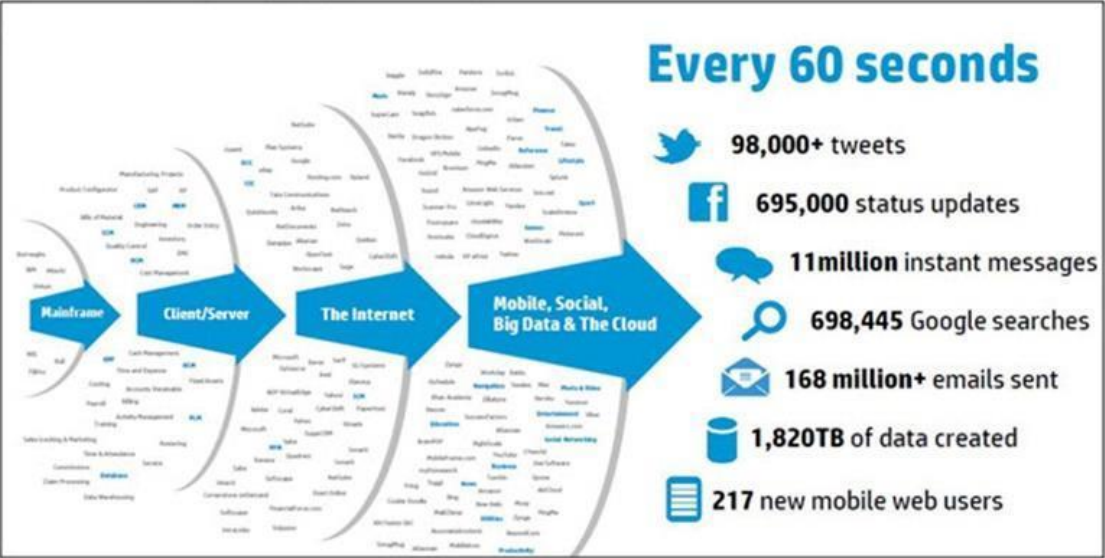
Source: Stanford Medicine 2017, IDC 2014



2013
153
EXABYTES



2020
2,314
EXABYTES

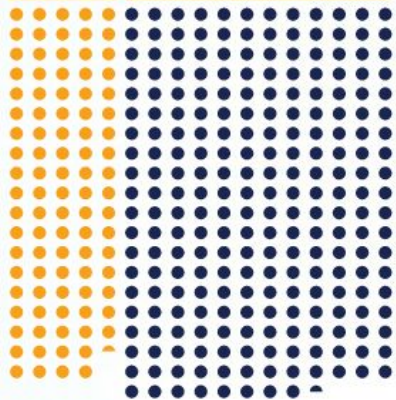


TO PUT THAT INTO PERSPECTIVE,
DATA CENTERS GLOBALLY
will only have enough room for an estimated
985 exabytes by 2020—
meaning that almost two and a half times this
capacity would be required to house all the
healthcare data.

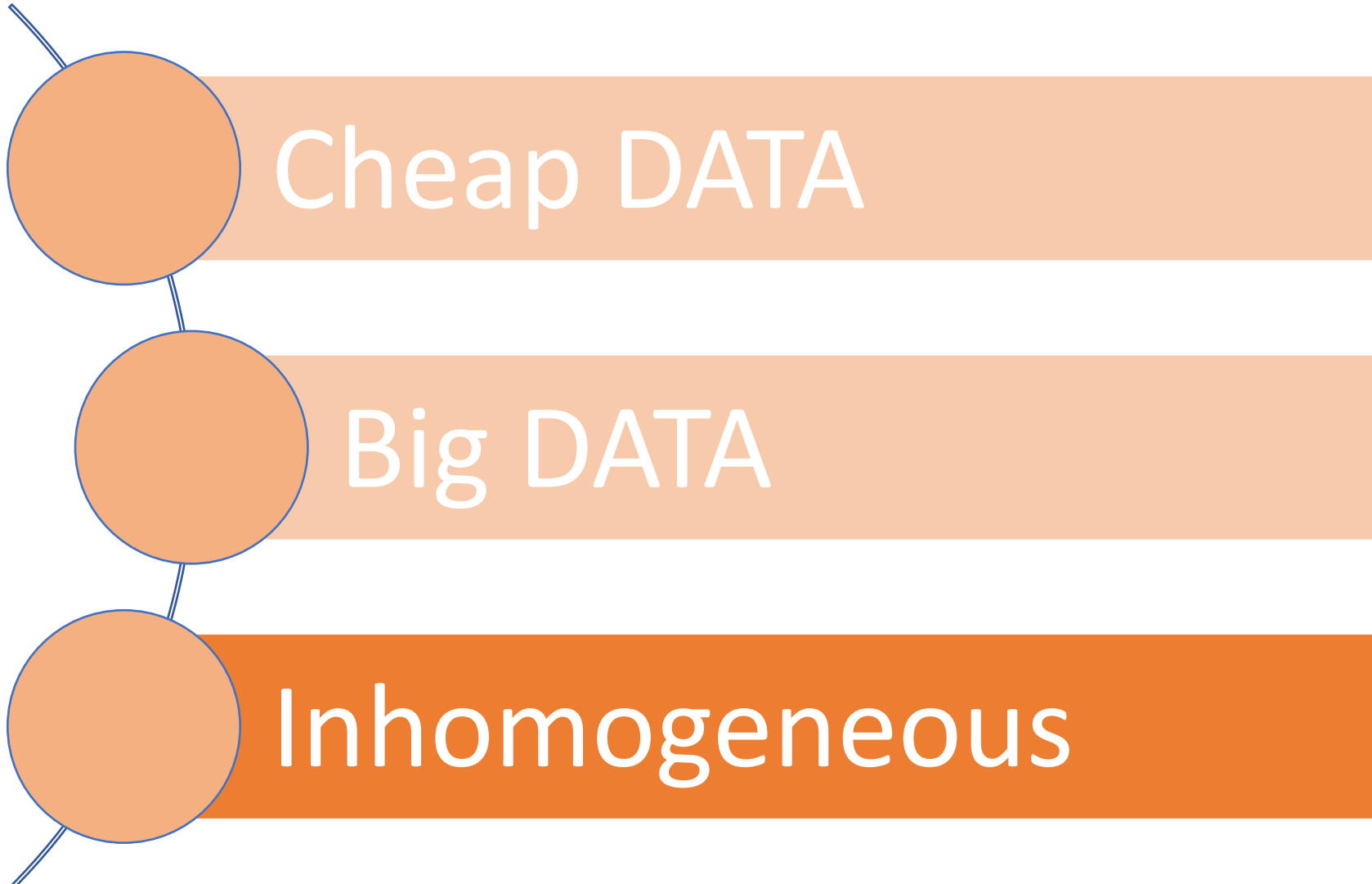
Source: Cisco Global Cloud Index 2016

DATA STORAGE VS MEDICAL DATA (2020)

STORAGE CAPACITY	MEDICAL DATA GENERATED
985 EXABYTES	2,314 EXABYTES



WHY DATA SCIENCE ?

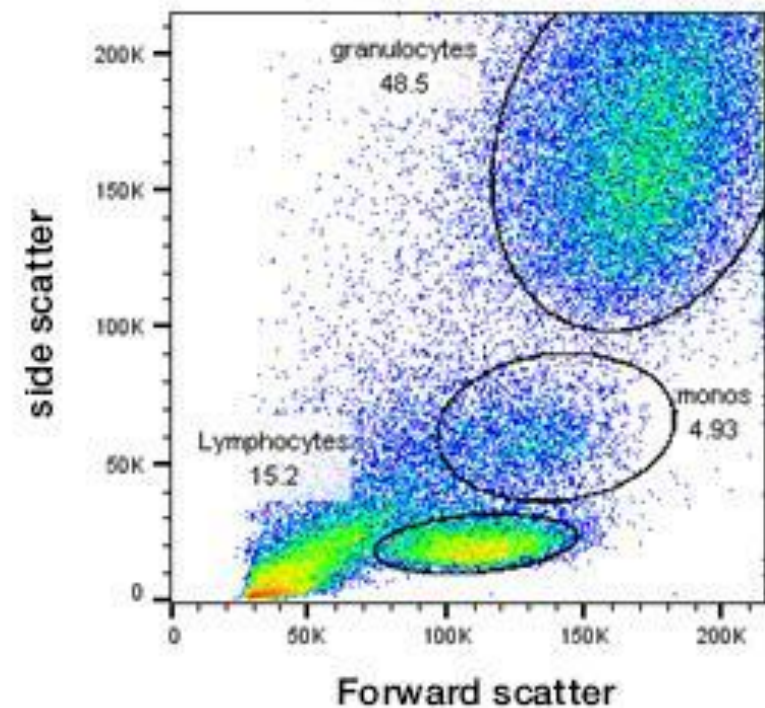
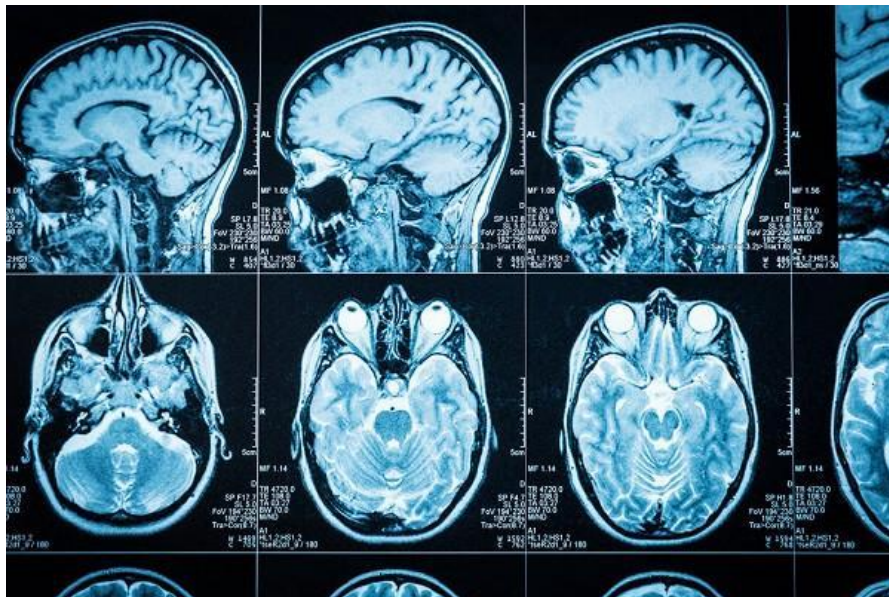
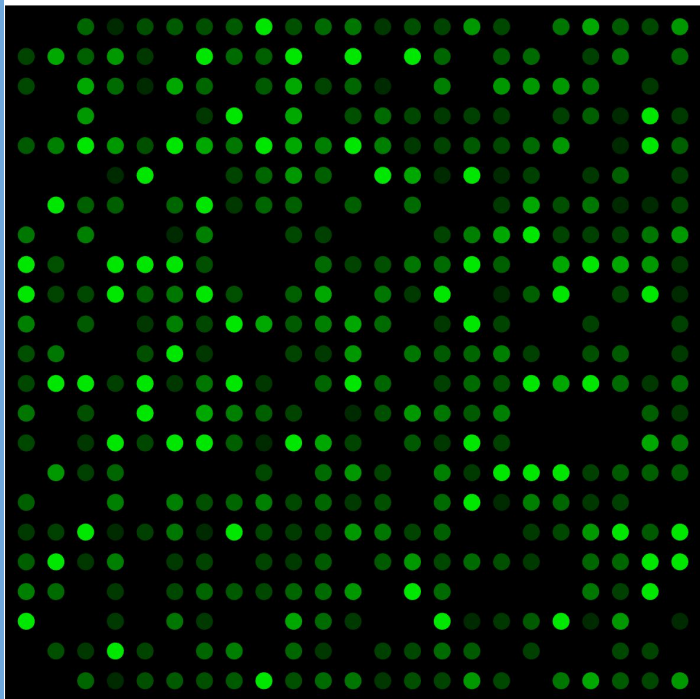


Cheap DATA

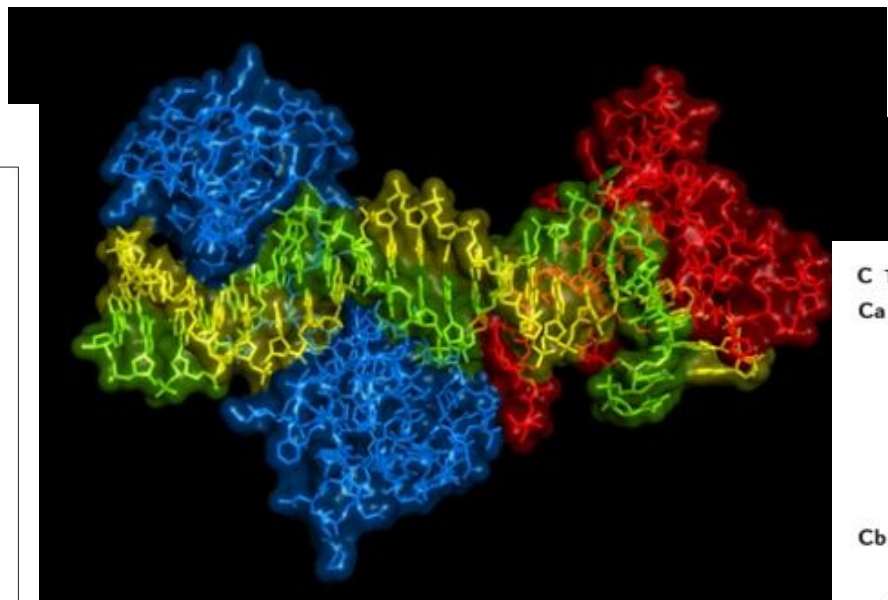
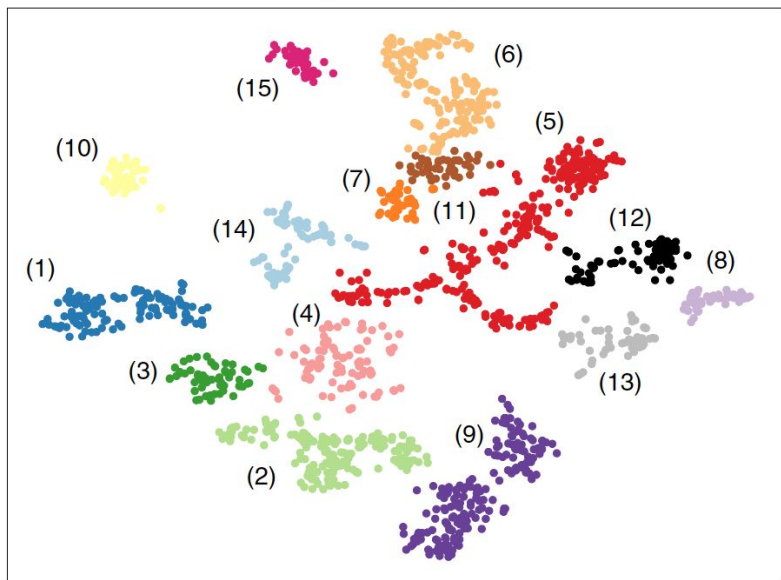
The diagram consists of three orange circles arranged vertically on the left side, connected by thin blue lines. Each circle is positioned to the left of a horizontal bar. The top bar is light orange and contains the text 'Cheap DATA'. The middle bar is also light orange and contains the text 'Big DATA'. The bottom bar is a darker orange and contains the text 'Inhomogeneous'. The circles have a blue outline and a thin blue line passing through them, suggesting a path or flow.

Big DATA

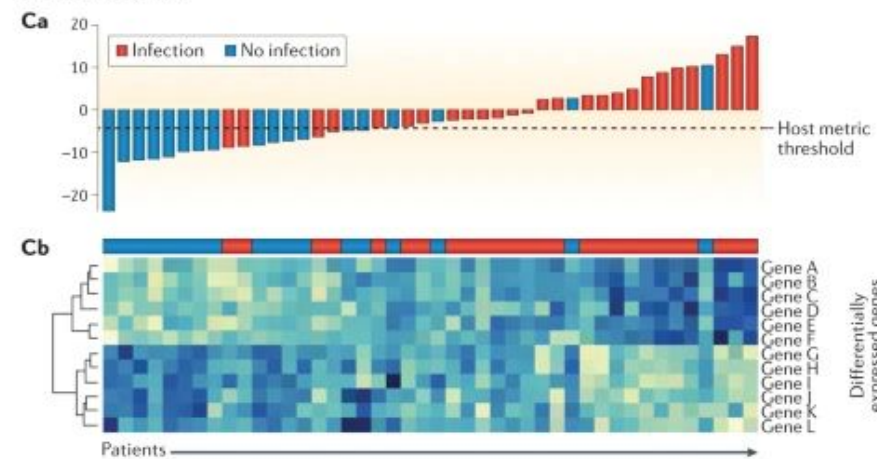
Inhomogeneous



a LHA^{Glut} (*Slc17a6*⁺) clusters



C Transcriptomics





BIG DATA





BIG DATA



RESEARCH

COLLECTION

DATABASE

STORAGE

RESOURCES

VOLUME

ANALYSIS



BIG DATA

RESEARCH COLLECTION

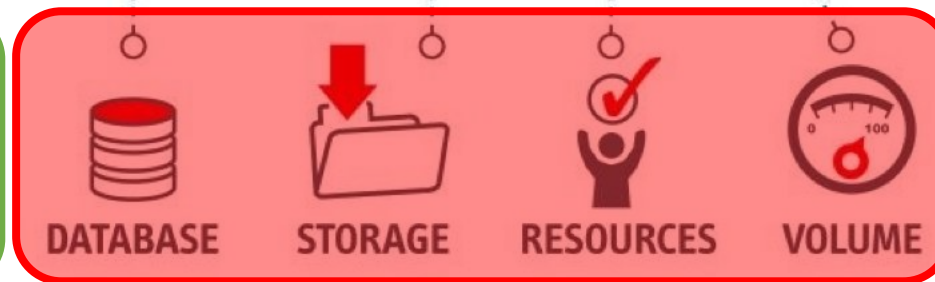
DATABASE STORAGE RESOURCES VOLUME

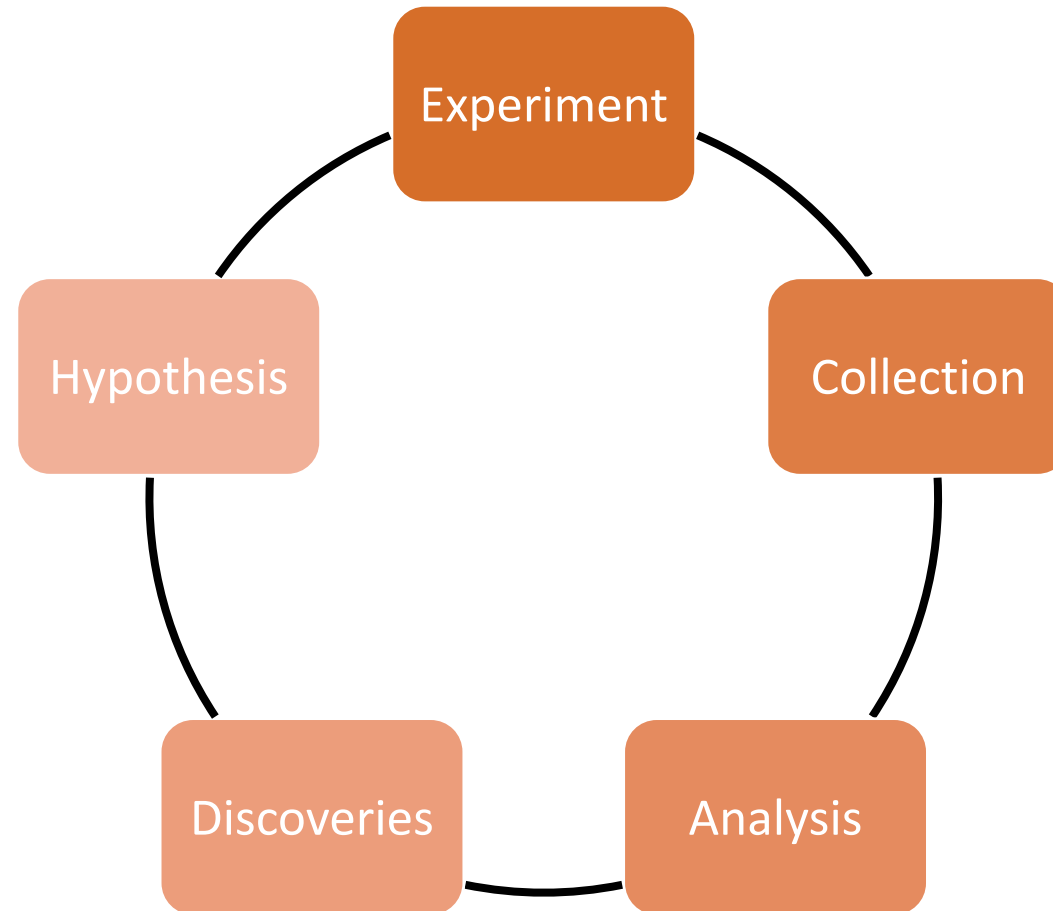


ANALYSIS

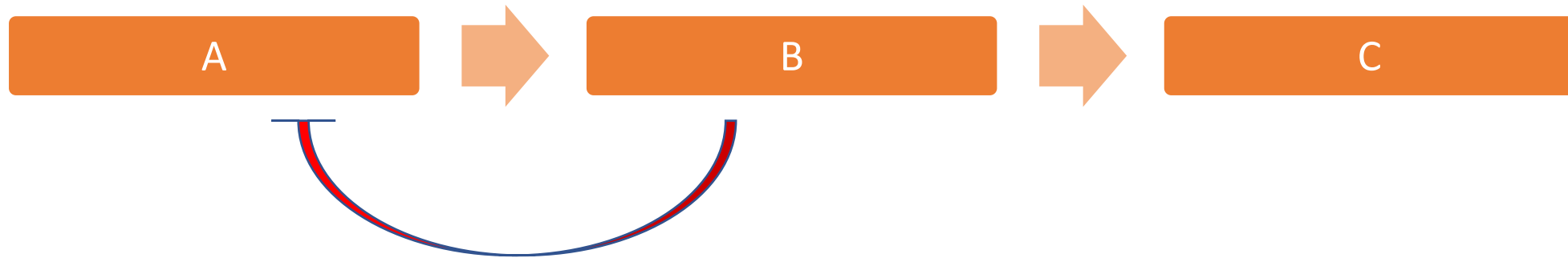
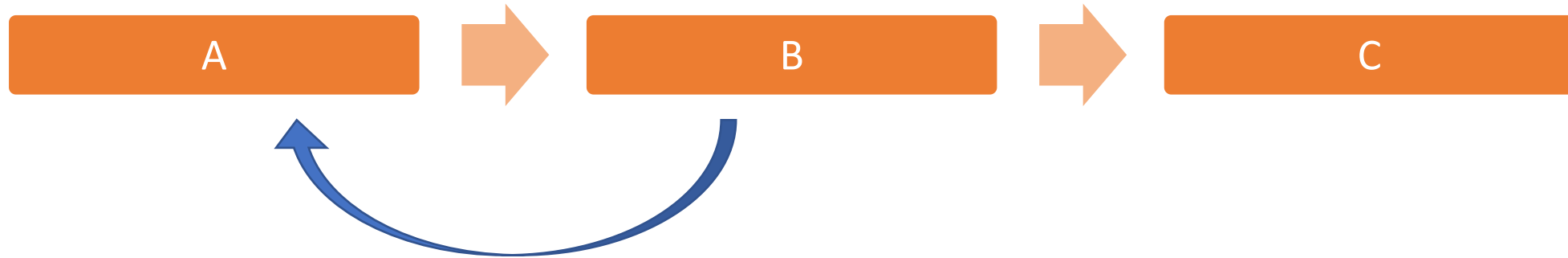


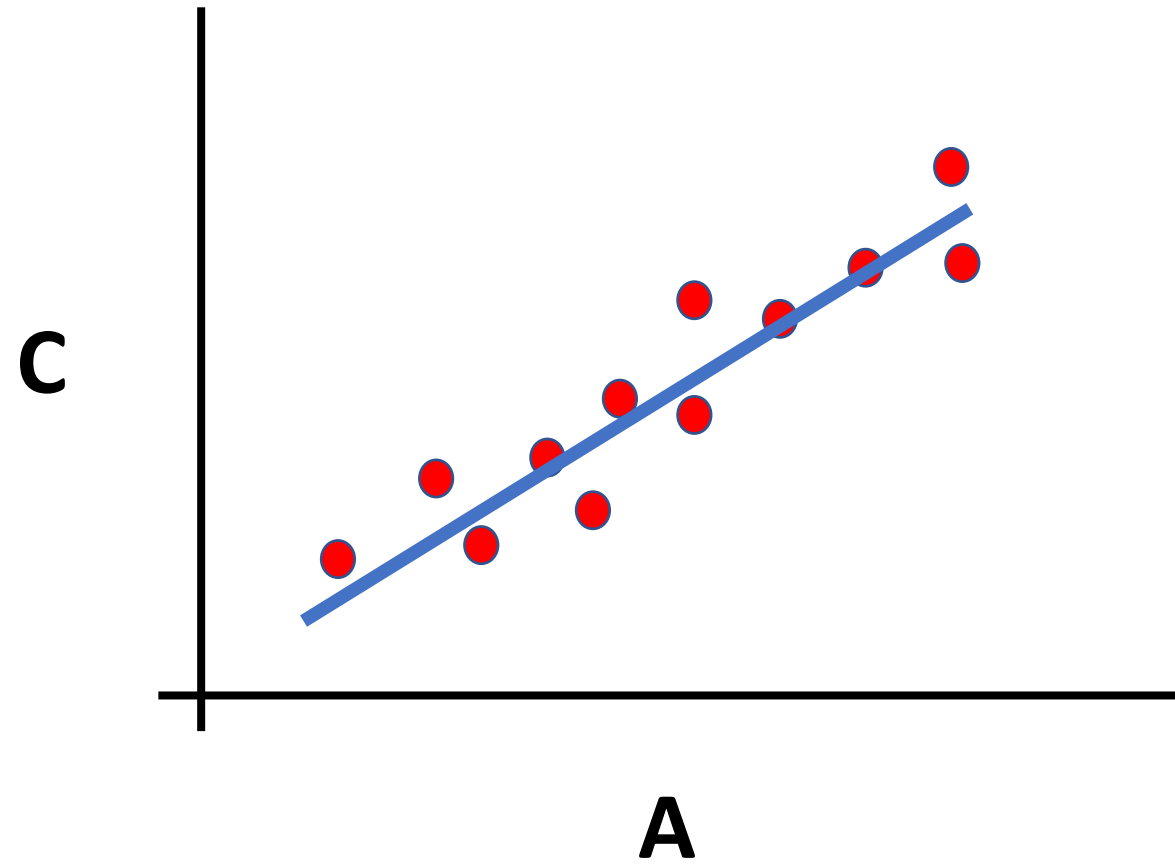
BIG DATA

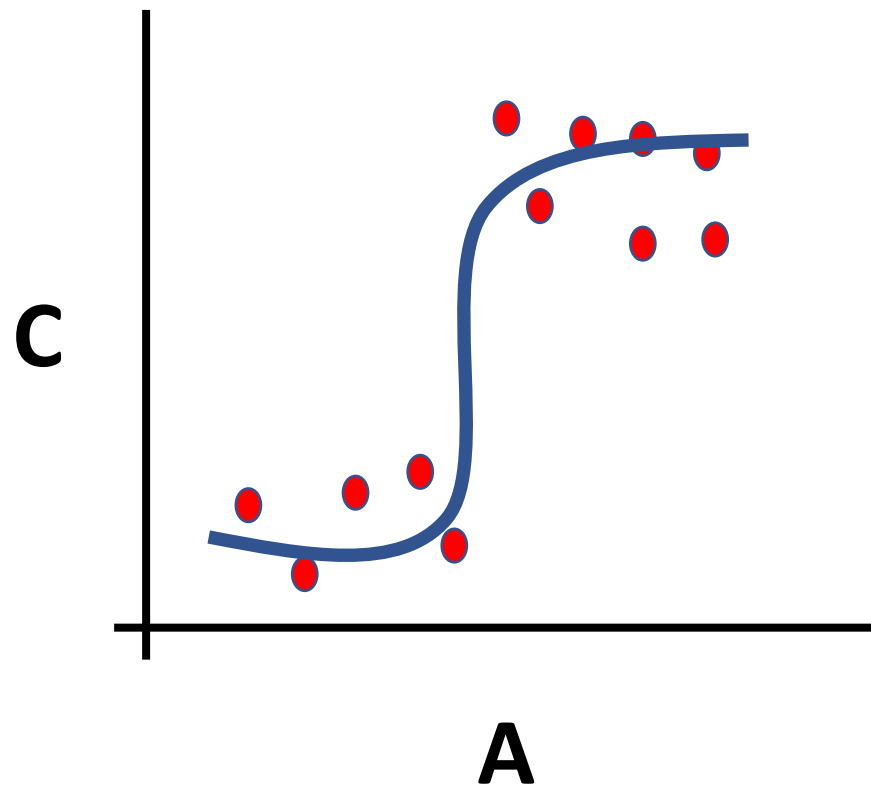
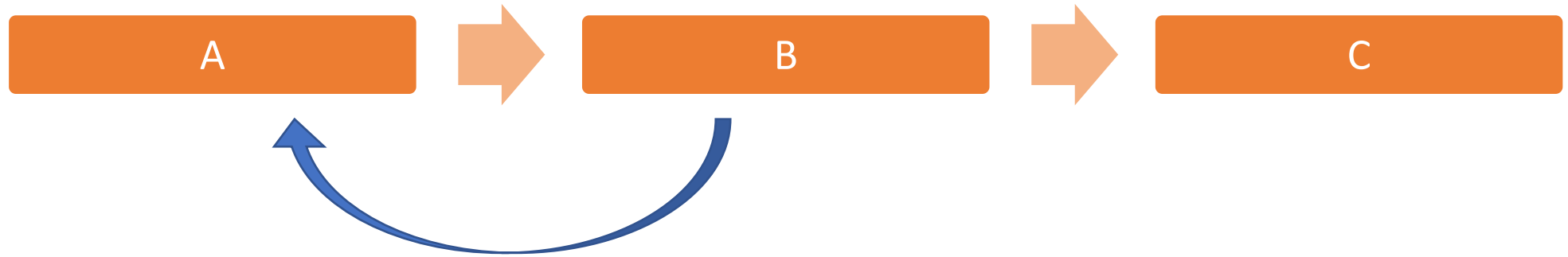


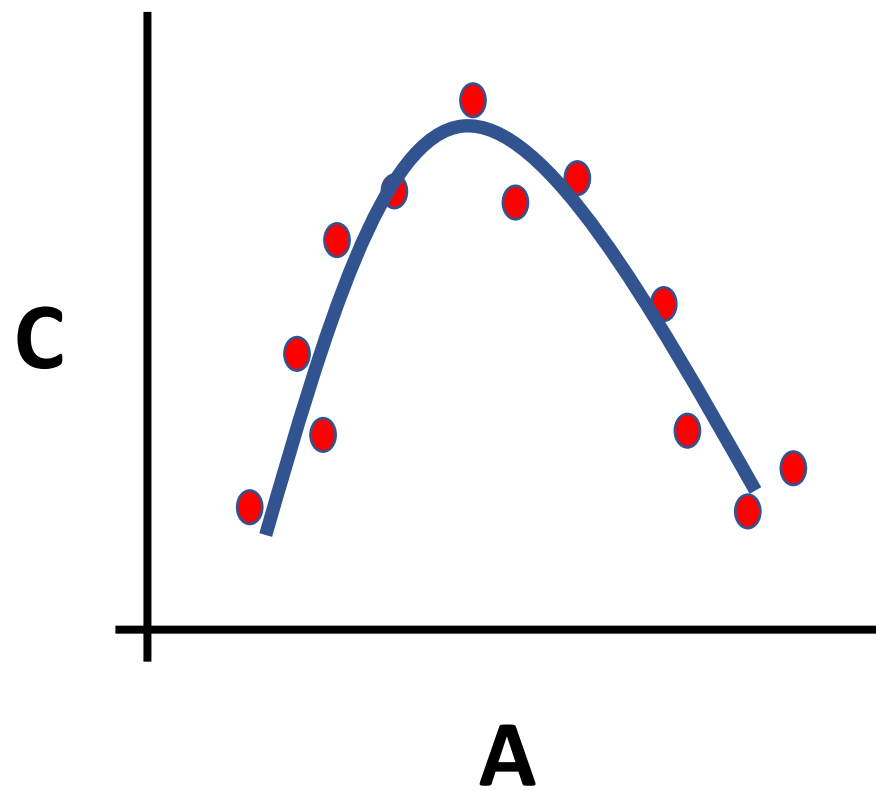
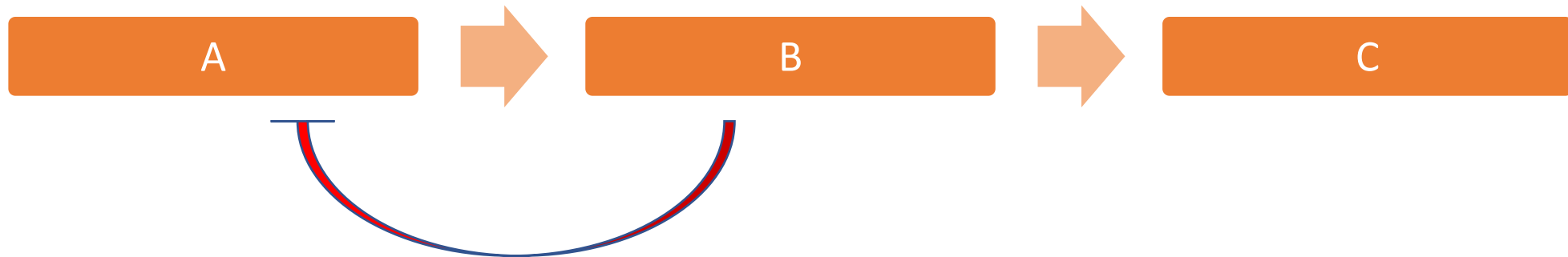


Phisolophical problem









Complex & Complicated

- Rats in India
- Nemo and fishes
- Pirates and corals
- El nino and revolution

Space Chem

- Videogame:
 - Small number of instruction
 - Sequential pathway
 - Building block with bottom up building
 - Multi scale

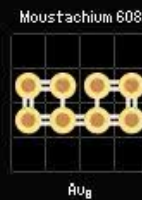
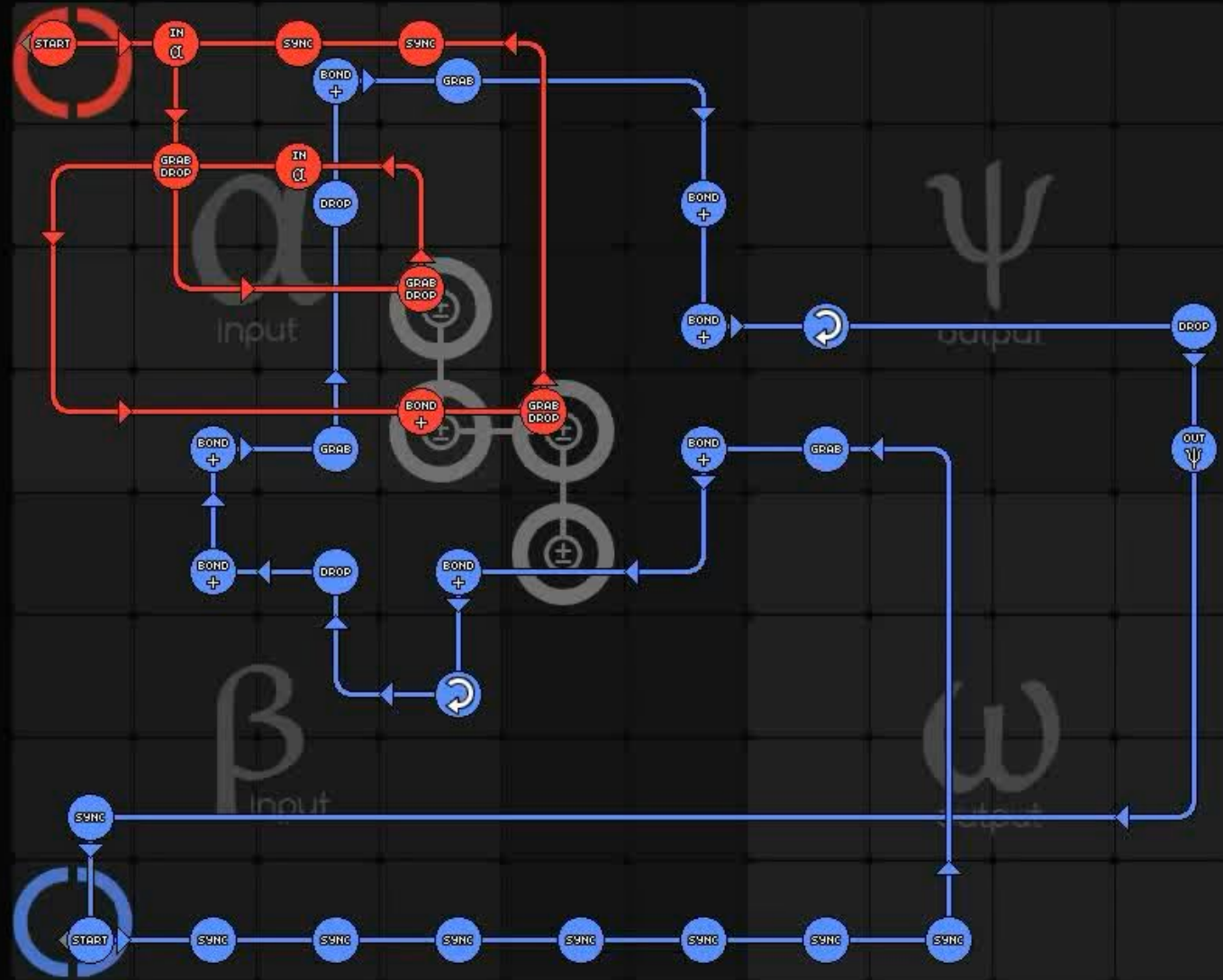
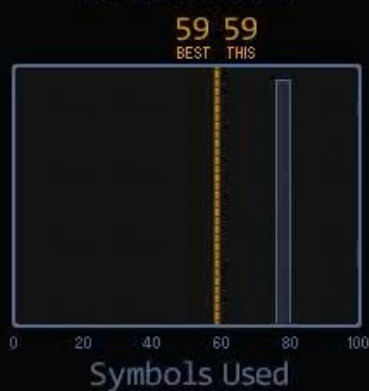
Videos of gameplay

Alkonost



Return to Menu



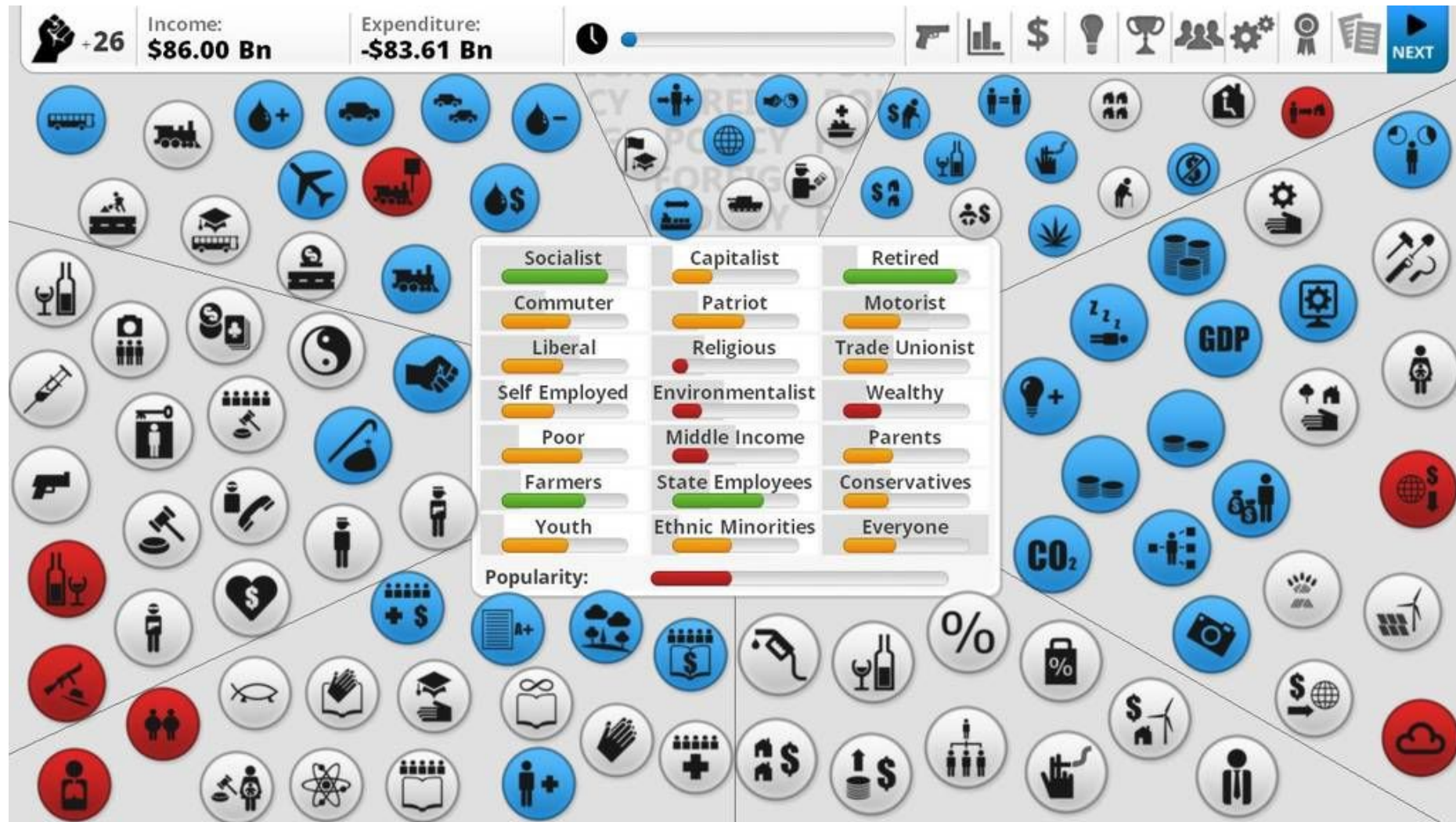


DISABLED
This output is disabled.

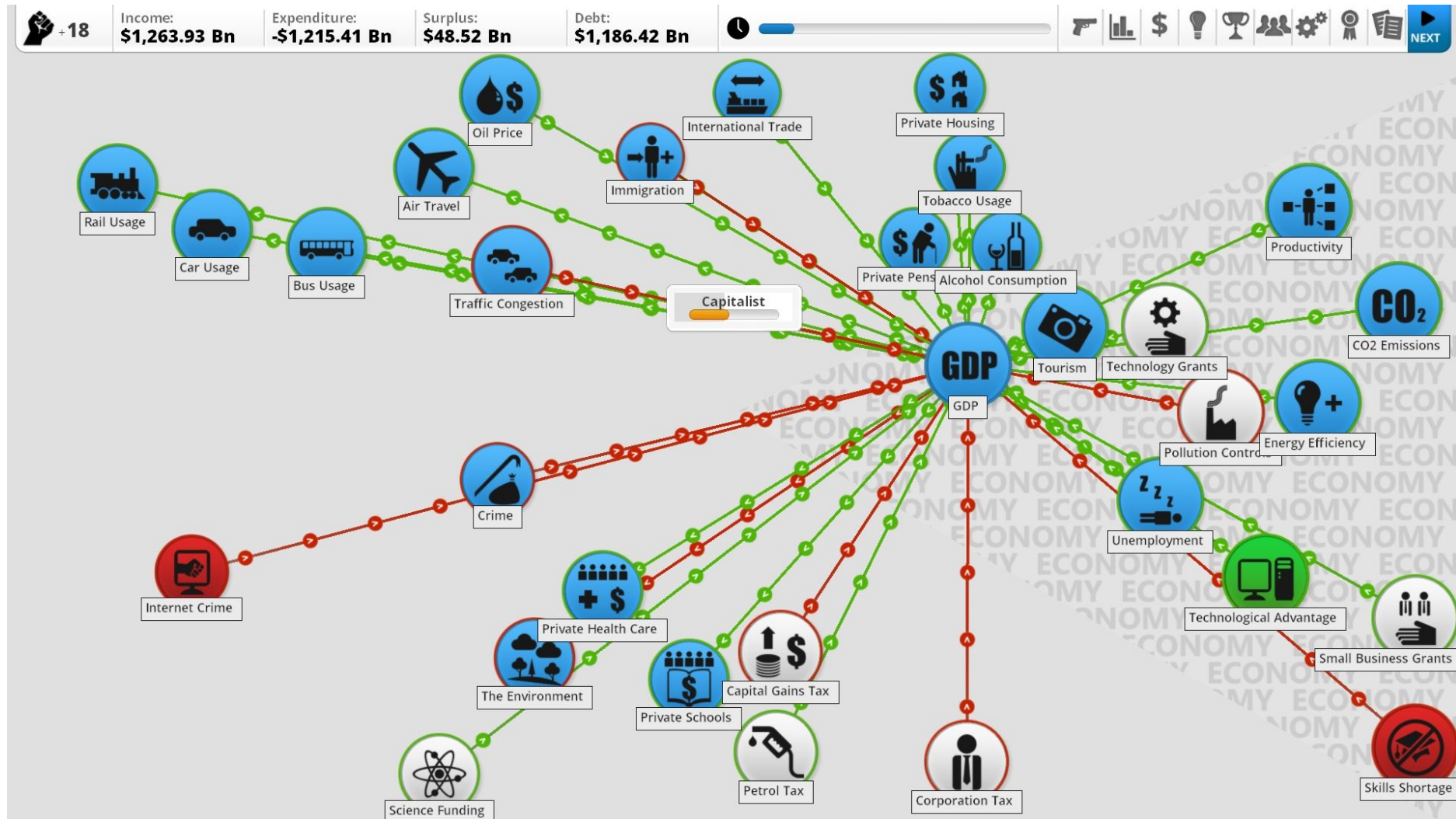
Democracy 3

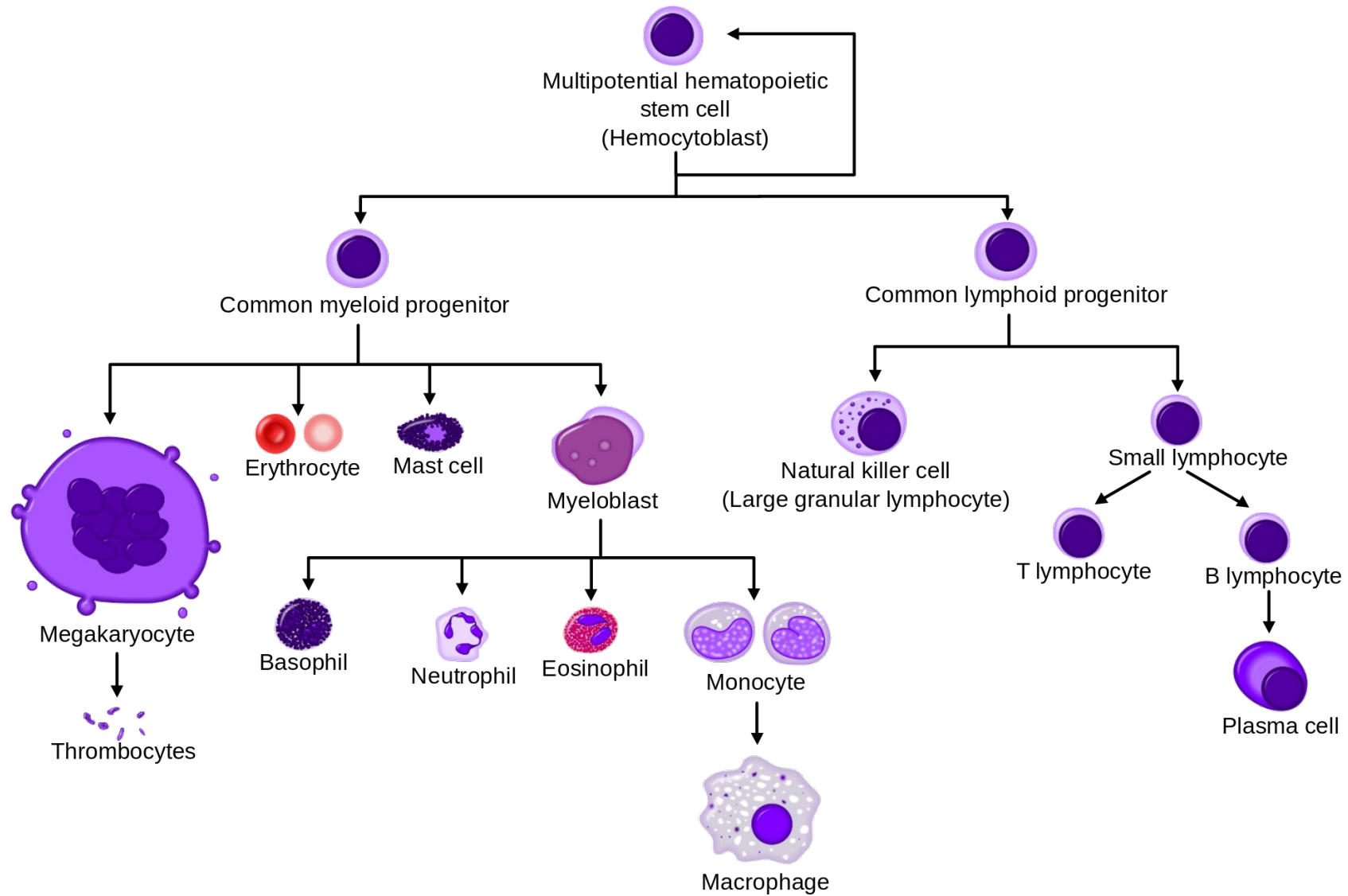


Democracy 3



Democracy 3

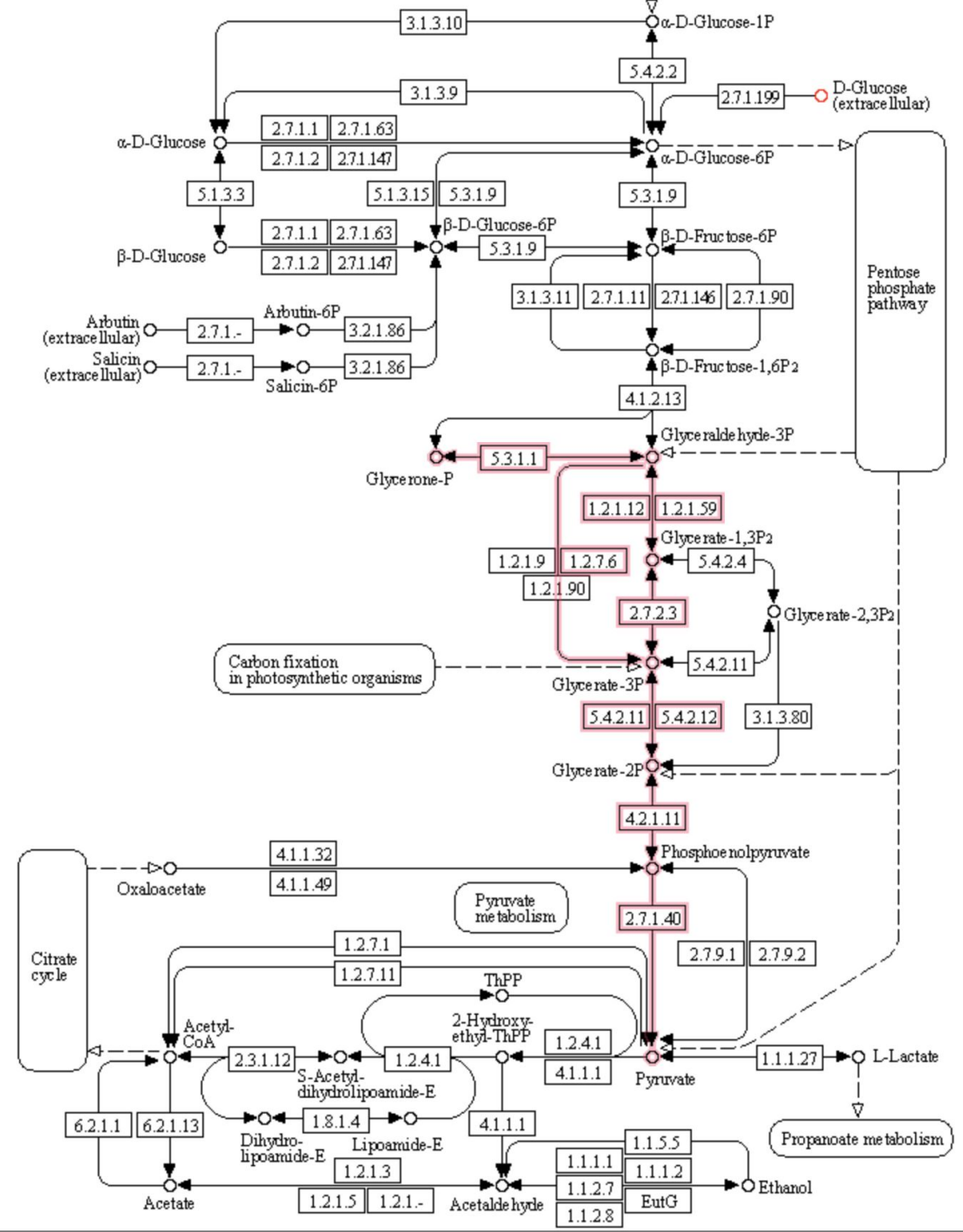


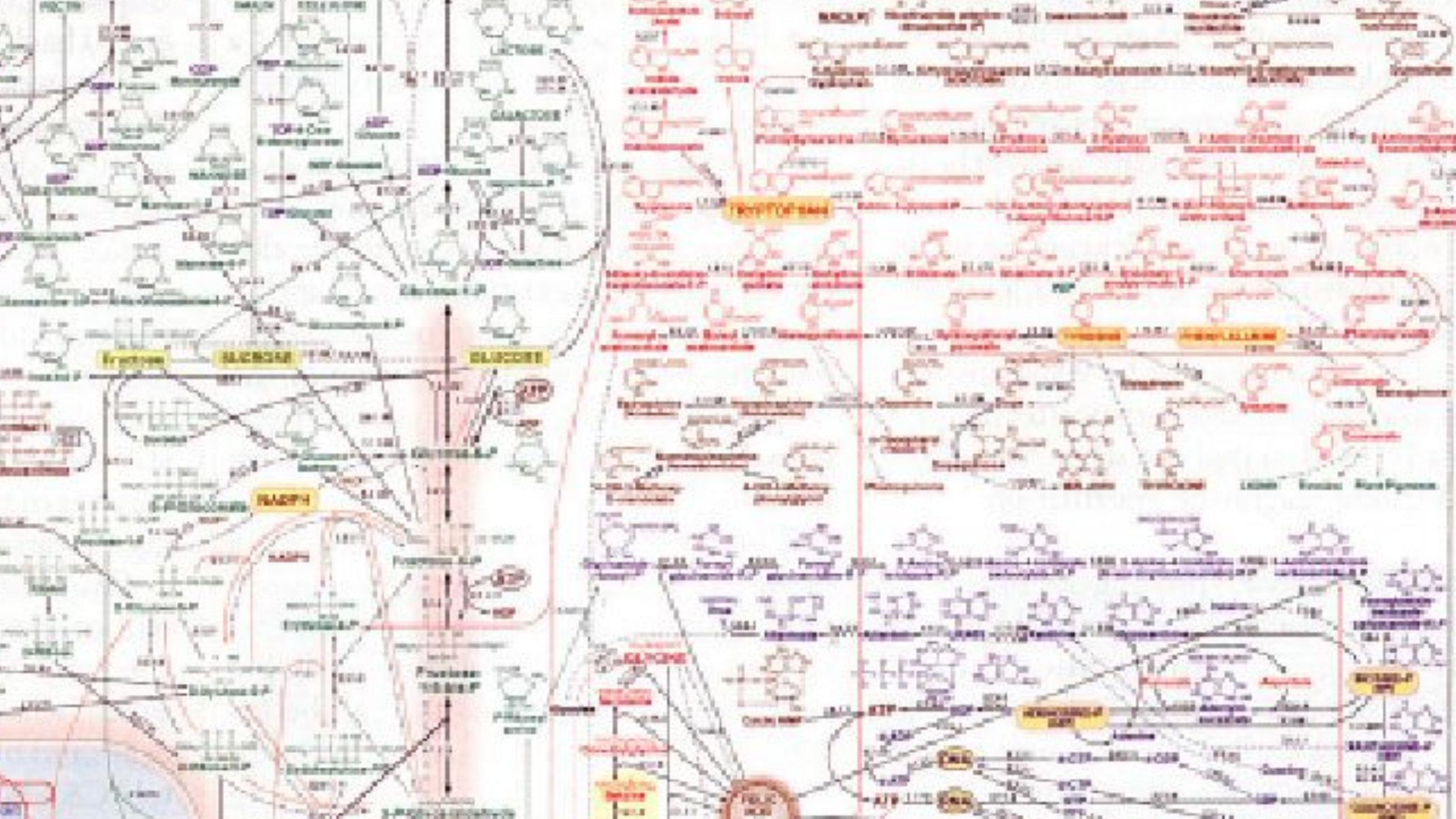


Building blocks

Metabolic scale

Scales up





Approaches in Biological Data Science

Descriptive

- Exploratory
- No ground truth

Machine Learning

- A lot of data
- Black box

Mechanistic

- Hard to build
- Structured data

DESCRIPTIVE METHODS

- **Dimension reduction**

- Projection in smaller space

- **Graph description**

- Human readable

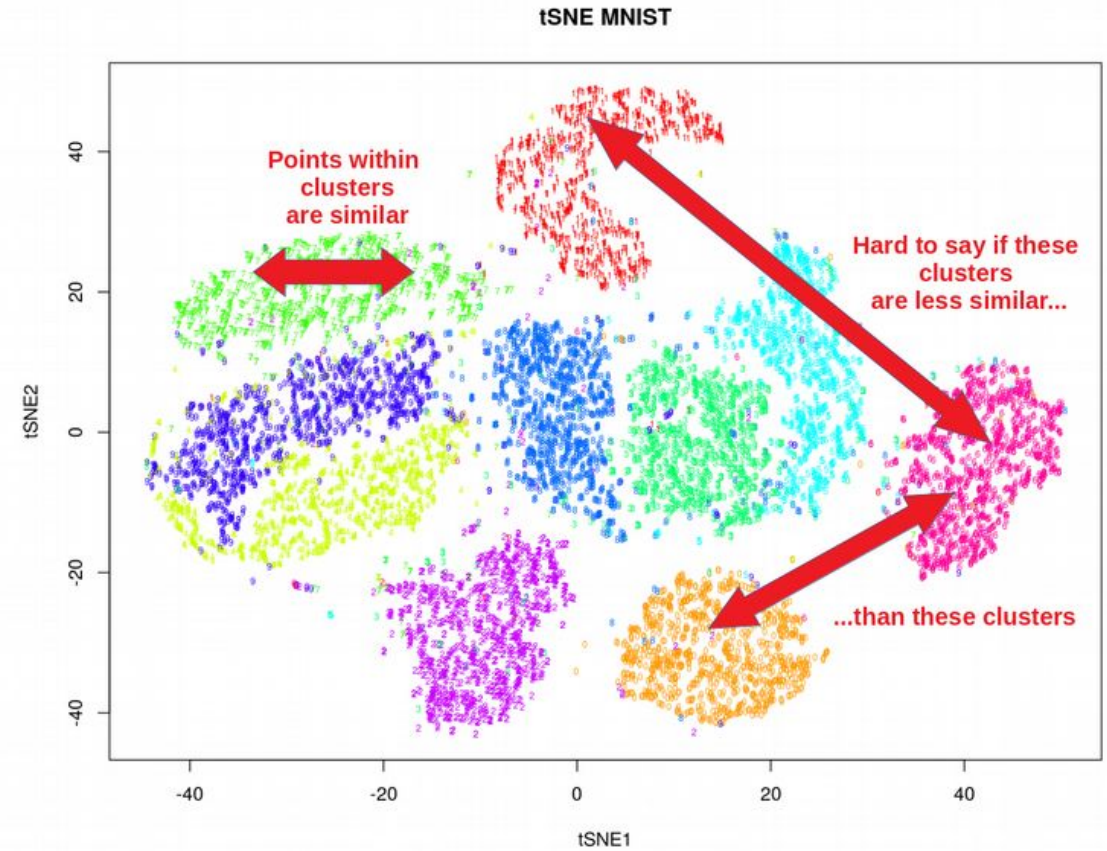
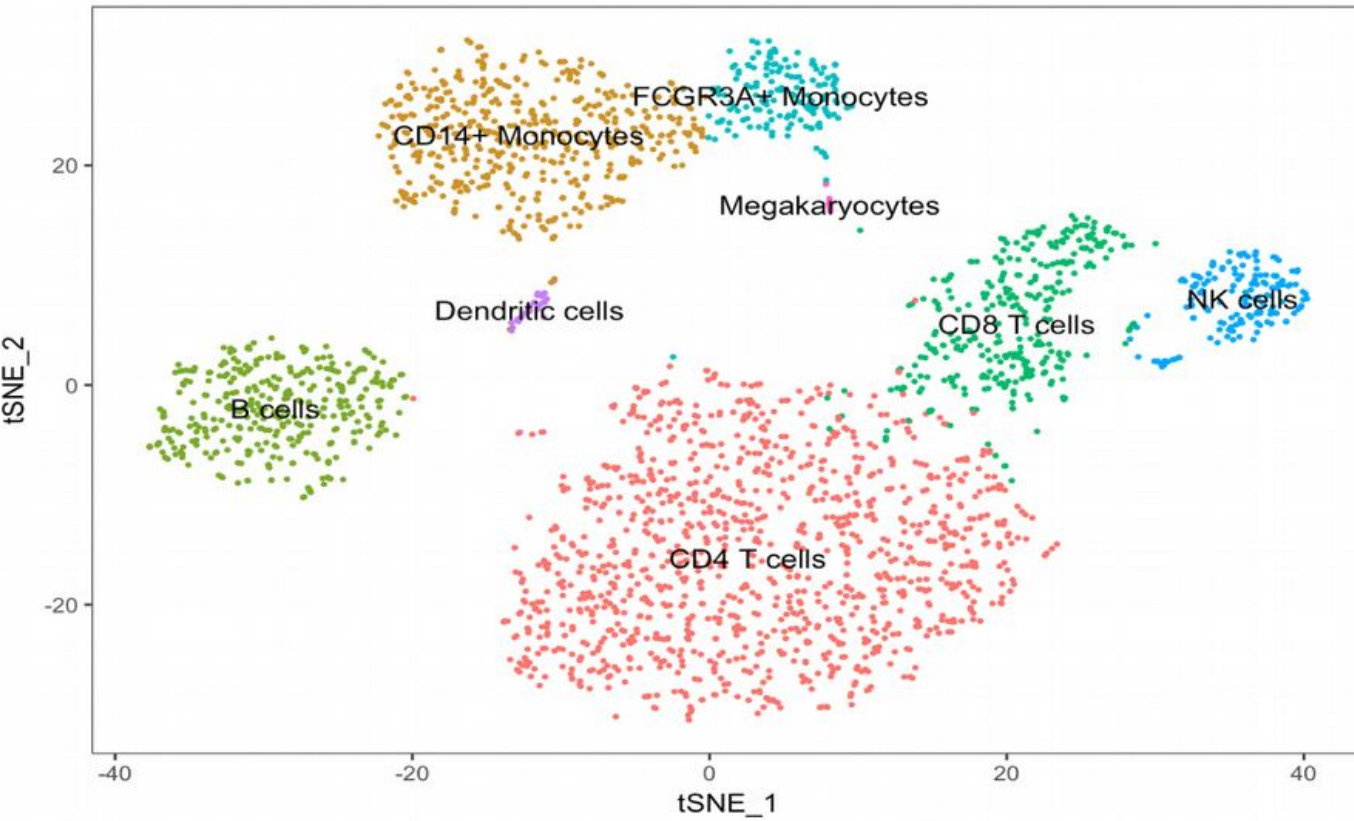
- **Clustering**

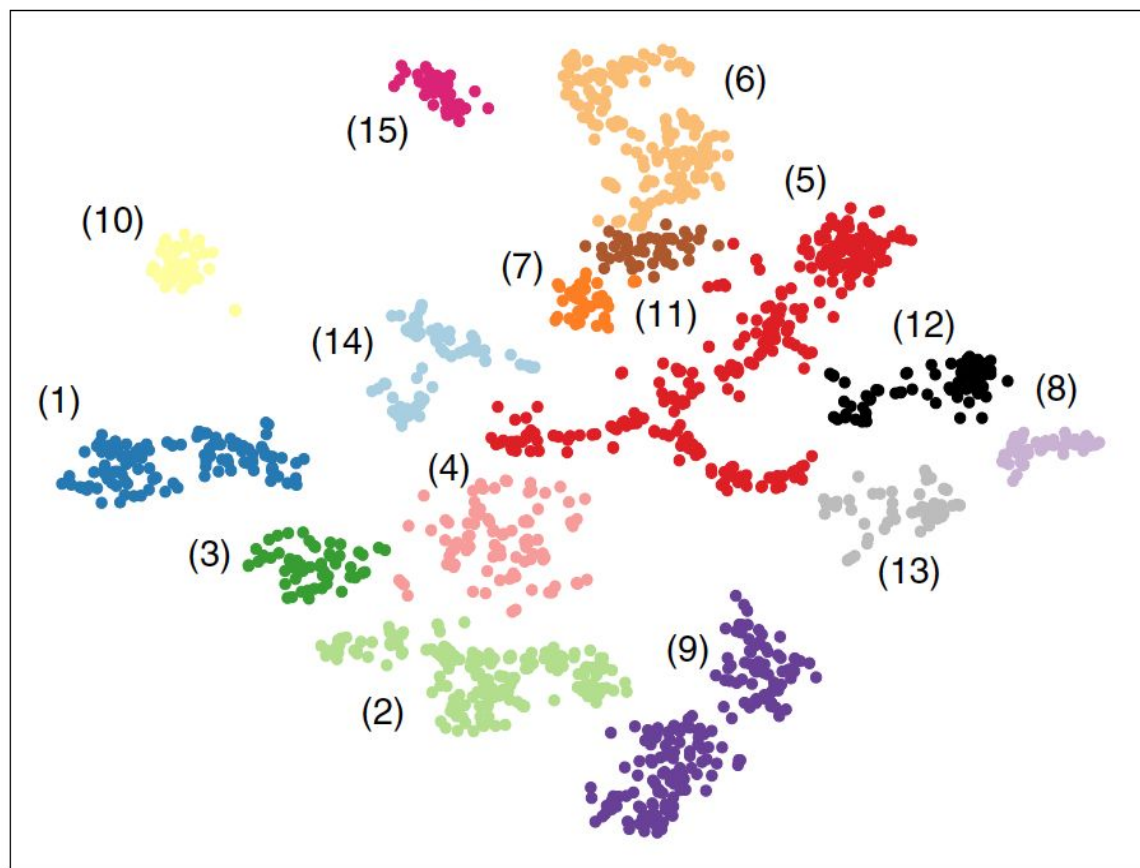
- Unsupervised learning / Data mining

PCA, MDS, t-SNE, UMAP

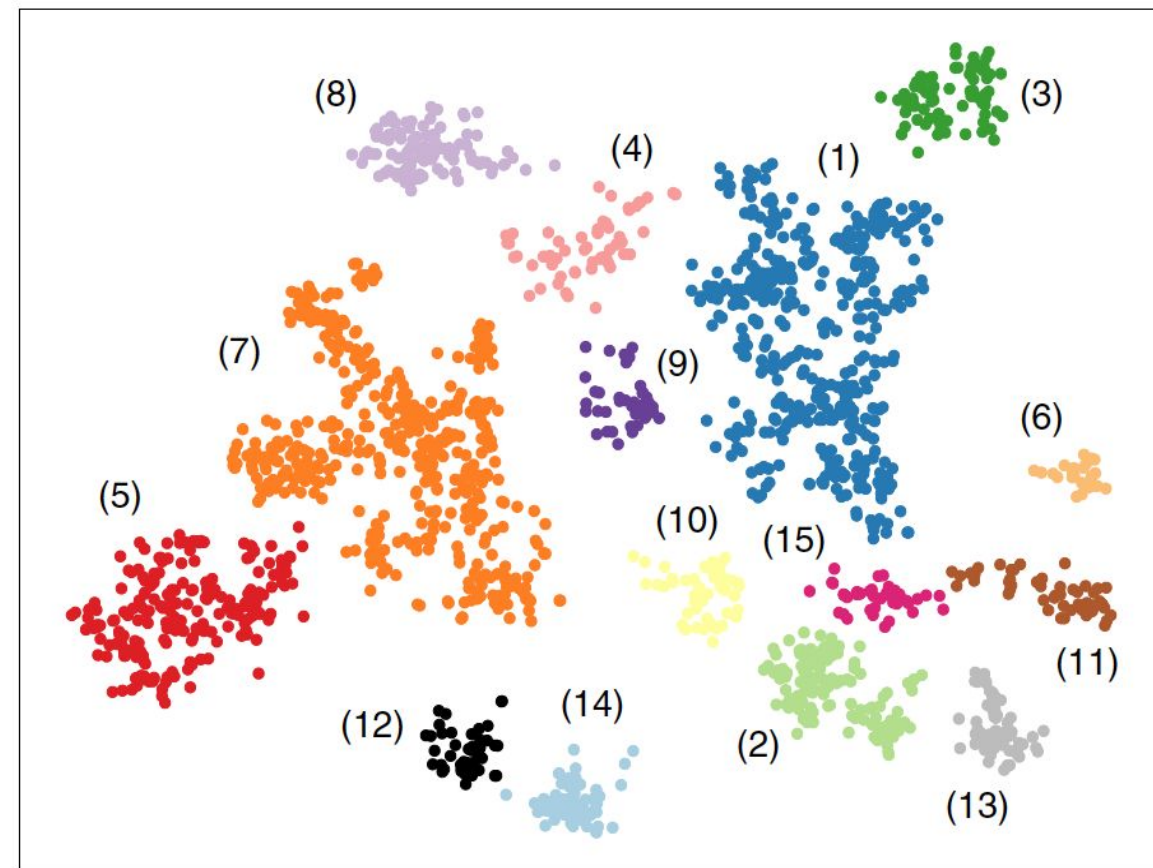
K-means, Spectral

Example: *t*-sne



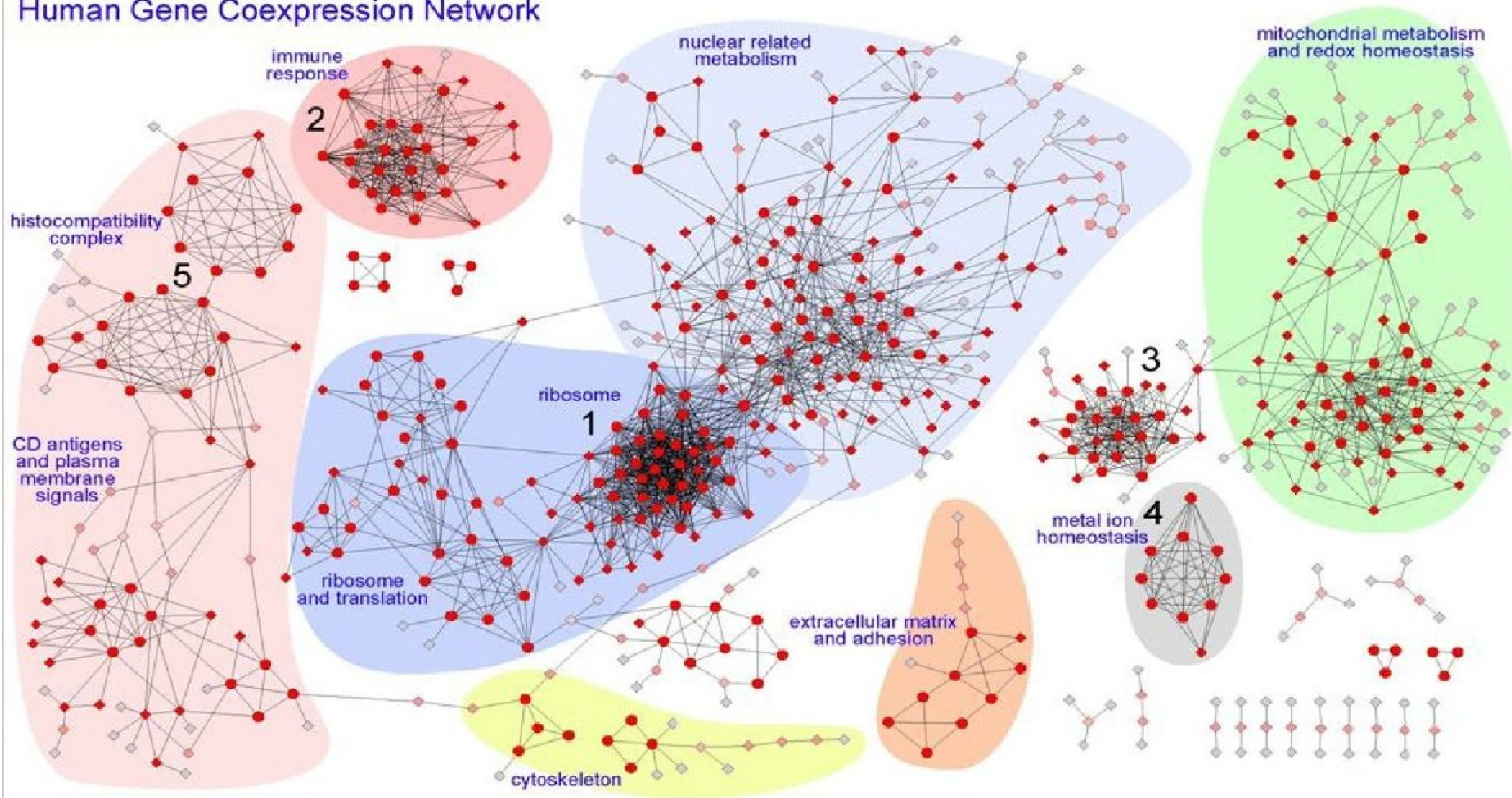
aLHA^{Glut} (*Slc17a6*⁺) clusters

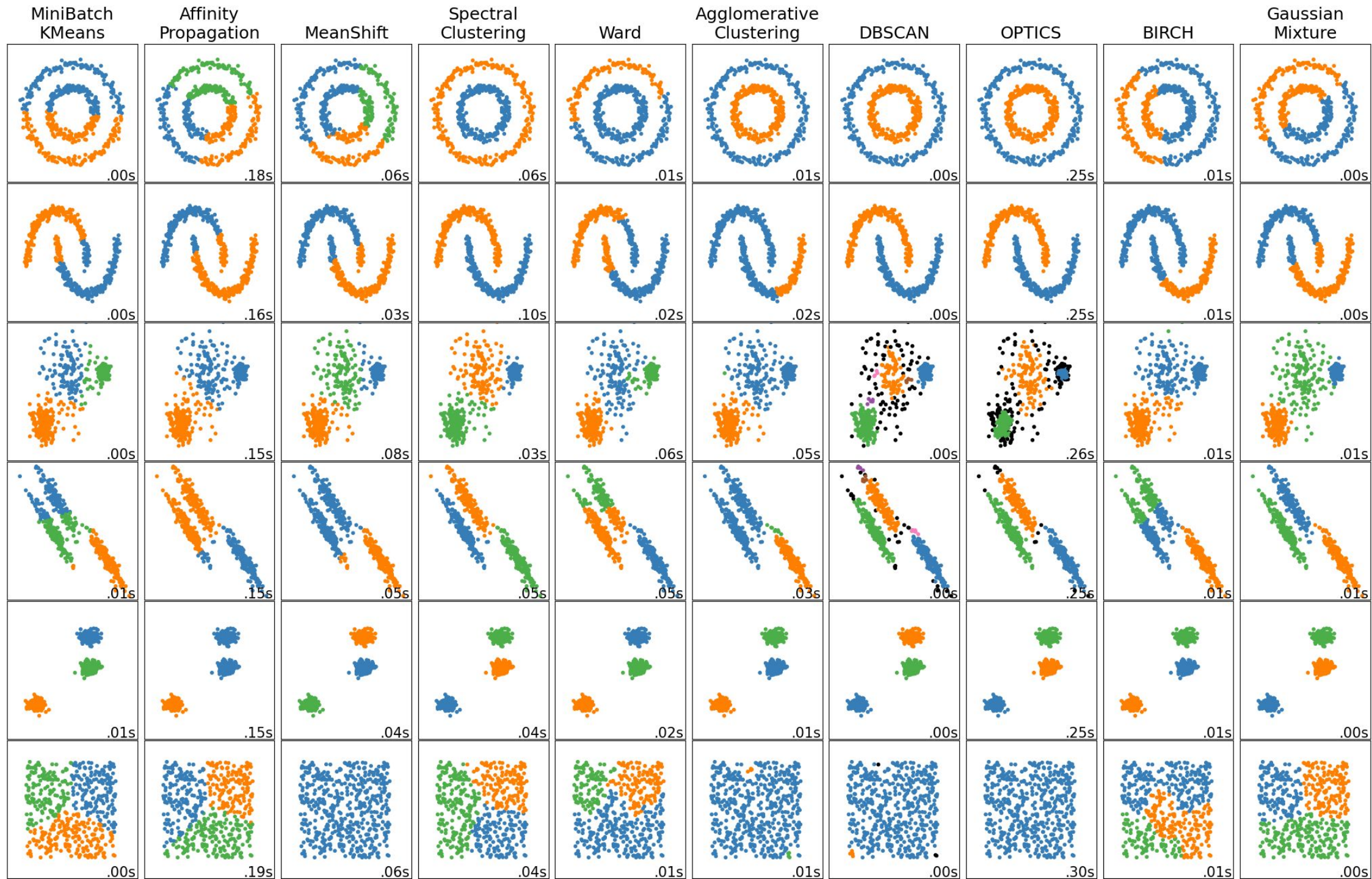
1	<i>Pmch + Gad1</i>	9	<i>Synpr + Gad1</i>
2	<i>Nrgn + Gda</i>	10	<i>Grp + Cck</i>
3	<i>Zic1 + Gad1</i>	11	<i>Calca + Col27a1</i>
4	<i>Tac1 + Pitx2</i>	12	<i>Trh + Syt2</i>
5	<i>Ebf3 + Otp</i>	13	<i>Syt2 + Meis2 + Gad1</i>
6	<i>Hcrt</i>	14	<i>Otp + Gpr101</i>
7	<i>Gpr101 + Tcf4</i>	15	<i>Sst</i>
8	<i>Trh + Cbln2</i>		

cLHA^{GABA} (*Slc32a1*⁺) clusters

1	<i>Gal + Dlk1</i>	9	<i>Cbln2 + Calb1</i>
2	<i>Npy + Npw</i>	10	<i>Sst + Meis2</i>
3	<i>Nts + Cartpt</i>	11	<i>Col25a1 + Otp</i>
4	<i>Unassigned</i>	12	<i>Th + Slc18a2</i>
5	<i>Meis2 + Calb2</i>	13	<i>Sst + Otp</i>
6	<i>Sst + Col25a1</i>	14	<i>Atp1a2</i>
7	<i>Tac2 + Serpini1</i>	15	<i>Calb1 + Cbln4</i>
8	<i>Lhx6 + Tcf4</i>		

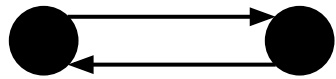
Human Gene Coexpression Network



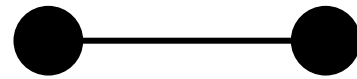


Metabolic networks

Networks : Objects (**nodes**) linked by links (**edges**)

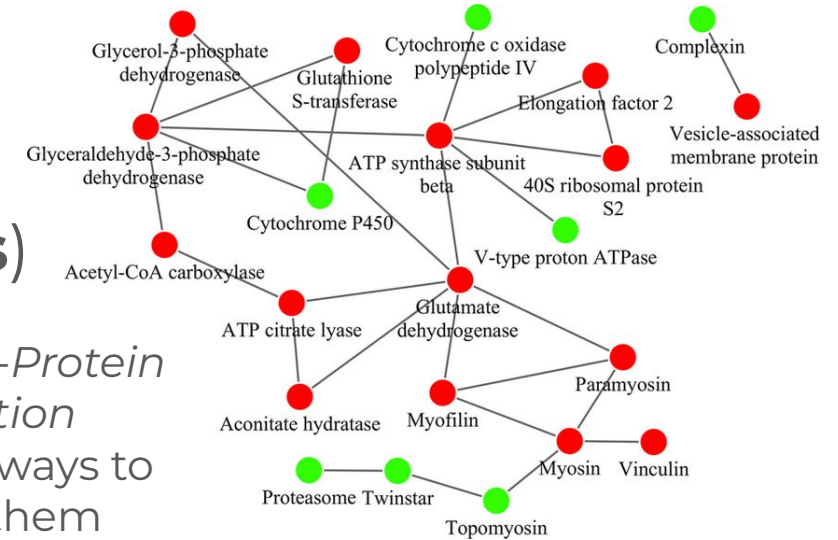


Directed



Undirected

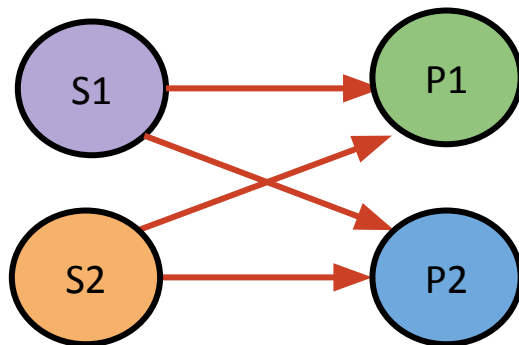
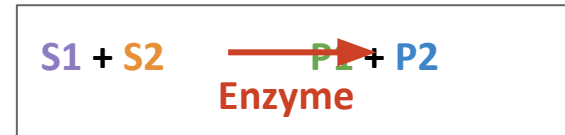
Ex : *Protein-Protein Interaction network*, 2 ways to identify them (colors)



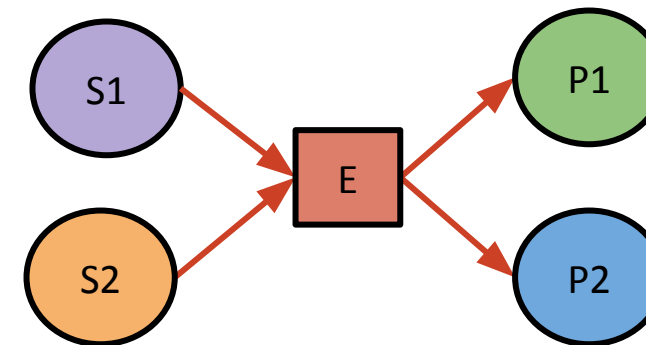
Wang, 2015

Metabolic networks : Representation of all chemical reactions

Eg : Different representations for reaction:

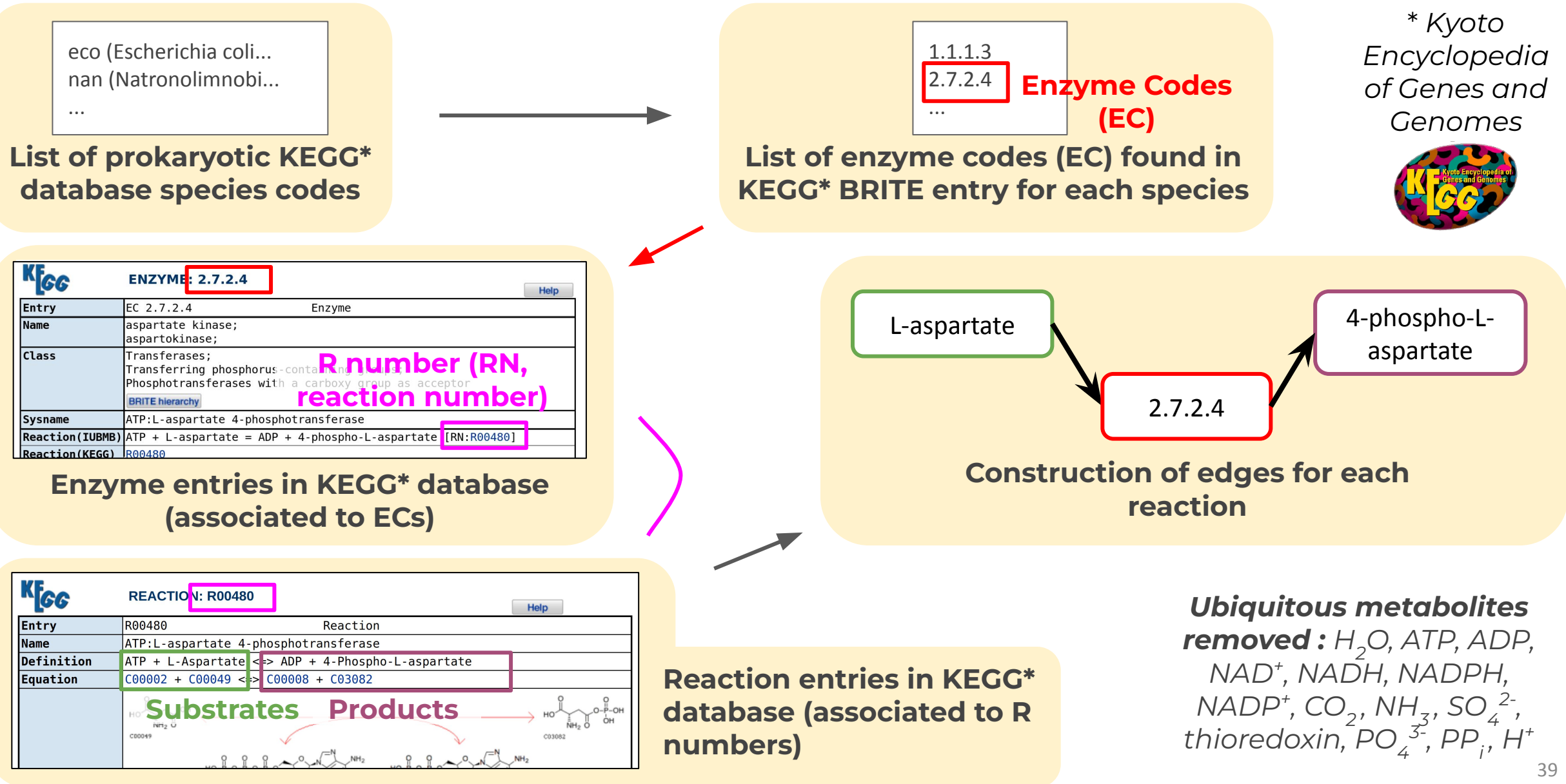


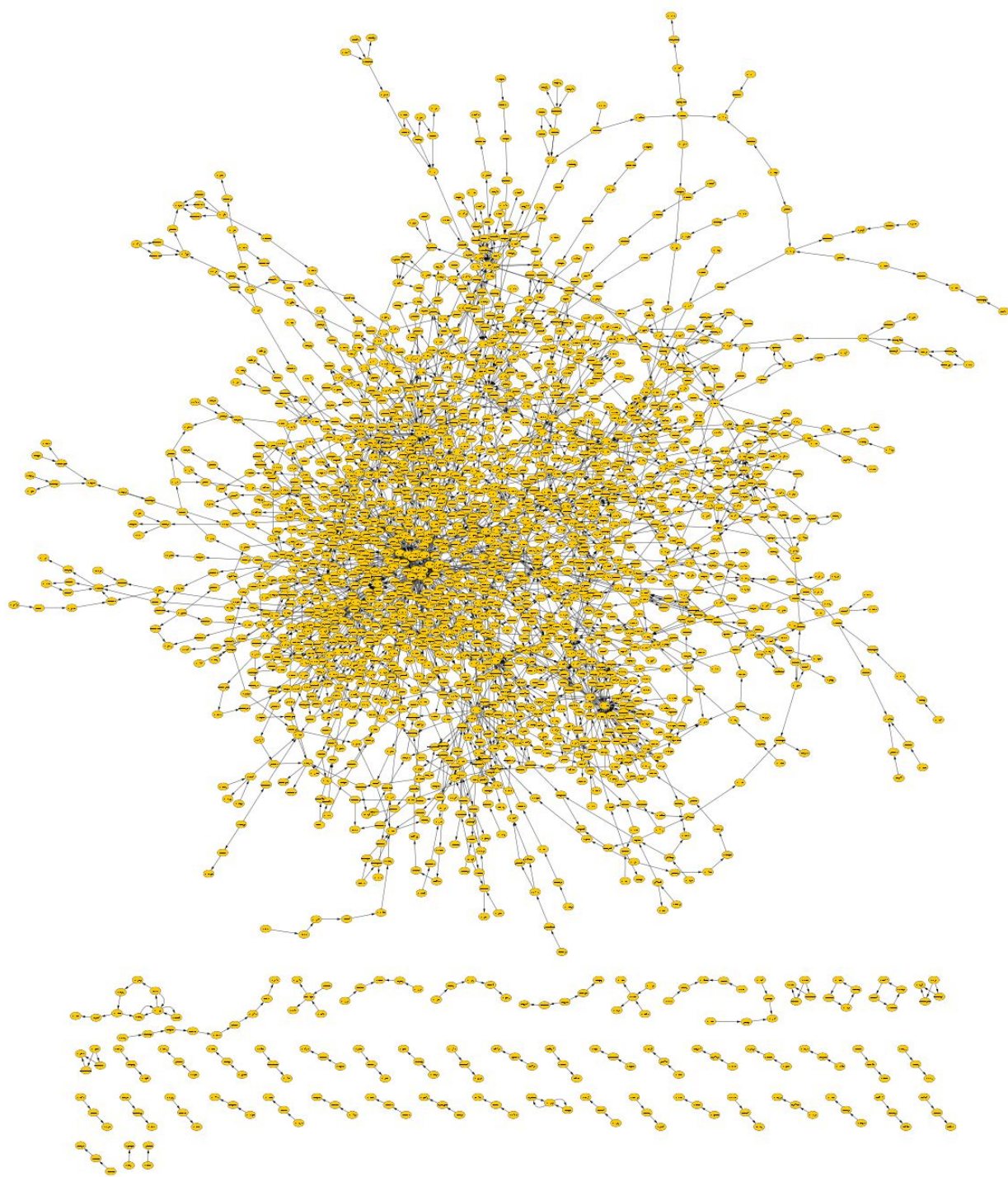
*Molecules (objects),
Chemical reactions (links)*



*Molecules and enzymes (objects),
Substrate/products (links)*

Metabolic network reconstruction





Escherichia
coli
K-12 MG16

Network properties

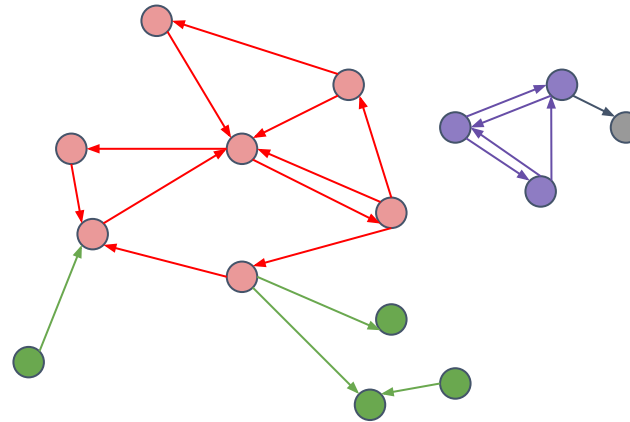
Strongly connected components:

group of nodes that can access all of the others in the group through *directed* edges (arrows) . ● & ●

Weakly connected components:

group of nodes that can access all of the others in the group through *undirected* edges (lines) .

● + ● & ● ●

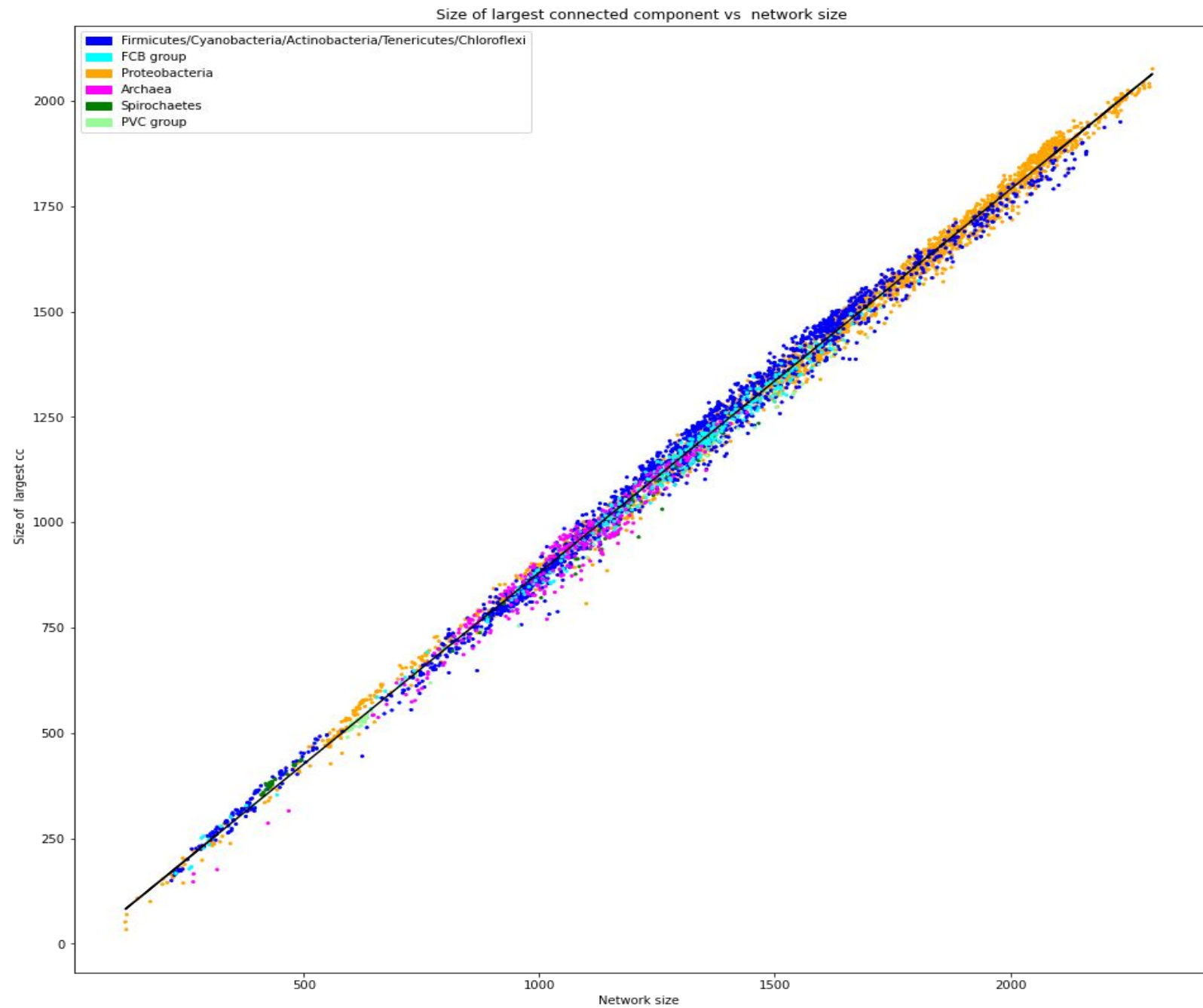


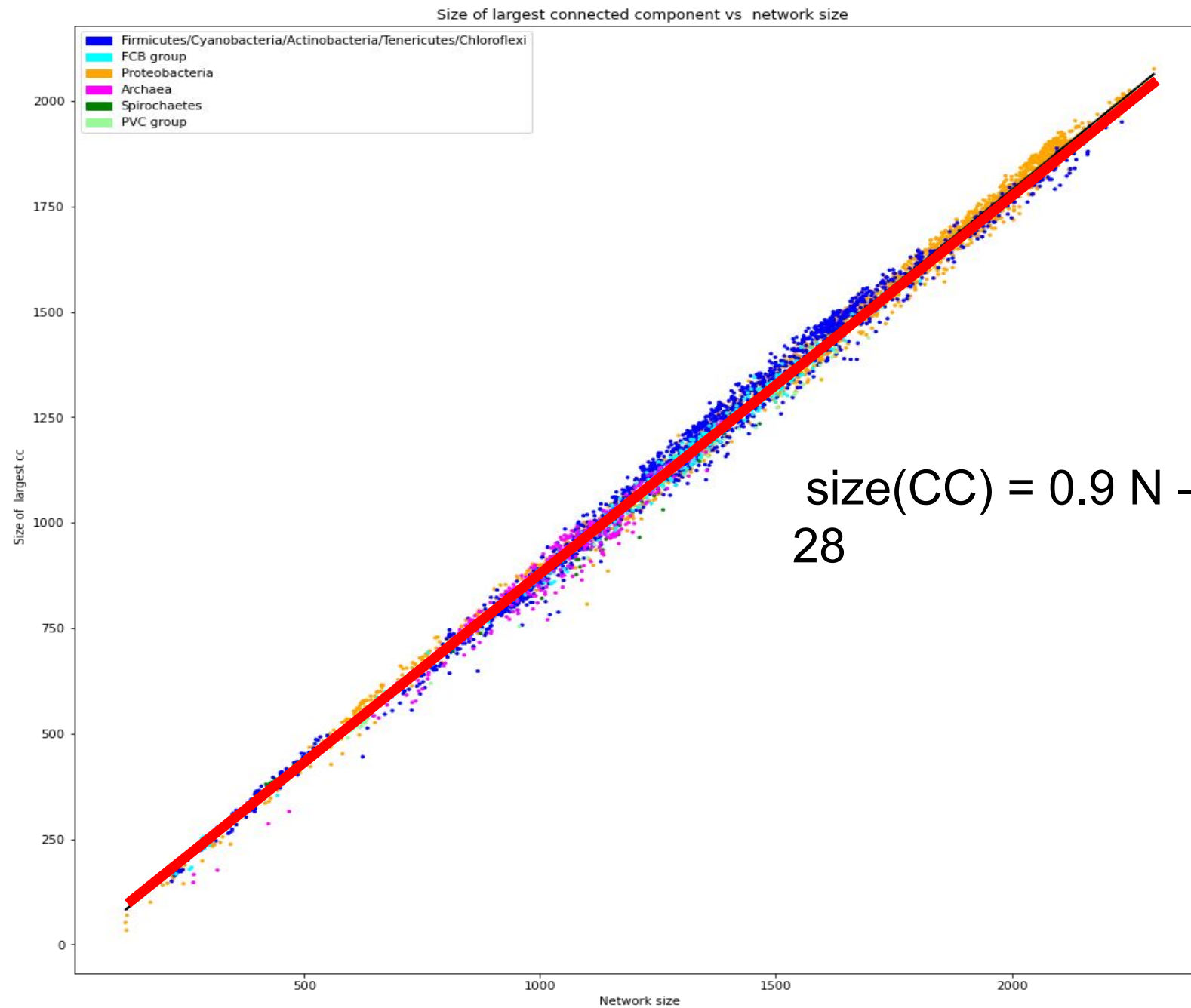
Node normalised size of largest strongly connected component:

$$\frac{\# \text{ (red nodes) }}{\# \text{ (red nodes)} + \# \text{ (green nodes)} + \# \text{ (blue nodes)} + \# \text{ (grey node)}}$$

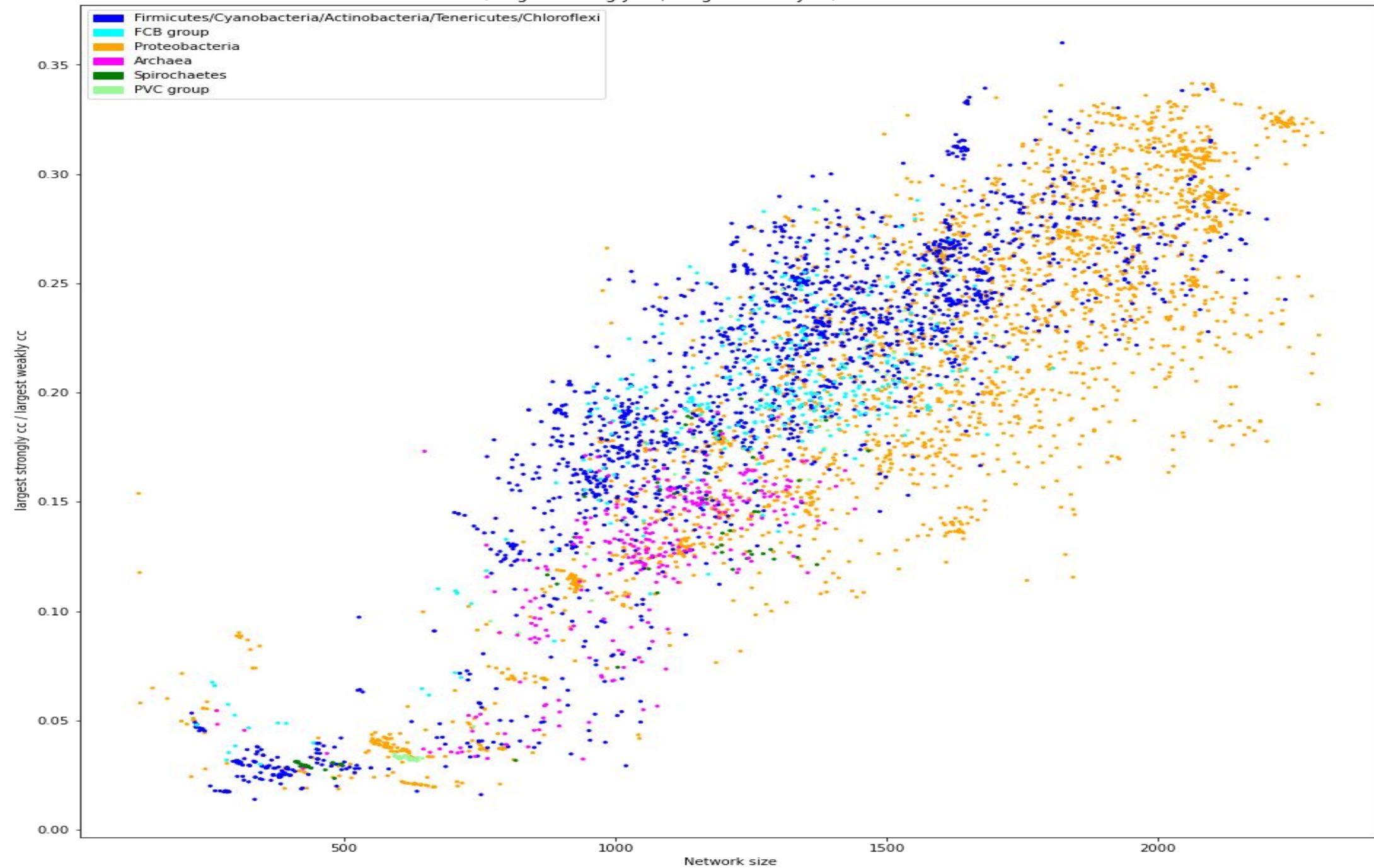
Node normalised flow hierarchy:

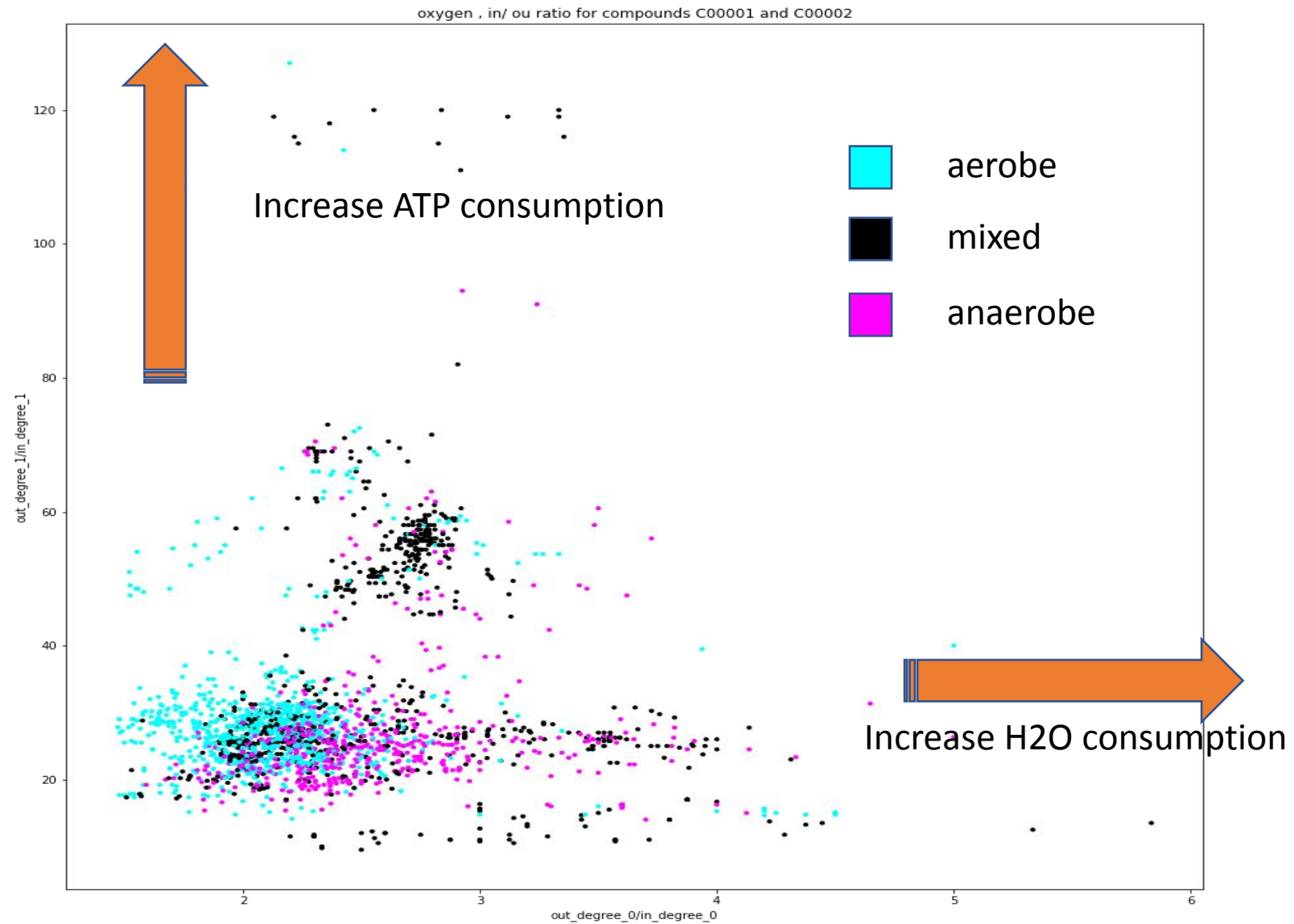
$$\frac{(\# \text{ (red nodes)} + \# \text{ (green nodes)} - \# \text{ (red nodes)})}{\# \text{ (red nodes)} + \# \text{ (green nodes)} + \# \text{ (blue nodes)} + \# \text{ (grey node)}}$$



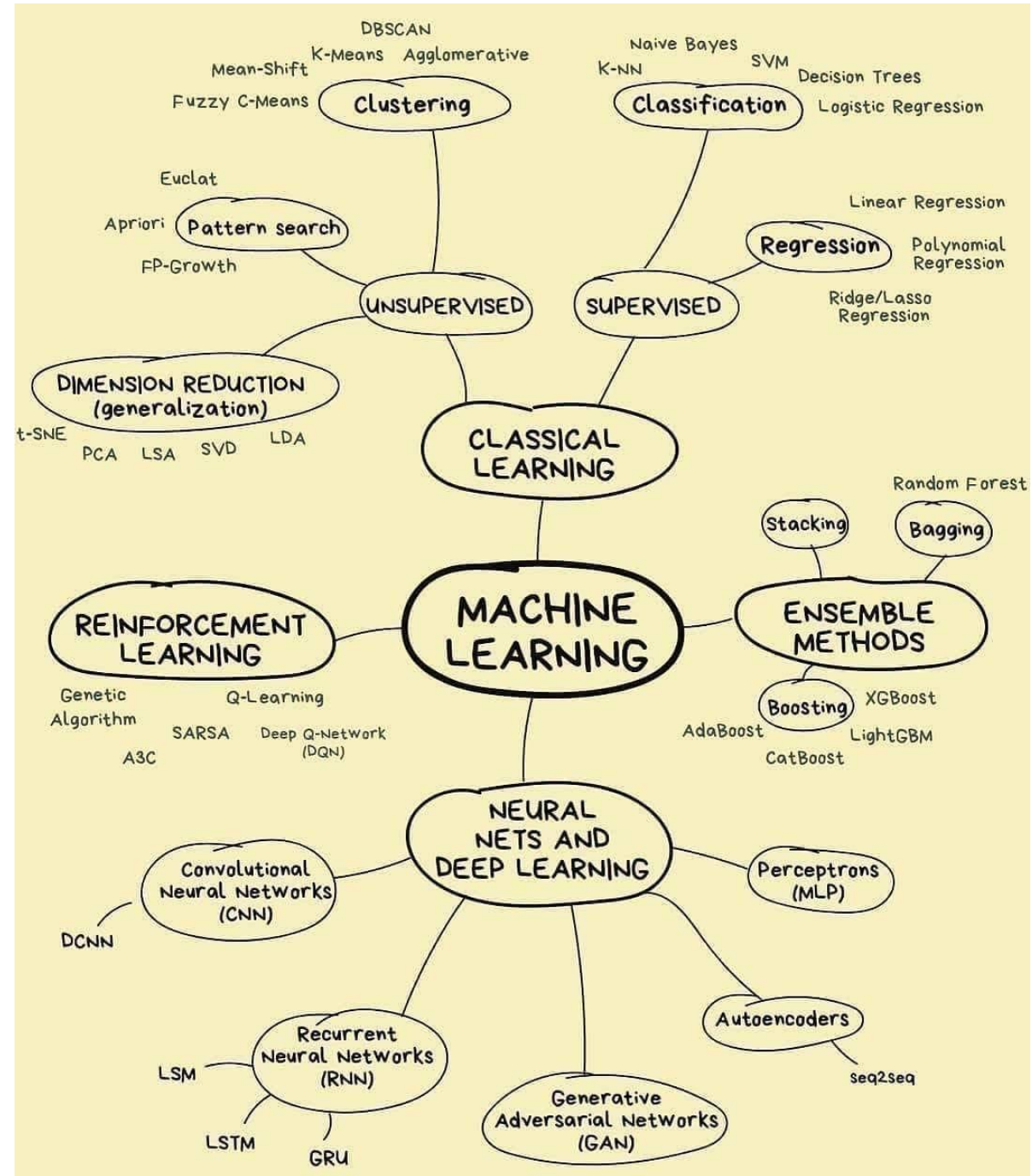


(Largest strongly cc / Largest weakly cc) vs network size



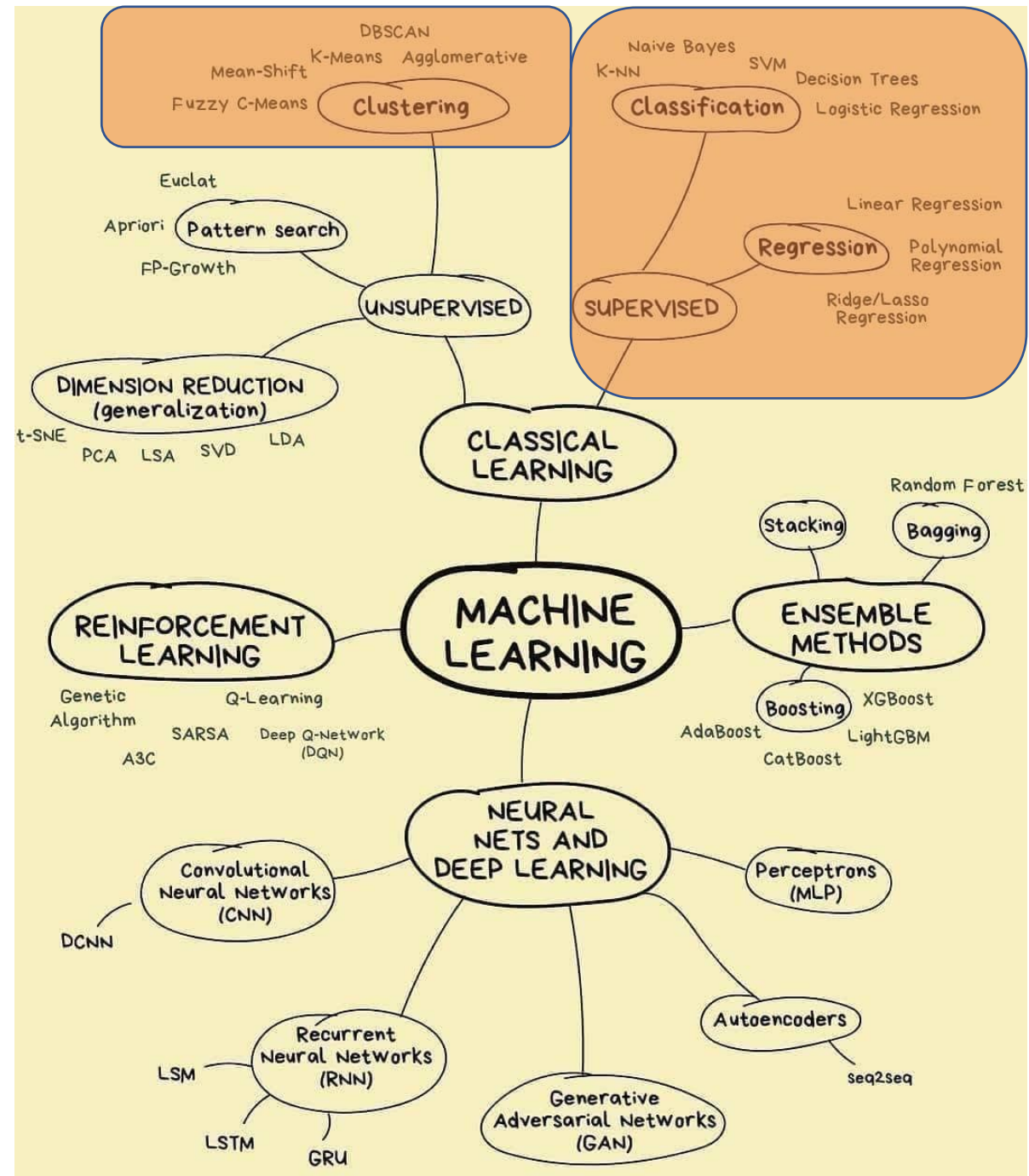


MACHINE LEARNING



MACHINE LEARNING

Covered in **PREVIOUS** slides

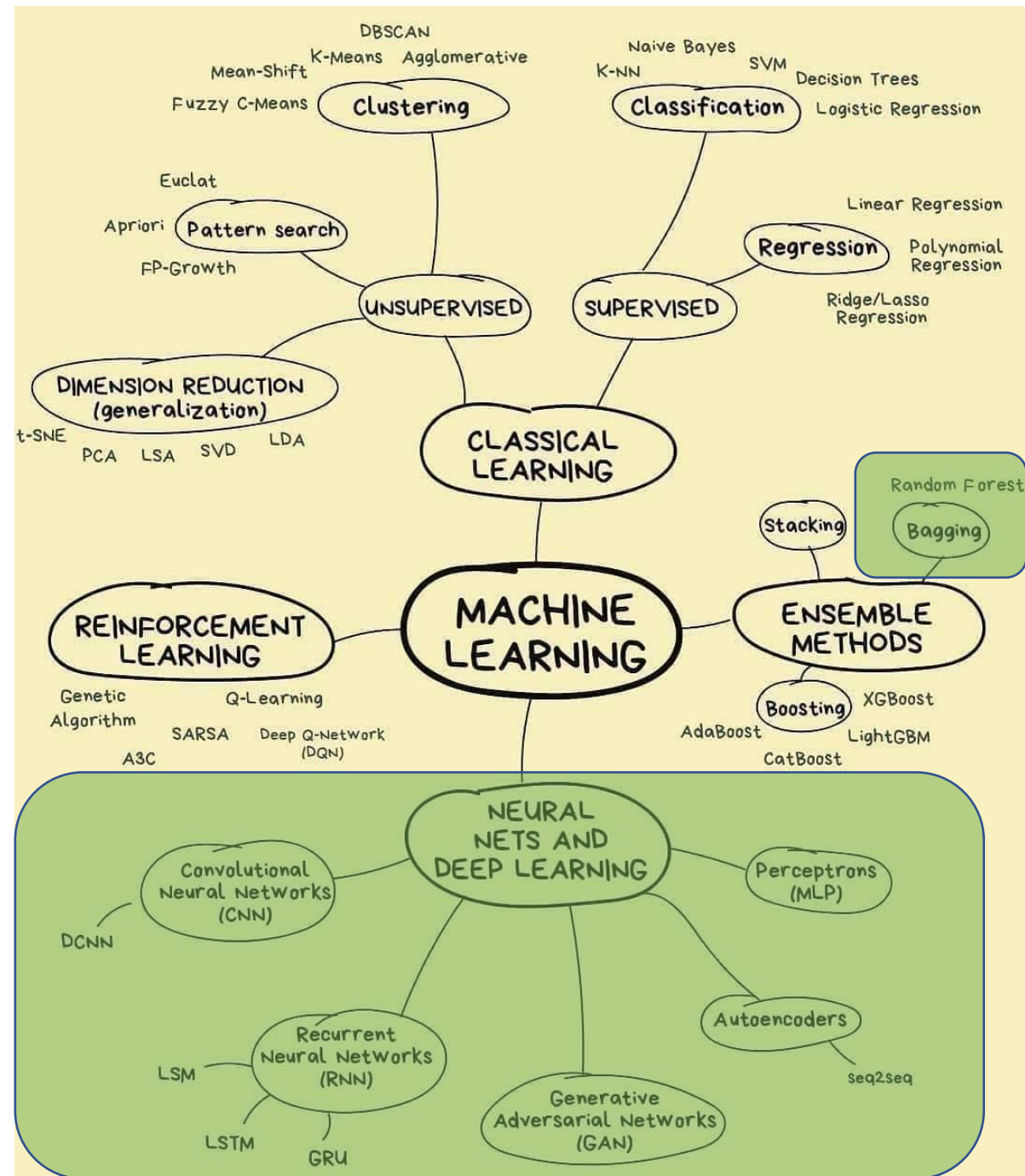


MACHINE LEARNING

Deals with **ANNOTATED** data :

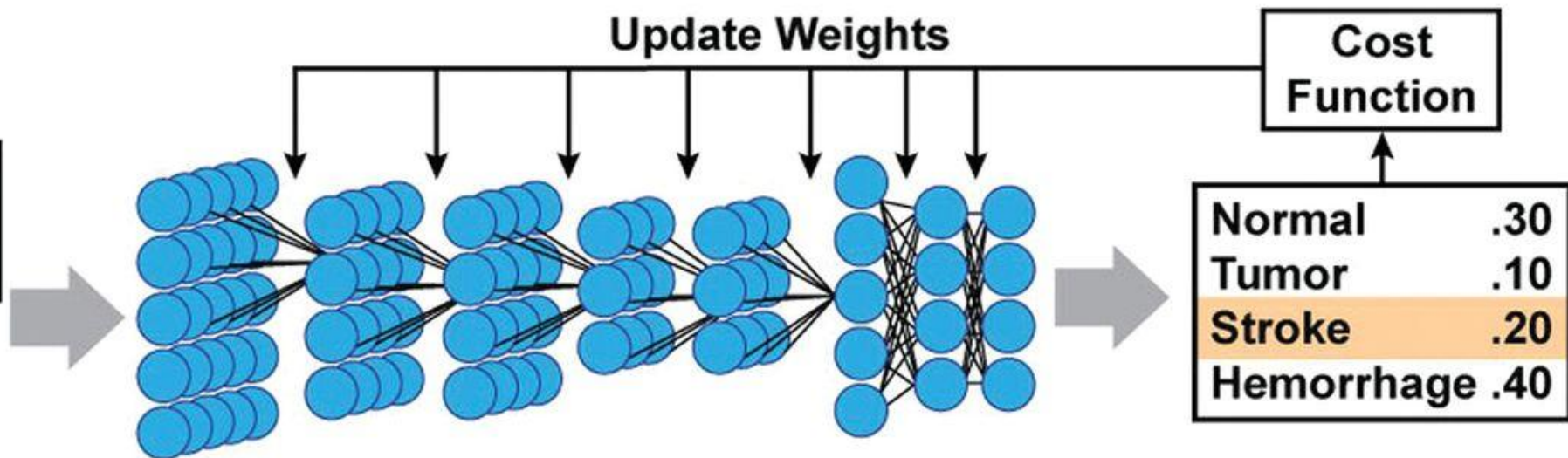
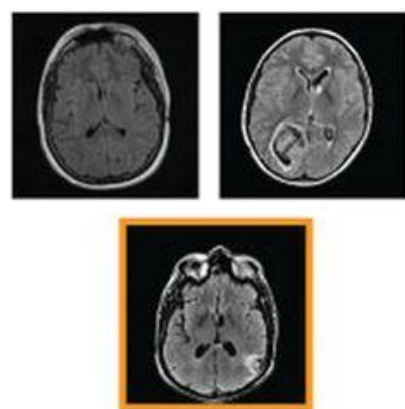
$$Y \sim X$$

Trains a **black box** to reproduce Y from X

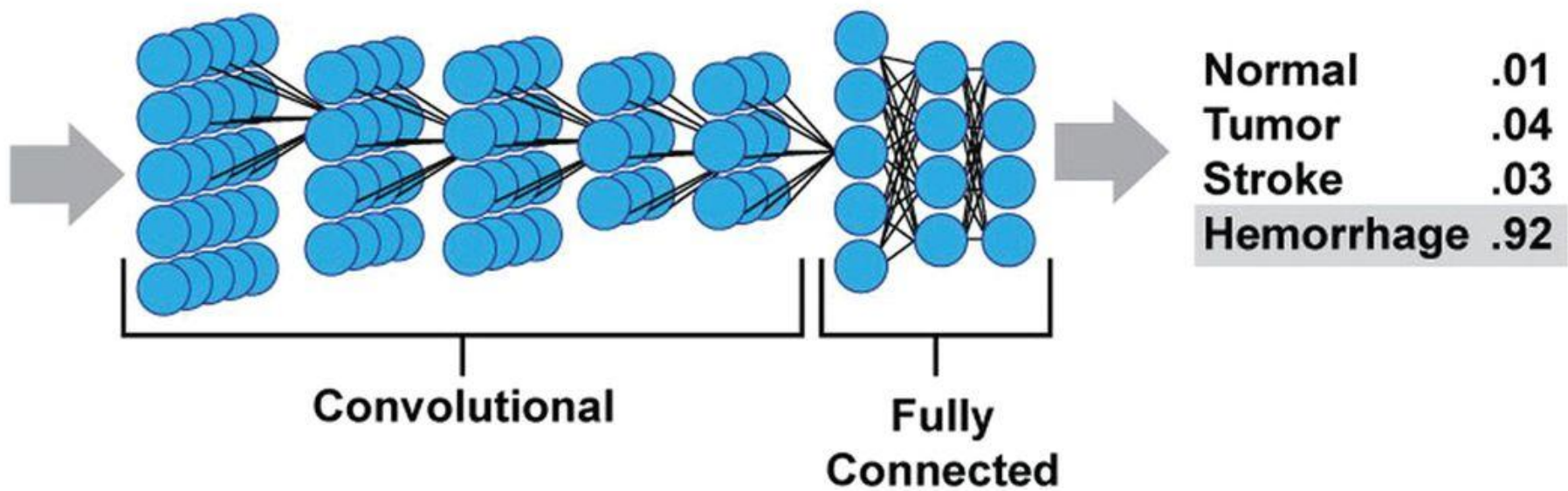
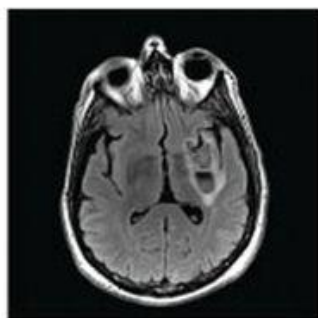


Convolutional Neural Networks

Train:

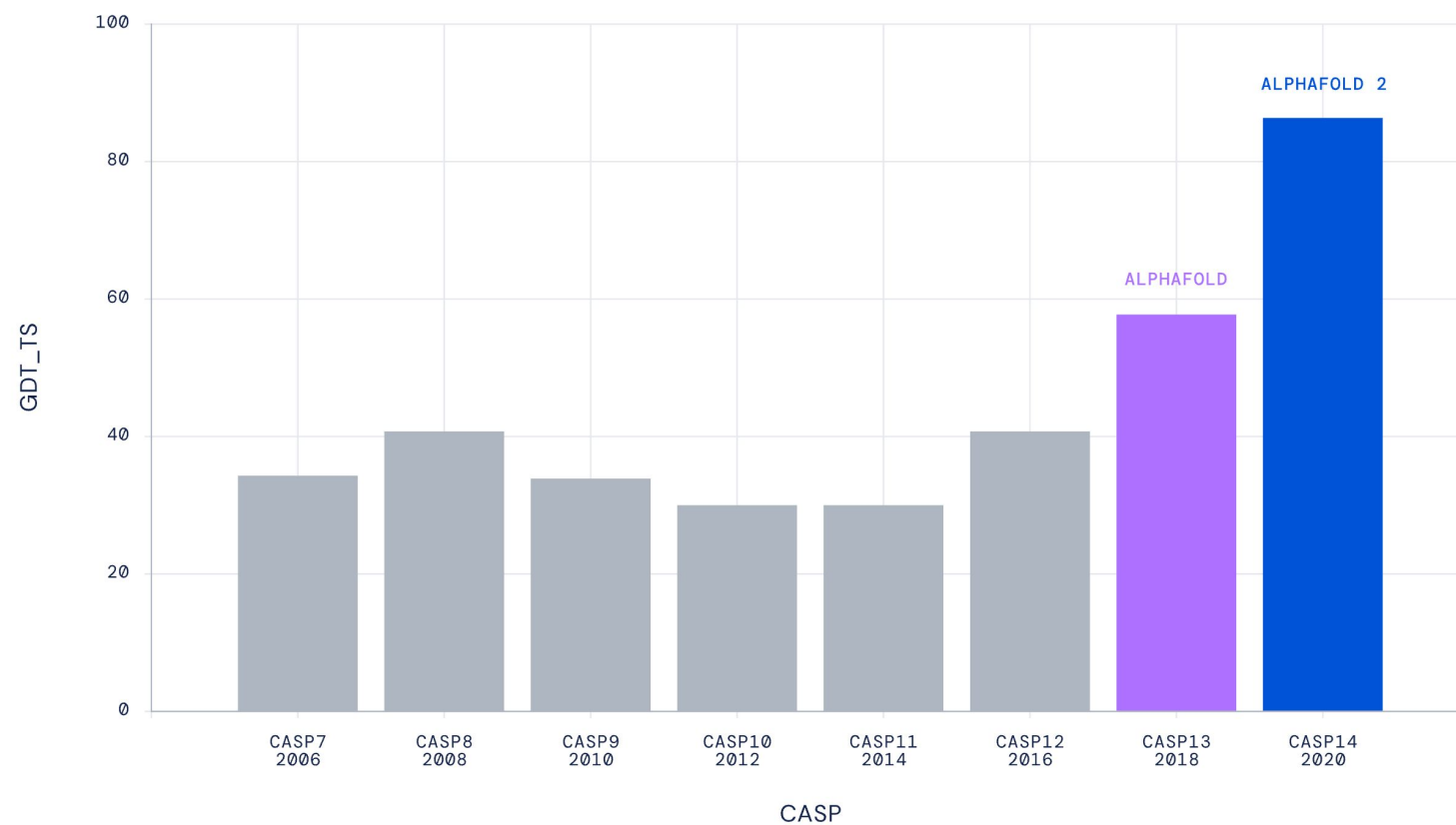


Deploy:

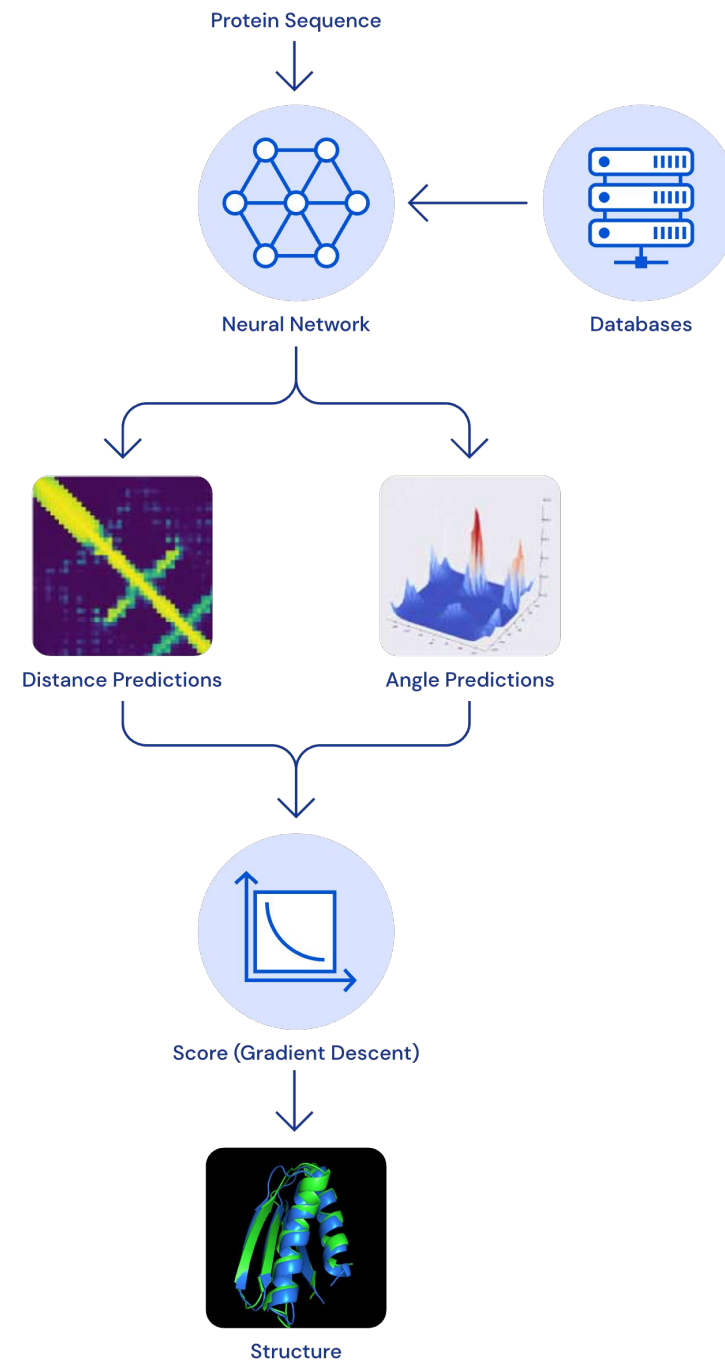


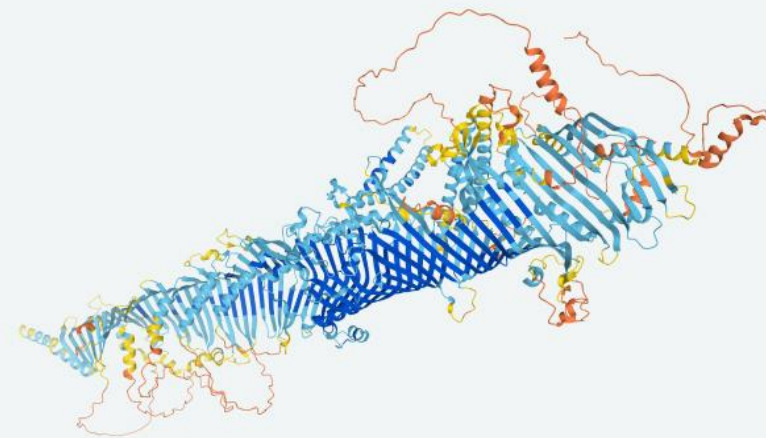
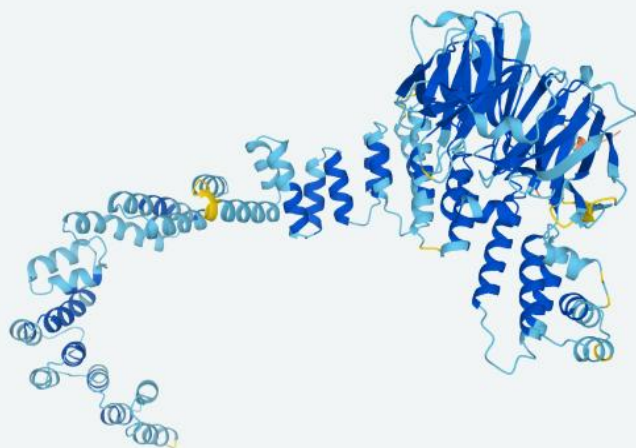
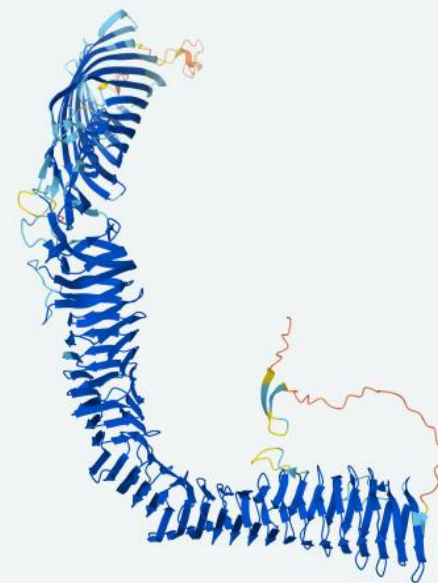
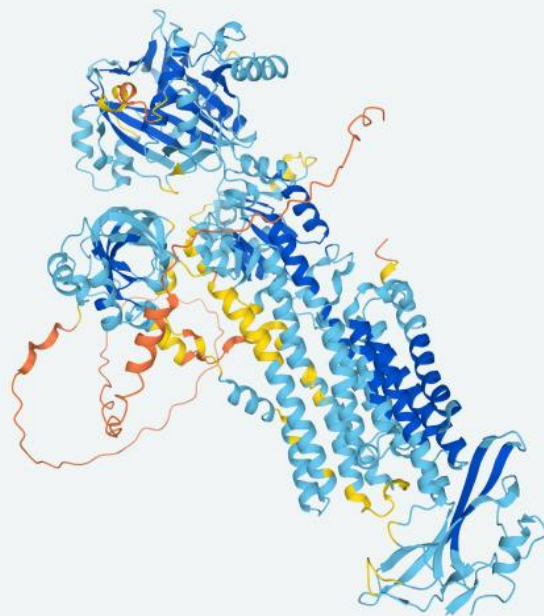
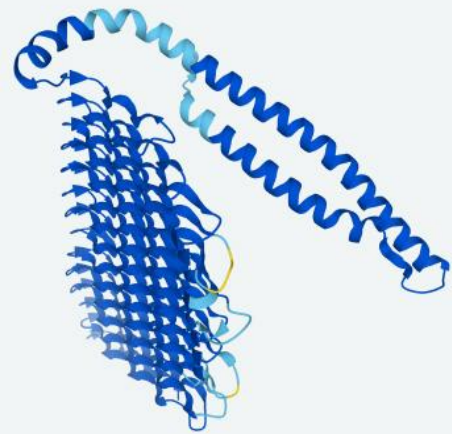
ALPHA FOLD

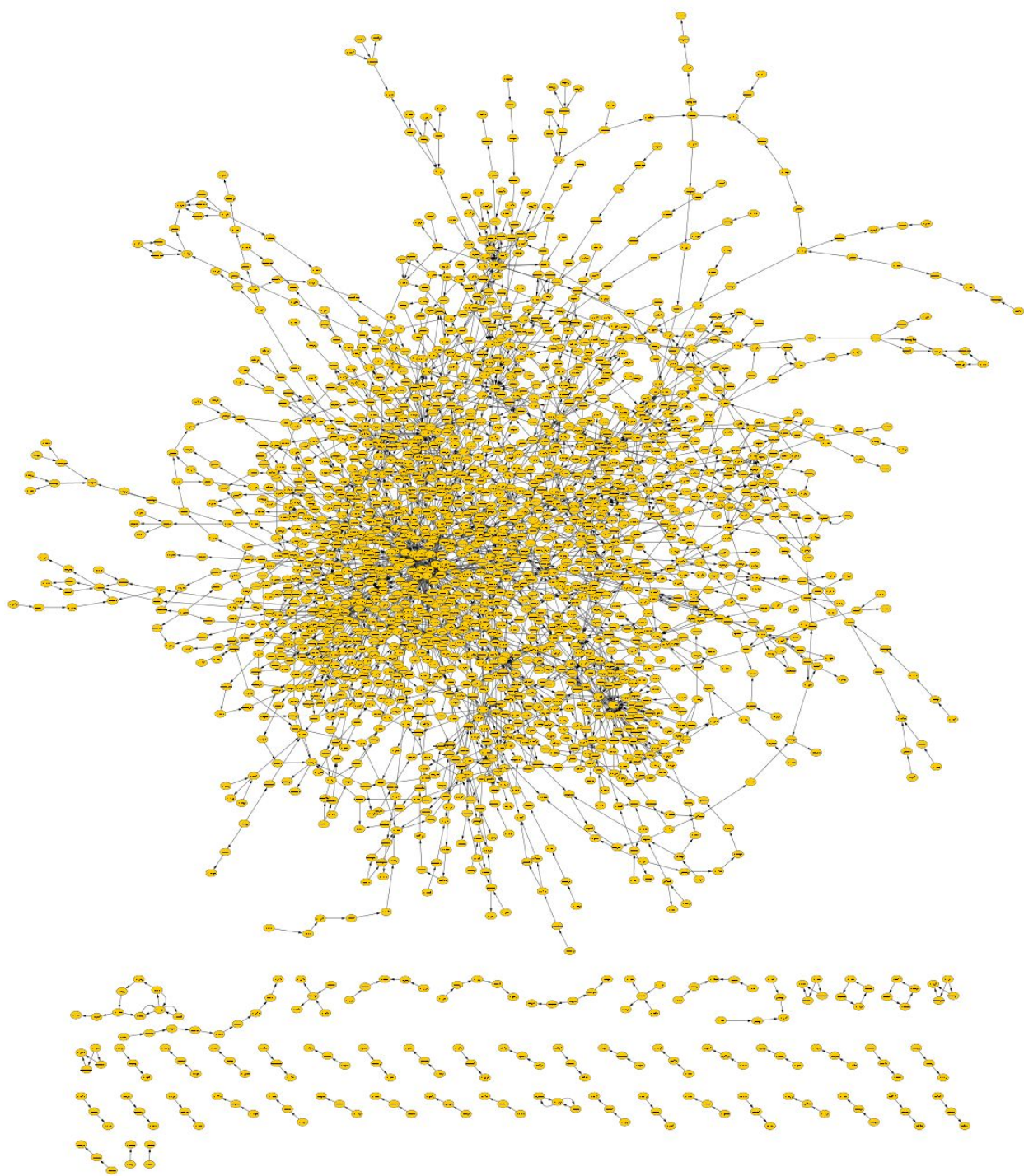
Median Free-Modelling Accuracy



SQETRKKCTEMKKKFKNCEVRCDESNHCVEVRCSDTKYTLC



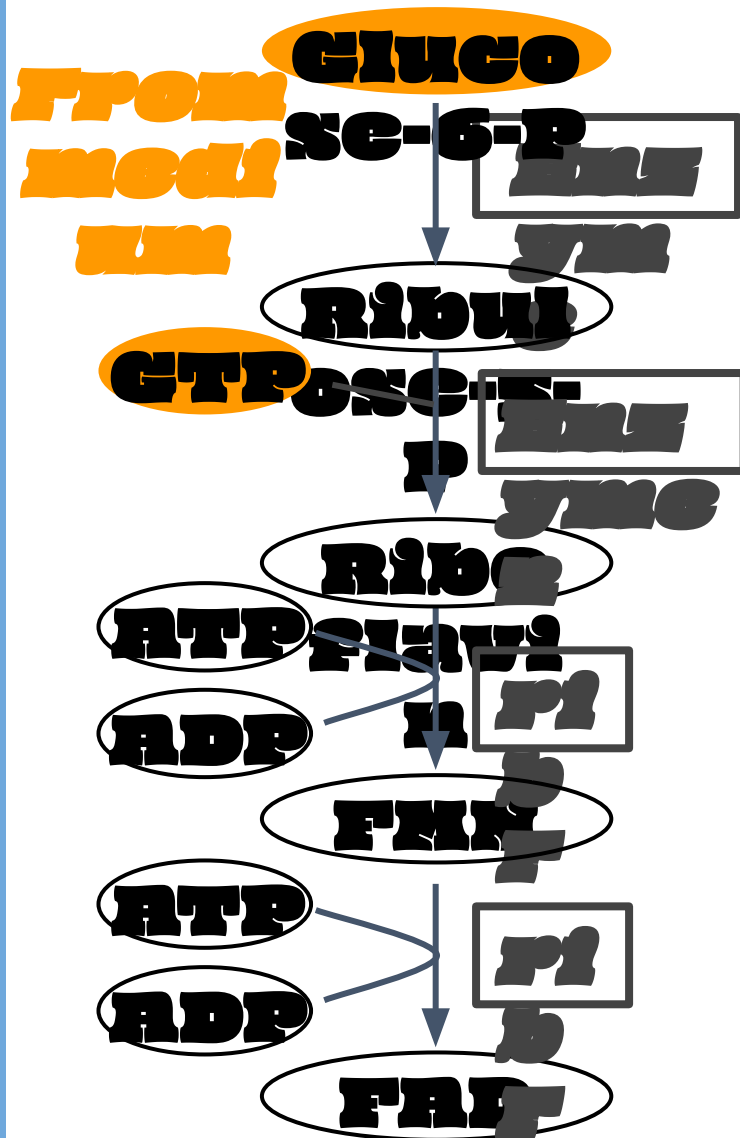




Escherichia
coli
K-12 MG16

Scope of a metabolic network

Handorf, Ebenhöf, and Heinrich, 2005

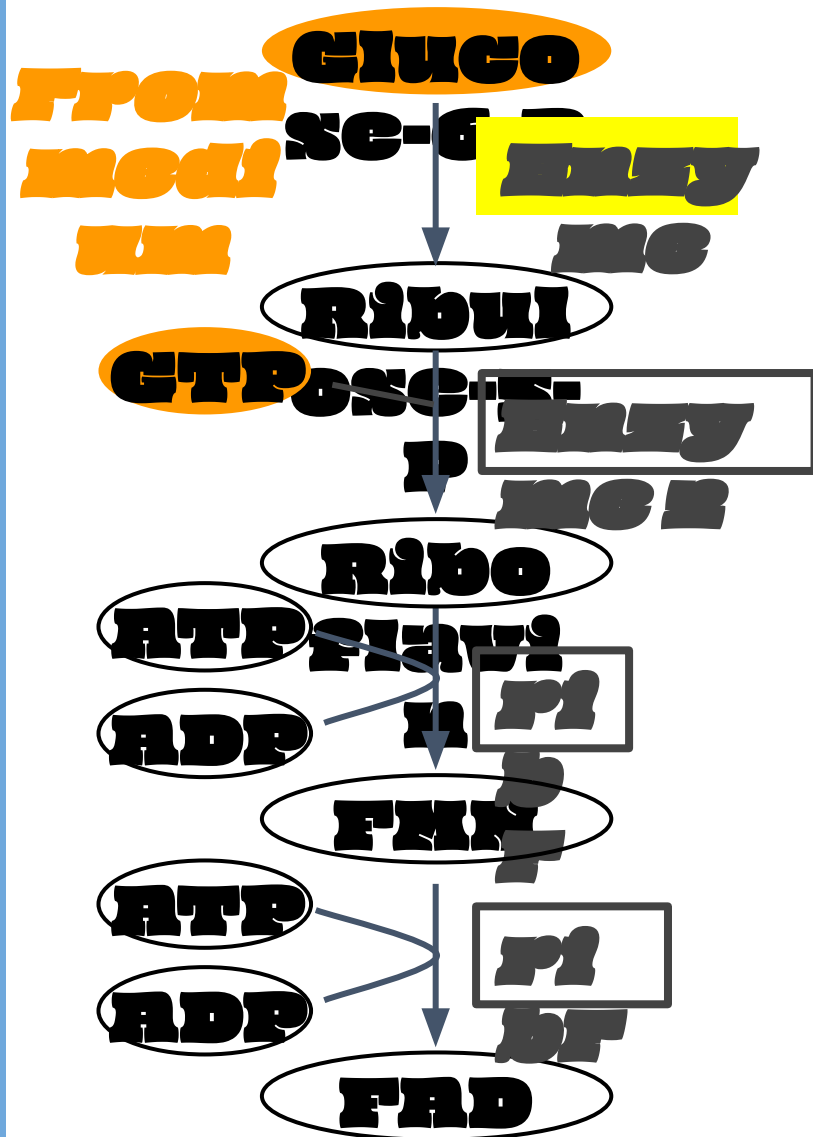


Scope of a network: nodes (metabolites and enzymes) reachable in the network from a set of input nodes from the environment.

Simplified riboflavin

Scope of a metabolic network

Handorf, Ebenhöf, and Heinrich, 2005

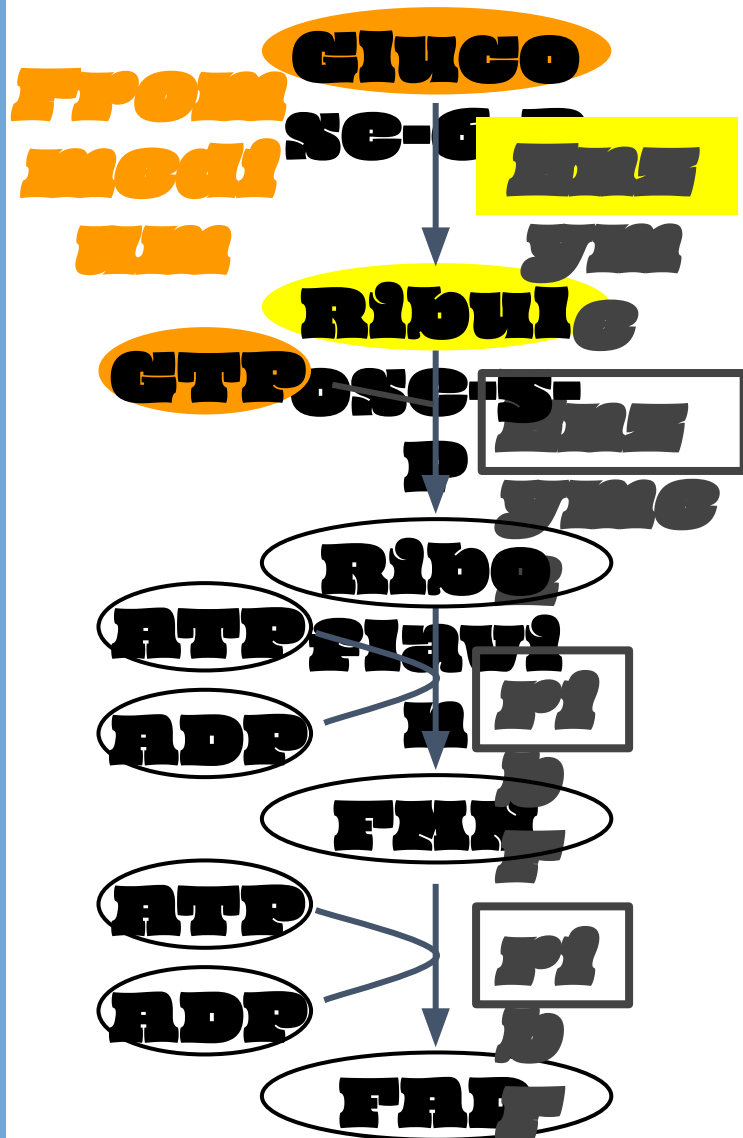


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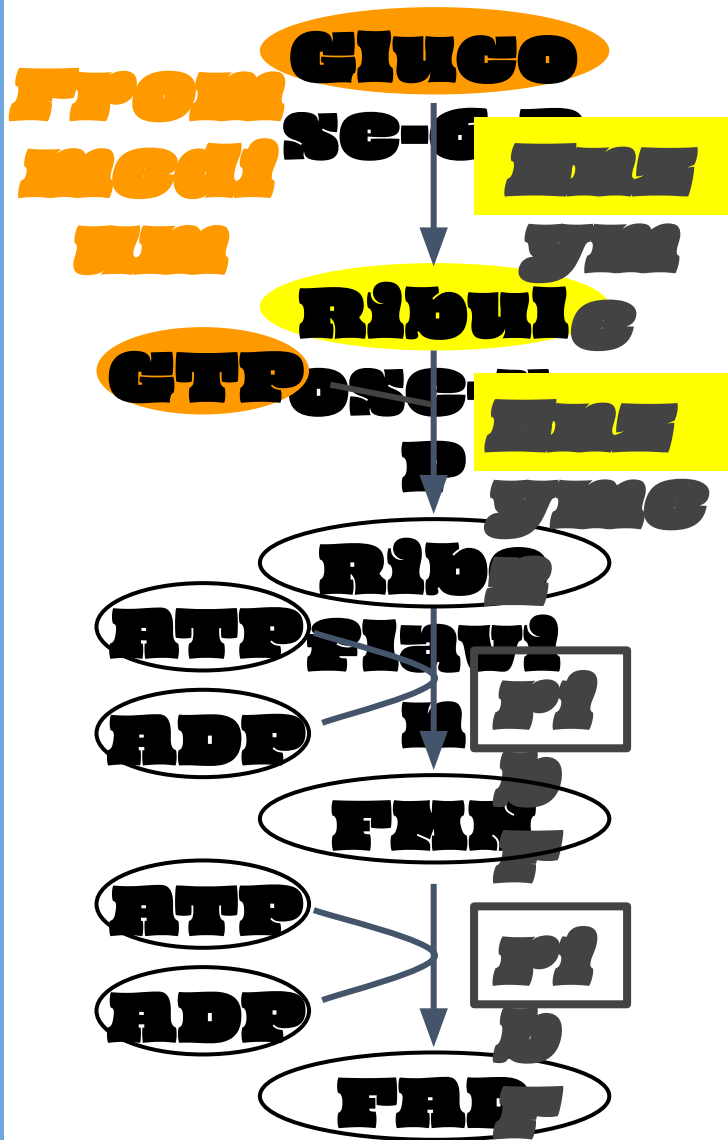


Scope of a network: nodes (metabolites and enzymes) reachable in the network from a set of input nodes from the environment.

FROM med1 TO

Handorf, Ebenhöf, and Heinrich, 2005

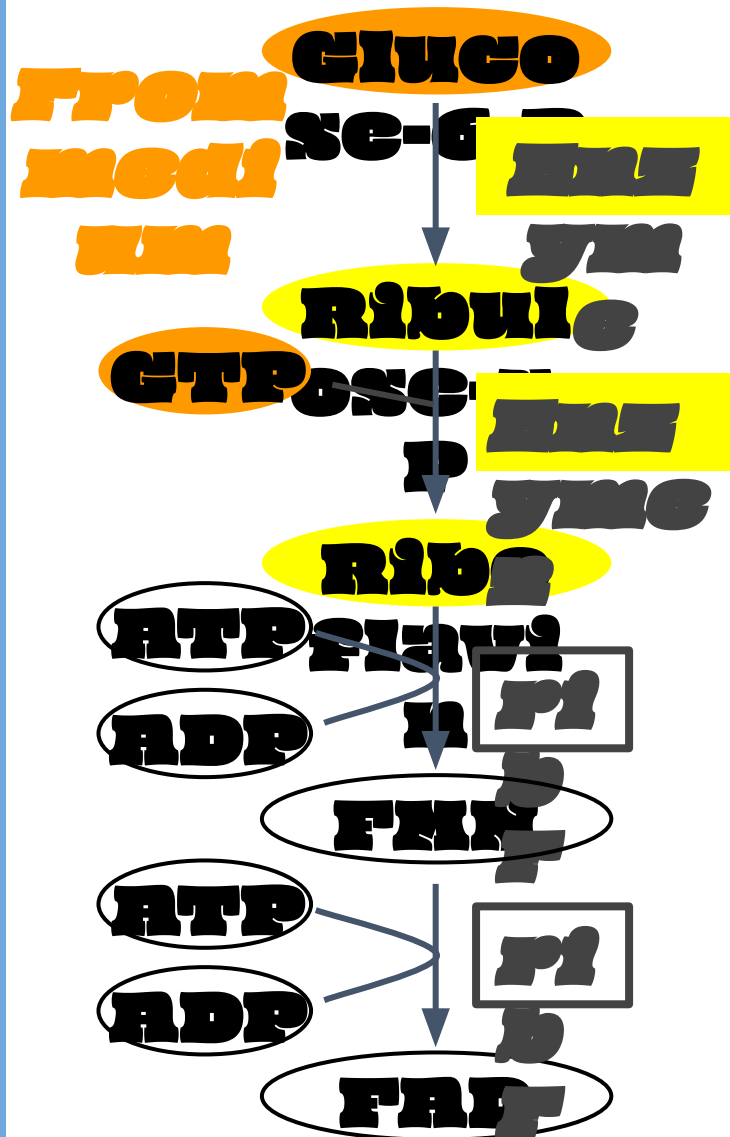
Scope of a network: nodes (metabolites and enzymes) reachable in the network from a set of input nodes from the environment.



Simplified riboflavin

Scope of a metabolic network

Handorf, Ebenhöf, and Heinrich, 2005



Scope of a network: nodes (metabolites and enzymes) reachable in the network from a set of input nodes from the environment.

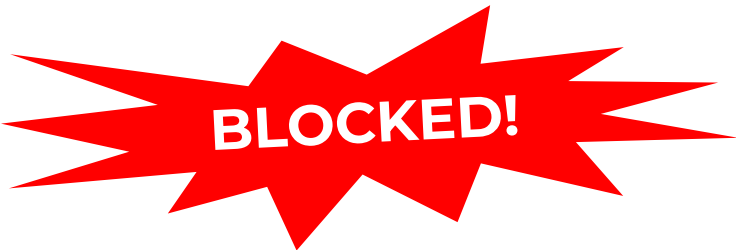
Simplified riboflavin

Scope of a metabolic network

Handorf, Ebenhöf, and Heinrich, 2005



Scope of a network: nodes (metabolites and enzymes) reachable in the network from a set of input nodes from the environment.



Substrate missing (ATP):
ATP, ADP, FMN, FAD, ribF not reached

➔ Scope (size 6) = { Glucose-6-P, Enzyme, Ribulose-5-P, Enzyme 2, Riboflavin, GTP }

Simplified riboflavin

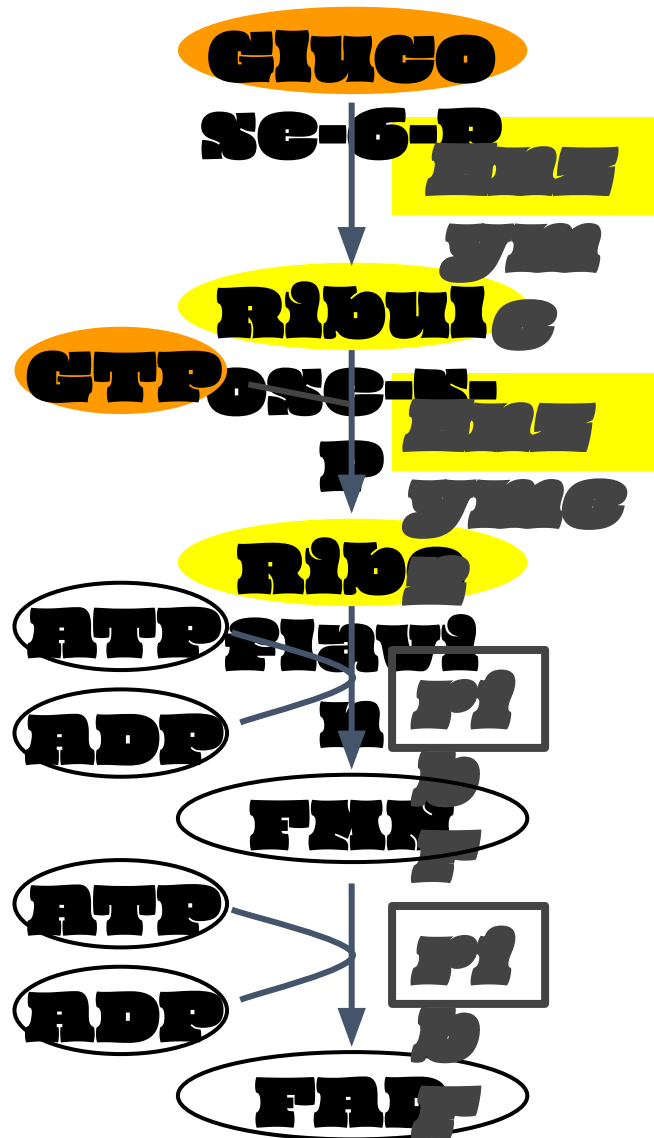
Scope of a metabolic network



Very medium
dependent!



Scope of a metabolic network



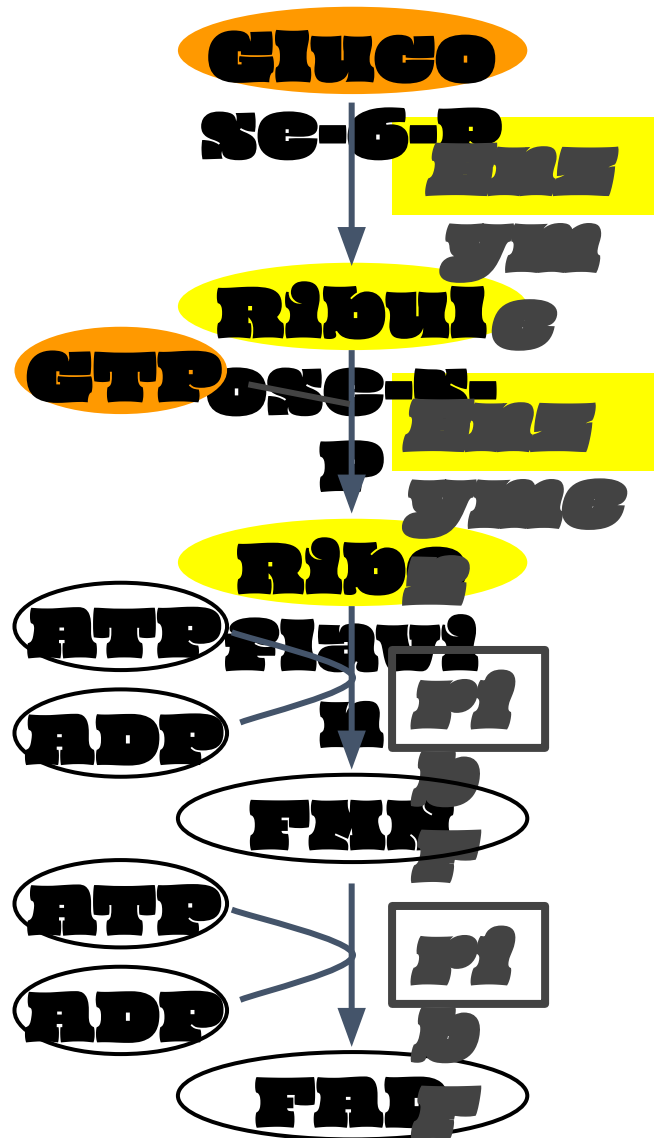
Very medium dependent!

Scope size for medium 1 = 6



Medium 1 = {Glucose-6-P, GTP}

Scope of a metabolic network



Very medium dependent!

Scope size for medium 1 = 6

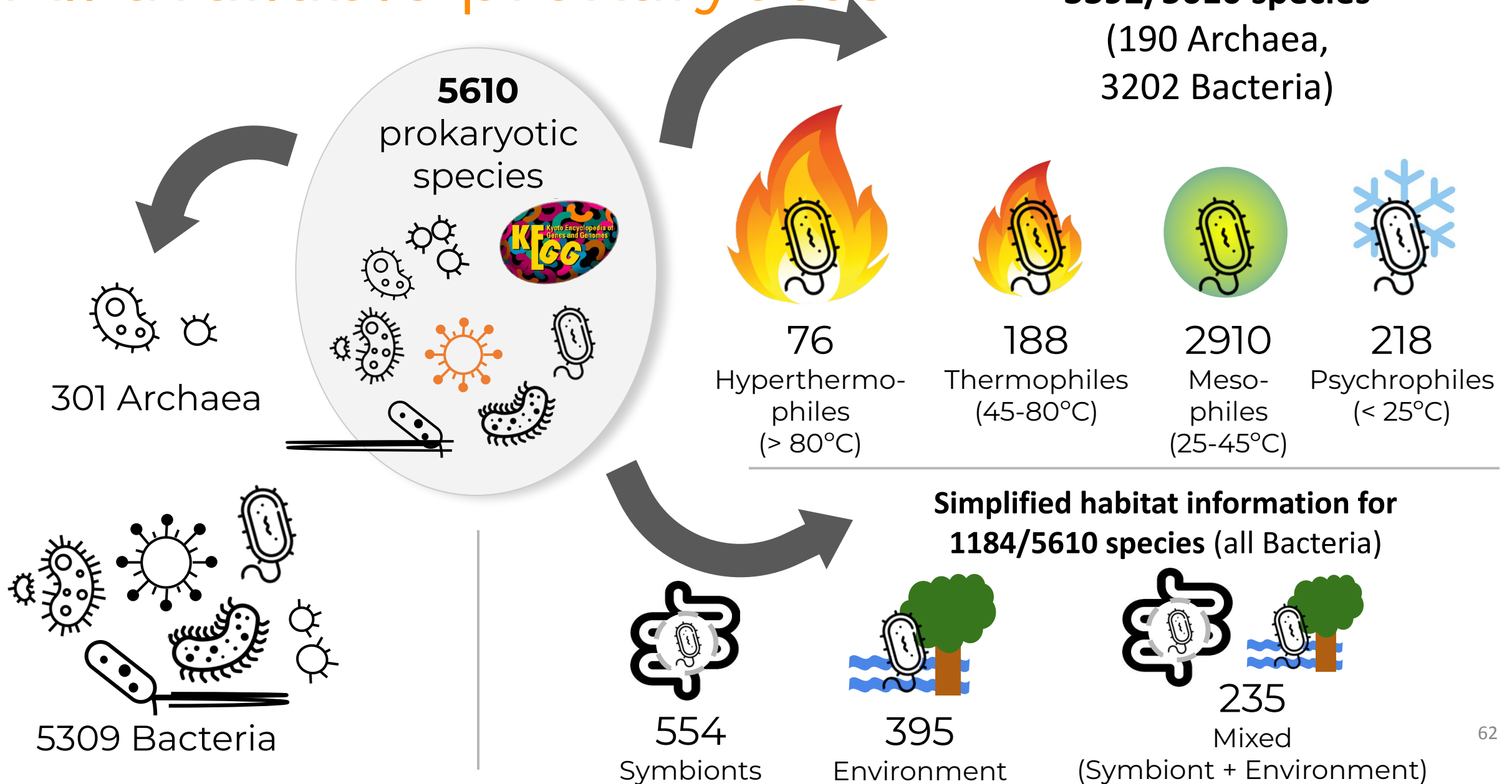
Scope size for medium 2 = 11



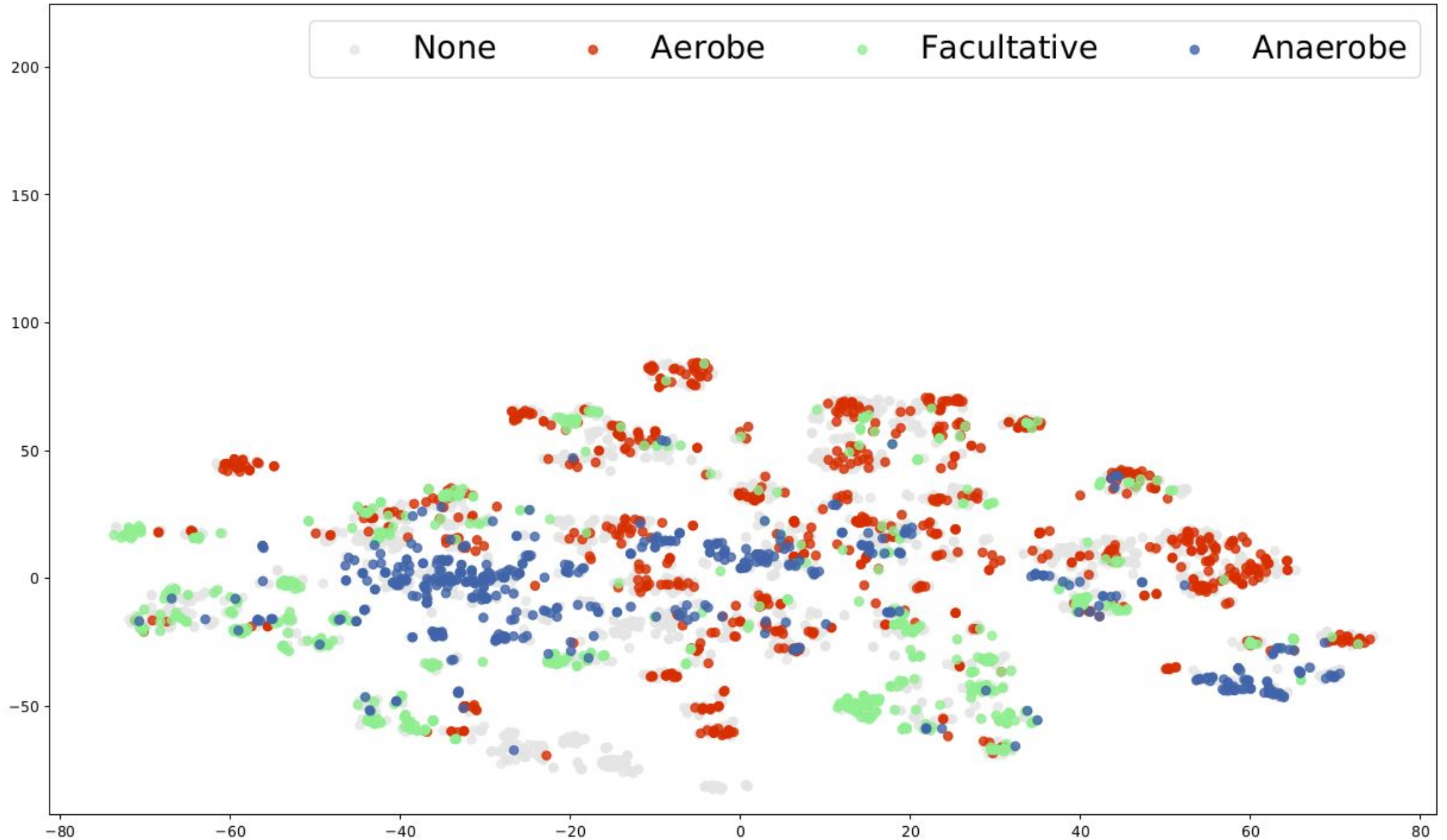
Medium 1 = {Glucose-6-P, GTP}

Medium 2 = {Glucose-6-P, GTP, ATP}

All available prokaryotes



tSNE: oxygen tolerance simplified



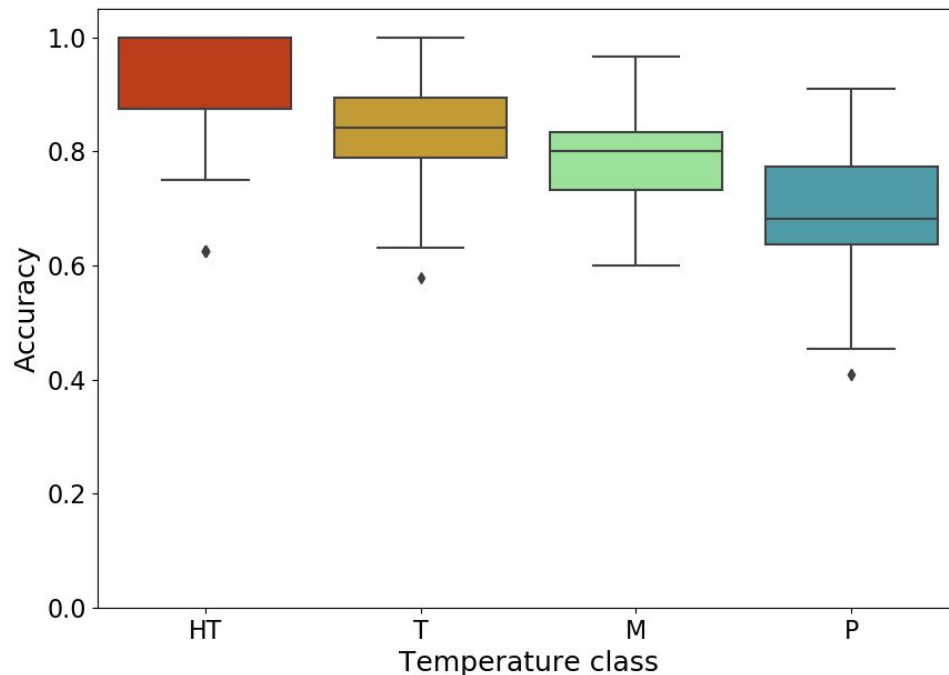
tSNE: temperature class



Temperature class classification: Random Forest

**Kept 300 mesophiles out of
2910 to balance classes
(76 HT, 188 T, 300 M, 218 P)**

Below: 300 different 300 mesophiles, each
with is train/test/validation splits

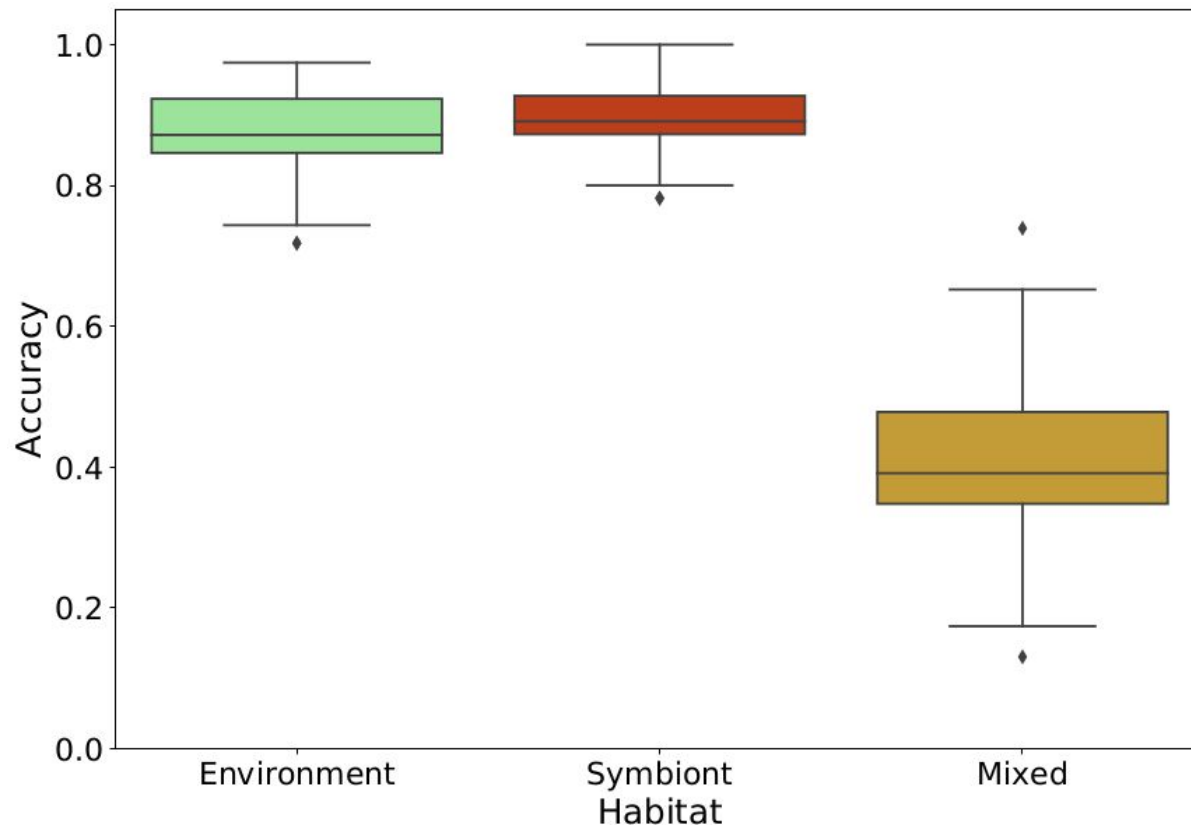


		Predicted			
					
Real		7.287 (91.08%)	0.71	0.003	0
		0.9767	15.673 (82.49%)	1.3967	0.9533
		0.0733	1.5367	23.6733 (78.91%)	4.7167
		0	1.3233	5.4733	15.2033 (69.10%)

Global accuracy = 78.27 ± 4.39 %
Mean of tree max depth: 23.7633

Habitat simplified classification: Random Forest (**1000 trees**)

Below: **300 cross-validations**, each with its
train/test/validation splits

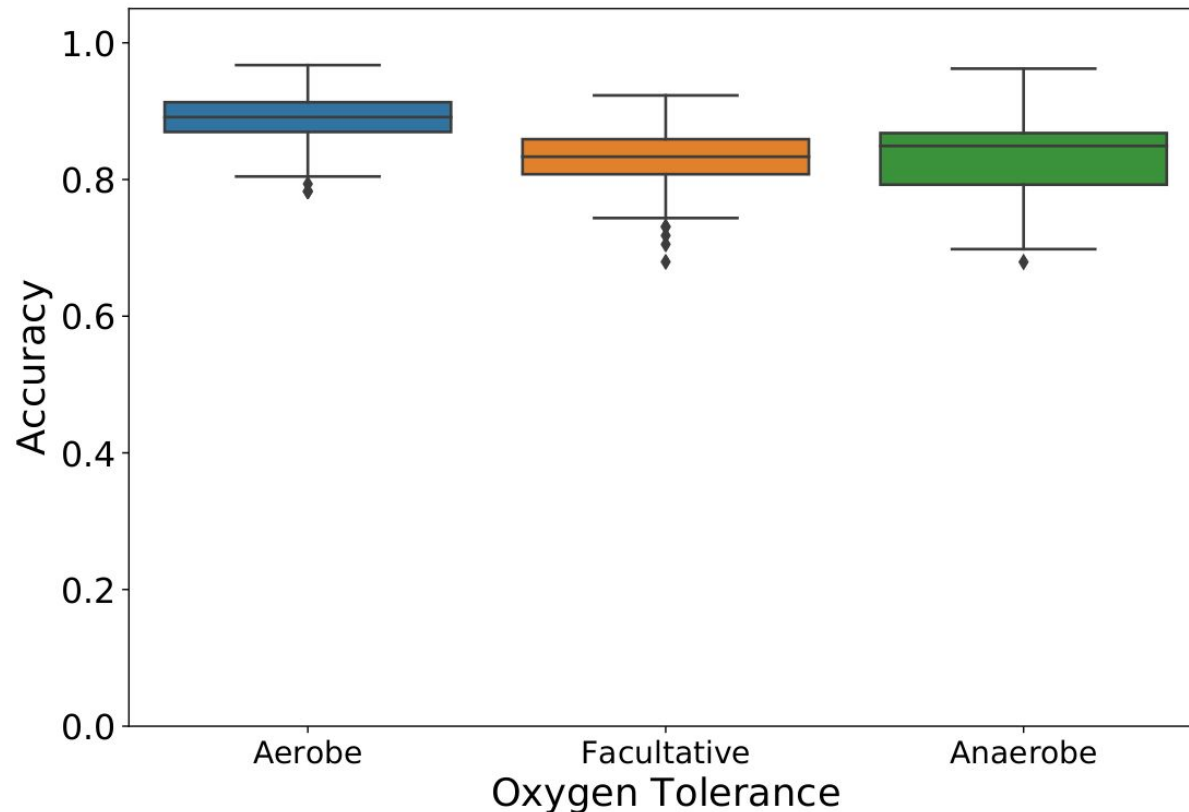


		Predicted		
		Environment	Symbiont	Mixed
Real	E	34.21 (87.71%)	1.98	2.81
	S	3.28	49.28 (89.61%)	2.44
	M	7.91	5.54	9.55 (41.52%)

Global accuracy = 79.52 ± 3.15 %
Mean of tree max depth = 28.28

Oxygen simplified classification: Random Forest

Below: **300 cross-validations**, each with is
train/test/validation splits

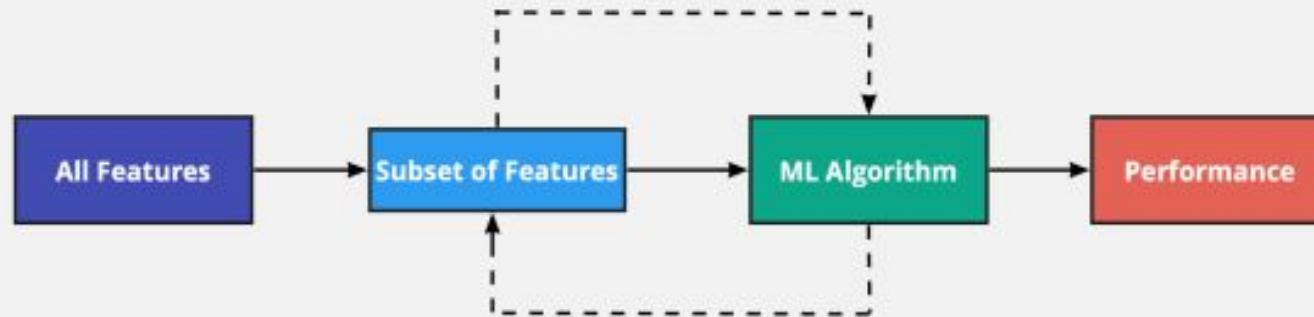


		Predicted		
		Aerobe	Facultative	Anaerobe
Real	A	81.76 (88.87%)	7.2133	3.02667
	F	10.37667	64.79 (83.06%)	2.8333
	An	4.87	3.87333	44.25667 (83.50%)

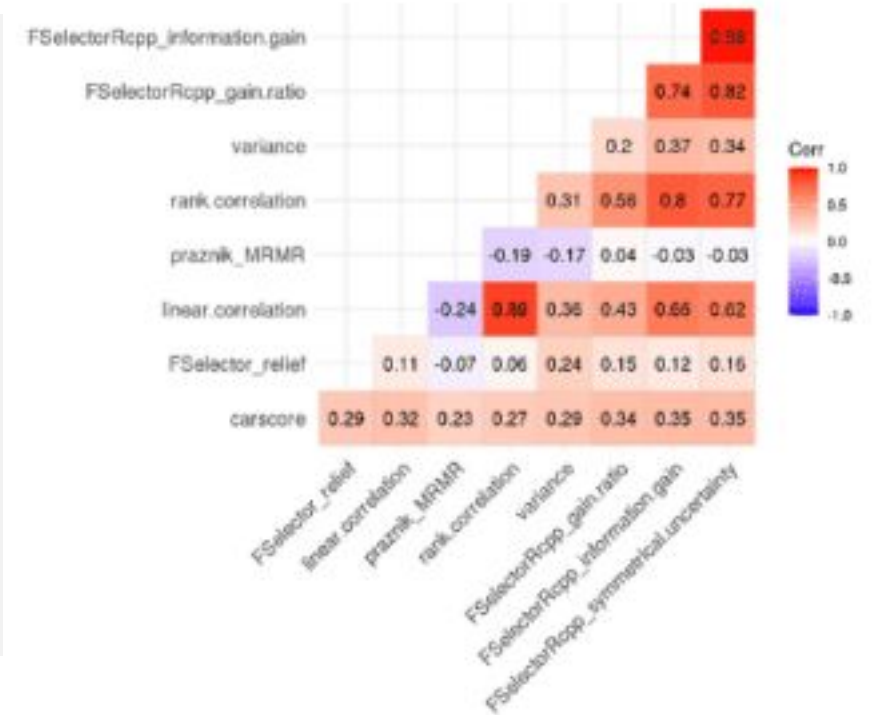
Global accuracy = 85.56 ± 2.35 %
Mean of tree max depth = 30.89

Features selections

Wrapper Feature Selection Method

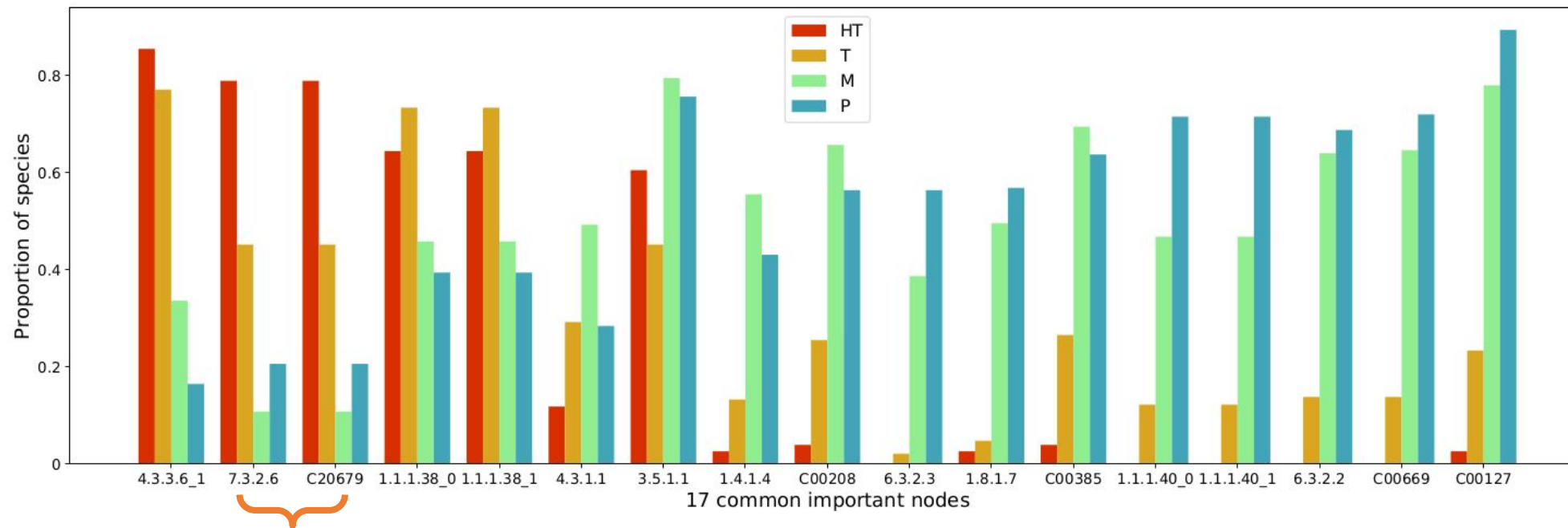


dataaspirant.com



Temperature class classification

17 nodes $\in$ “50 most important” in more than 90% of models



Tungstate: archaea appear to be obligately tungsten-dependent

Glutathione
metabolism: 5

Microbial metabolism
in diverse
environments: 4

Pyruvate metabolism:
4

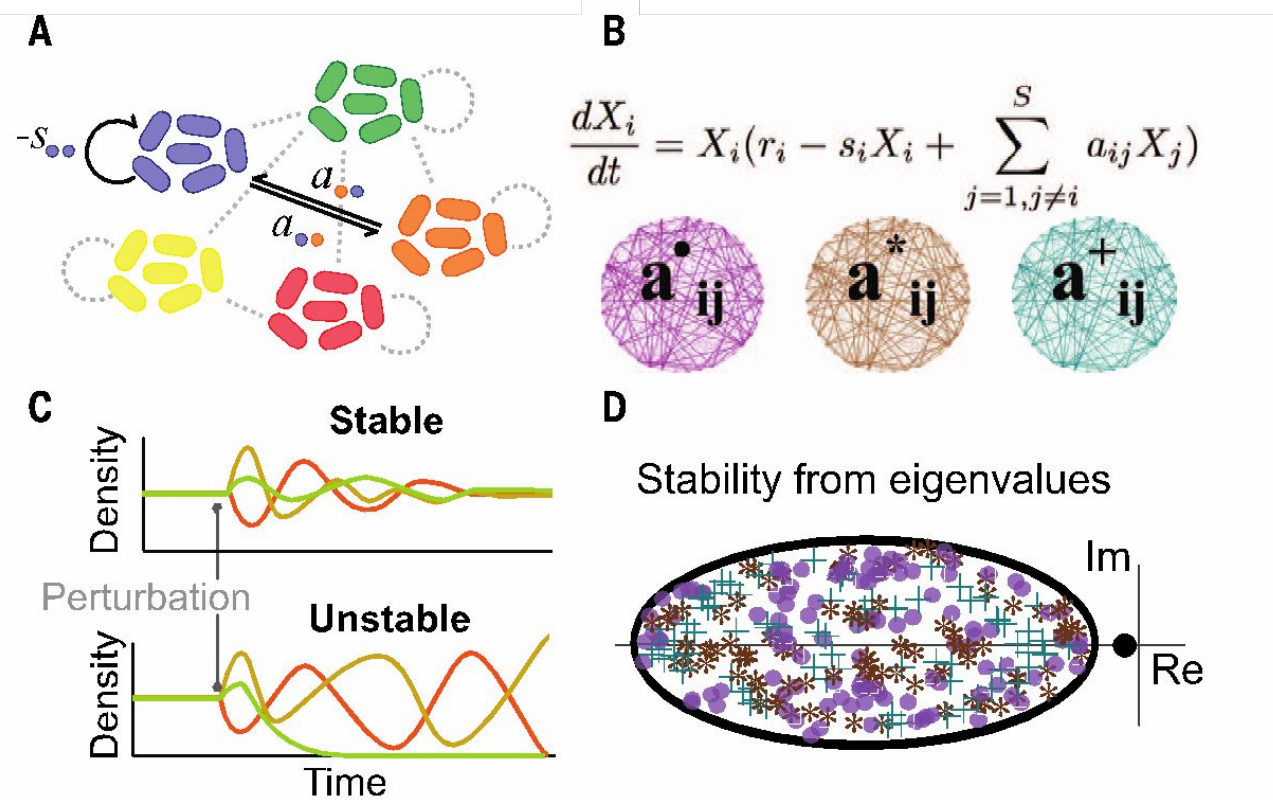
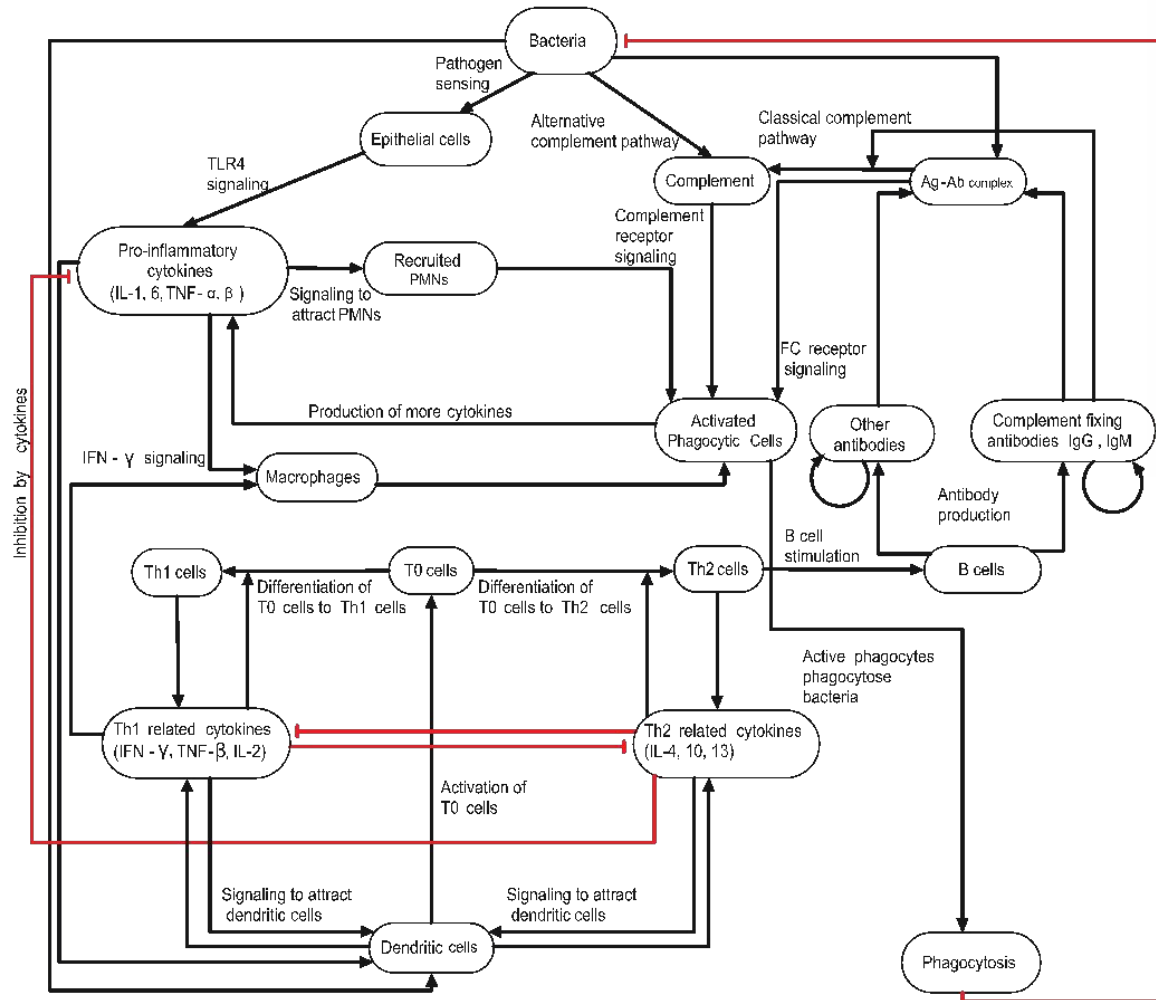
Alanine, aspartate
and glutamate
metabolism: 3

Carbon fixation in
photosynthetic
organisms: 2

Ferroptosis: 2

Cysteine and
methionine
metabolism: 2

Dynamical system



Mechanical description of phenomenon

Less data but more focused

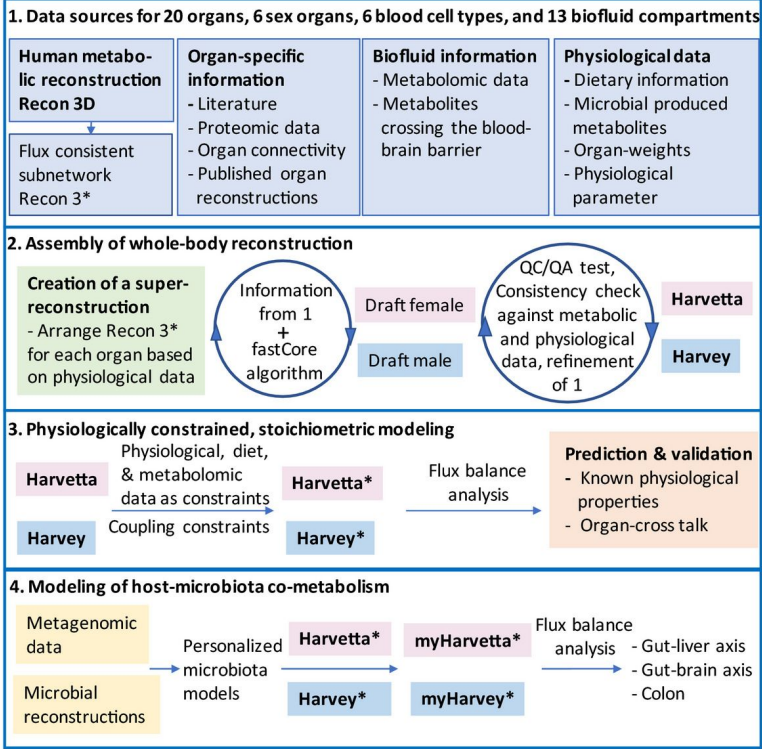
Atmospheric CO₂
enrichmentClimate
warmingPrecipitation
changeNitrogen
depositionLand-use and land
cover changes*Vegetation type**Plant traits**Soil attributes*Physiological &
Phenological
propertiesStructural &
chemical
characteristics*Scalars:*
Temperature
Water
Oxygen
NutrientsCarbon
allocationCarbon
transfersCarbon
agesGross primary
productivityCarbon use
efficiencyTotal environmental
ScalarBaseline carbon
residence timeNet primary
productivityCarbon
residence timeCarbon
chasing timeNet primary
productivityCarbon storage
capacityCarbon storage
potential

Carbon storage dynamic

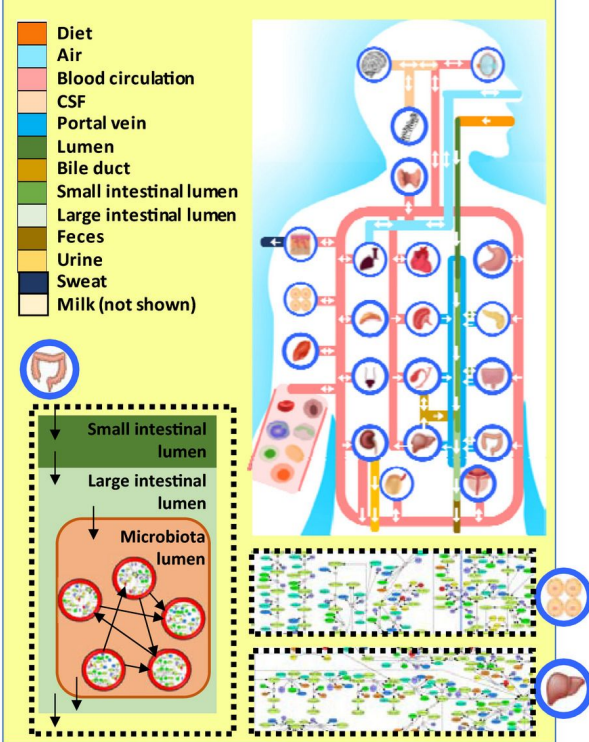
Example of whole human body model

Reconstruction with human data

A Overview of the study



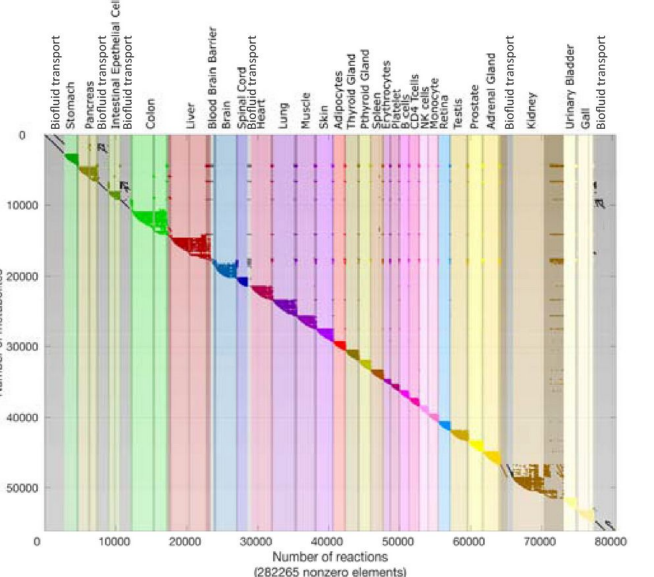
B Organ connections in Harvey



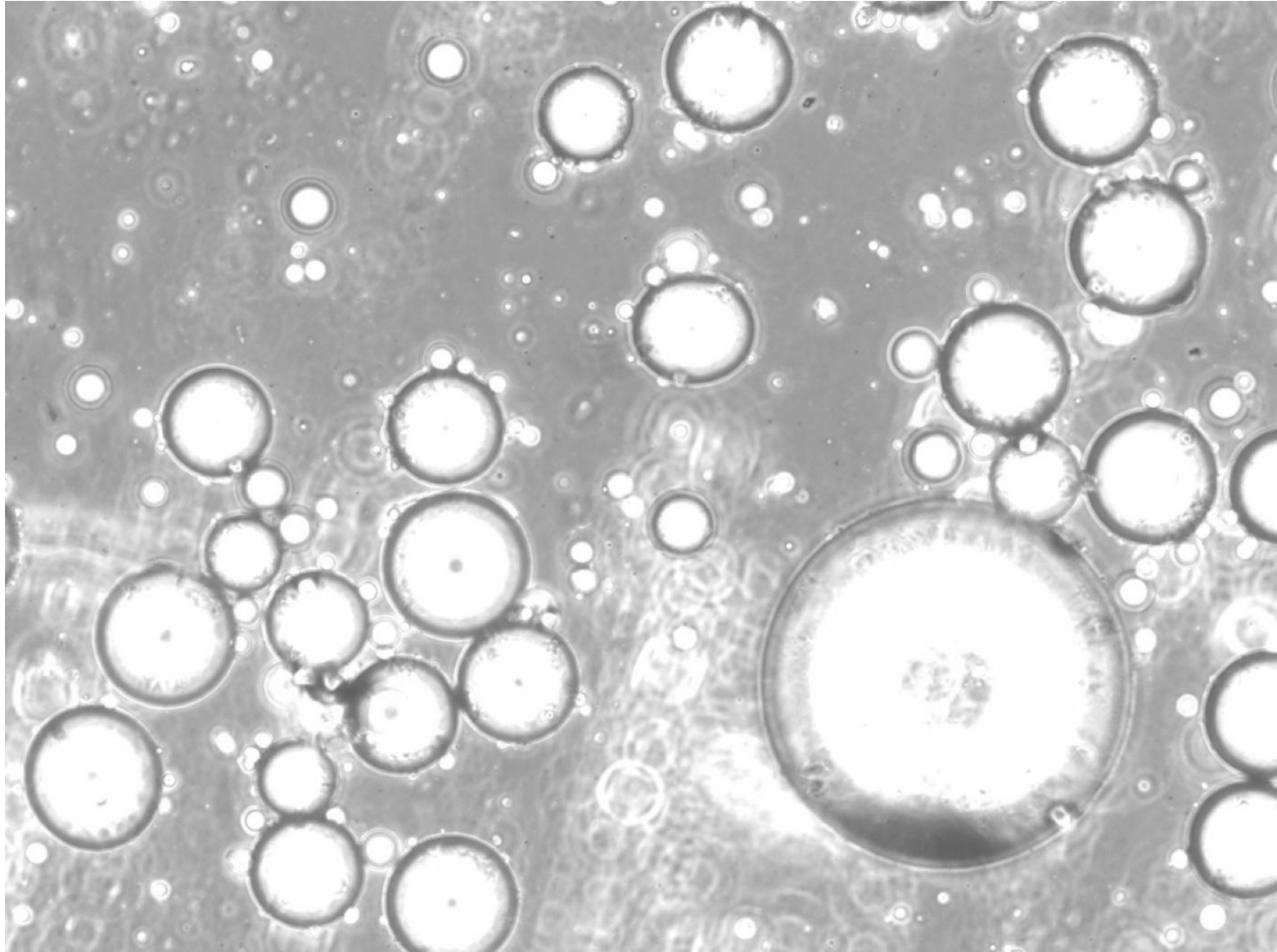
C Characteristics of the WBM reconstructions

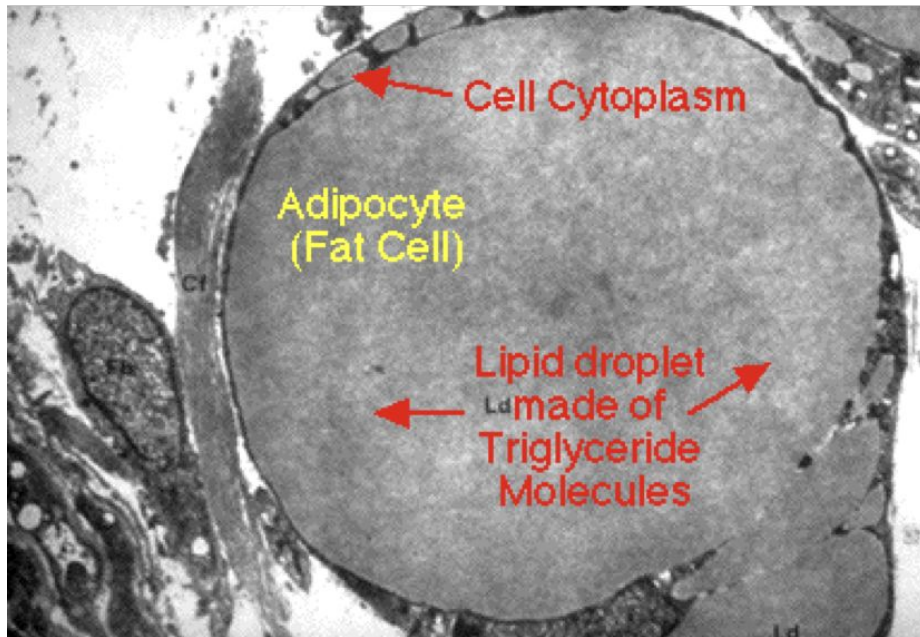
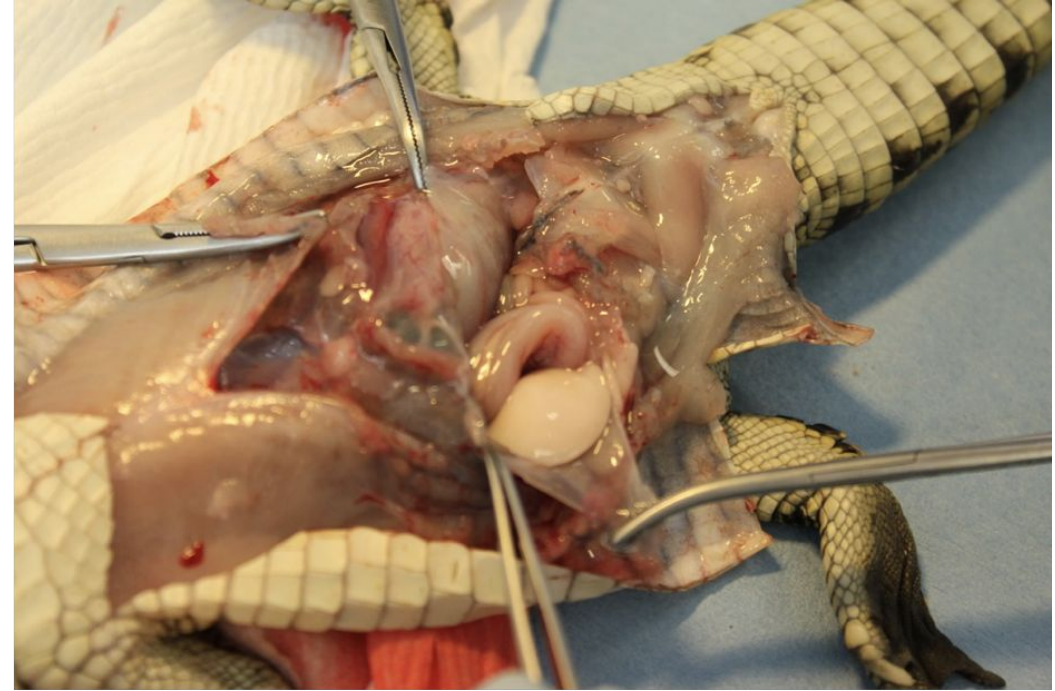
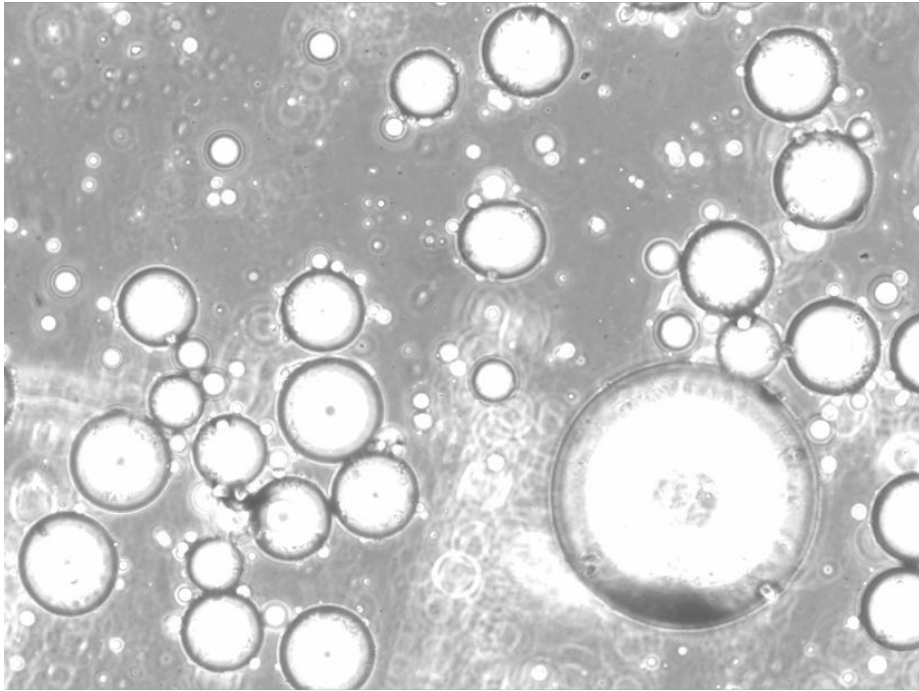
	Harvey	Harvetta
Sex	Male	Female
Sex-independent organs	20	20
Sex-specific organs	2 (testis, prostate)	4 (breast, cervix, uterus, ovary)
Blood cell types	6	6
Biofluid compartments	13	13
Overall reactions	81,094	83,521
Overall metabolites	56,452	58,851
Overall genes	1,681	1,681
Overall metabolic subsystems	104	104
Intracellular compartments	2-8	2-8
Fecal metabolites	1,400	1,398
Dietary metabolites	1,392	1,391
Blood circulation metabolites	979	978
Urine metabolites	920	923
CSF metabolites	517	508
Portal vein metabolites	488	487
Bile duct metabolites	261	262
Breast milk metabolites	--	84
Sweat metabolites	57	57
Air metabolites	3	3
Organ weight based on 70 kg (male) & 58 kg (female)	58.8 kg (84%)	49.5 kg (85%)
Organ weight plus skeleton and connective tissue (11kg)	69.3 kg (99%)	57.5 kg (99%)

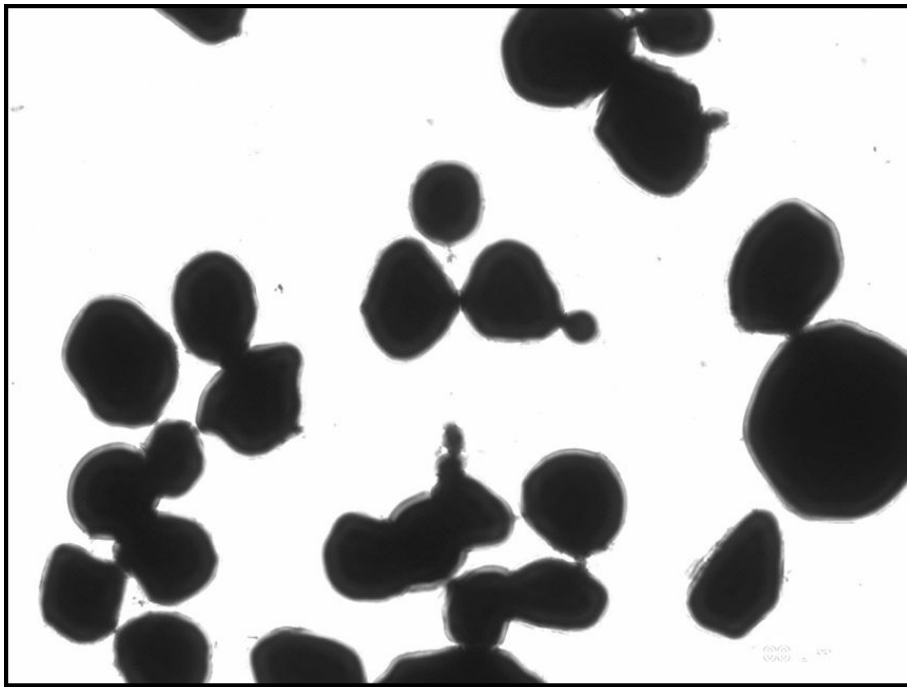
D Visualization of Harvey's stoichiometric matrix

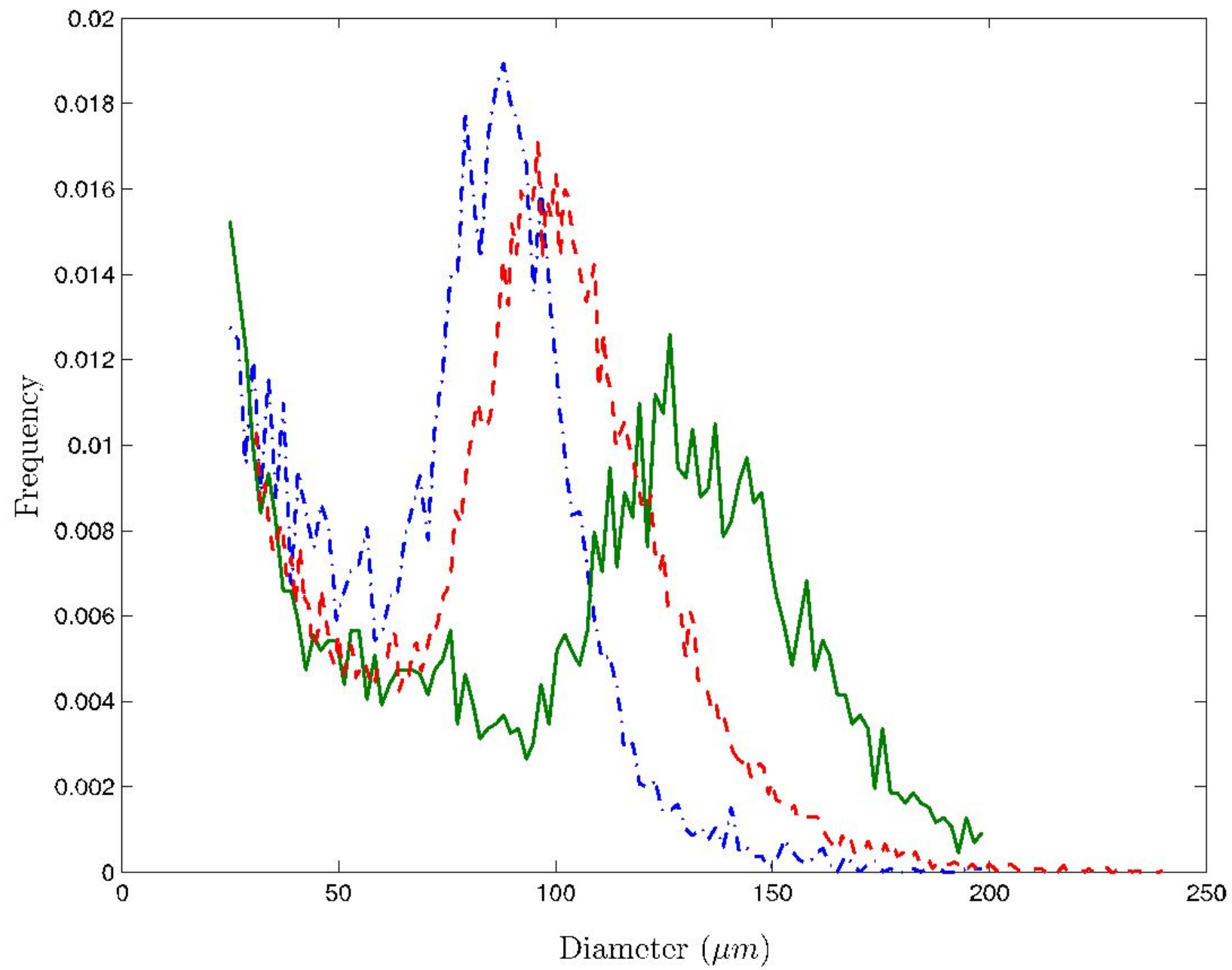


Modeling adipocytes





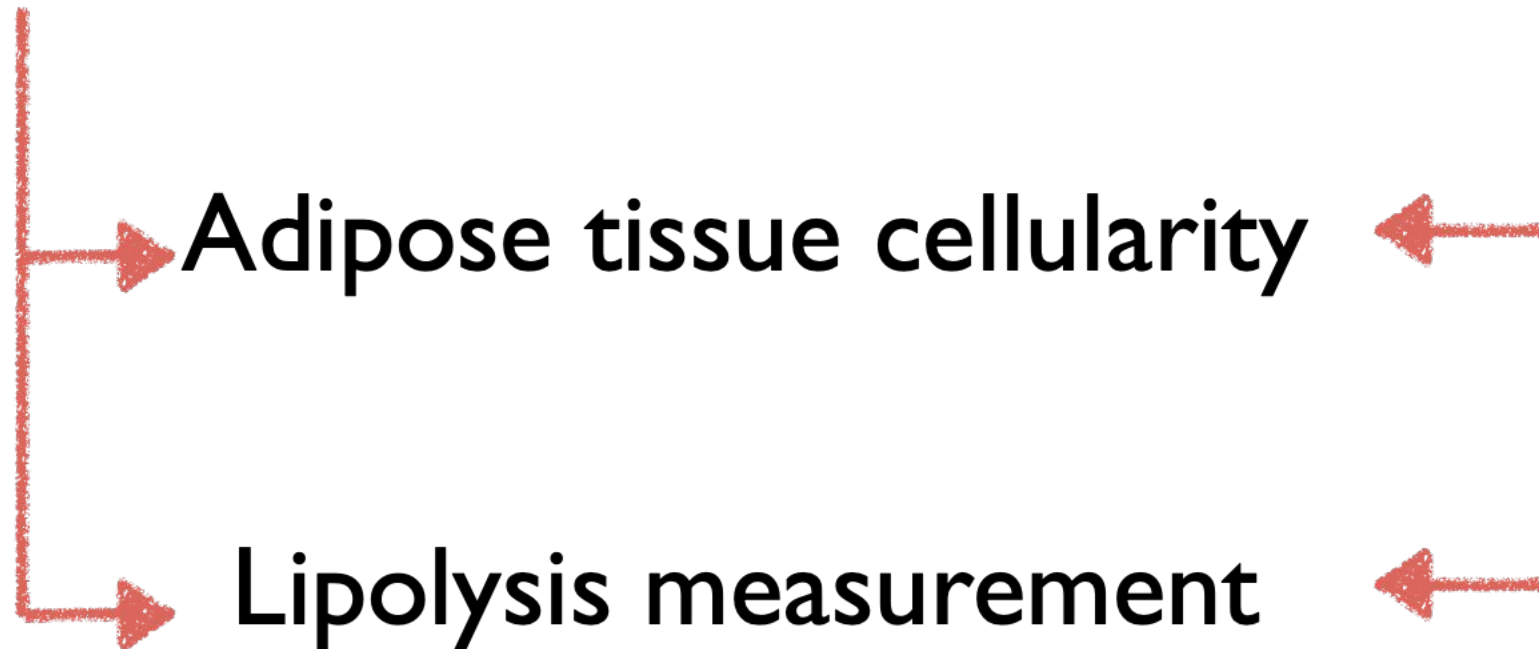


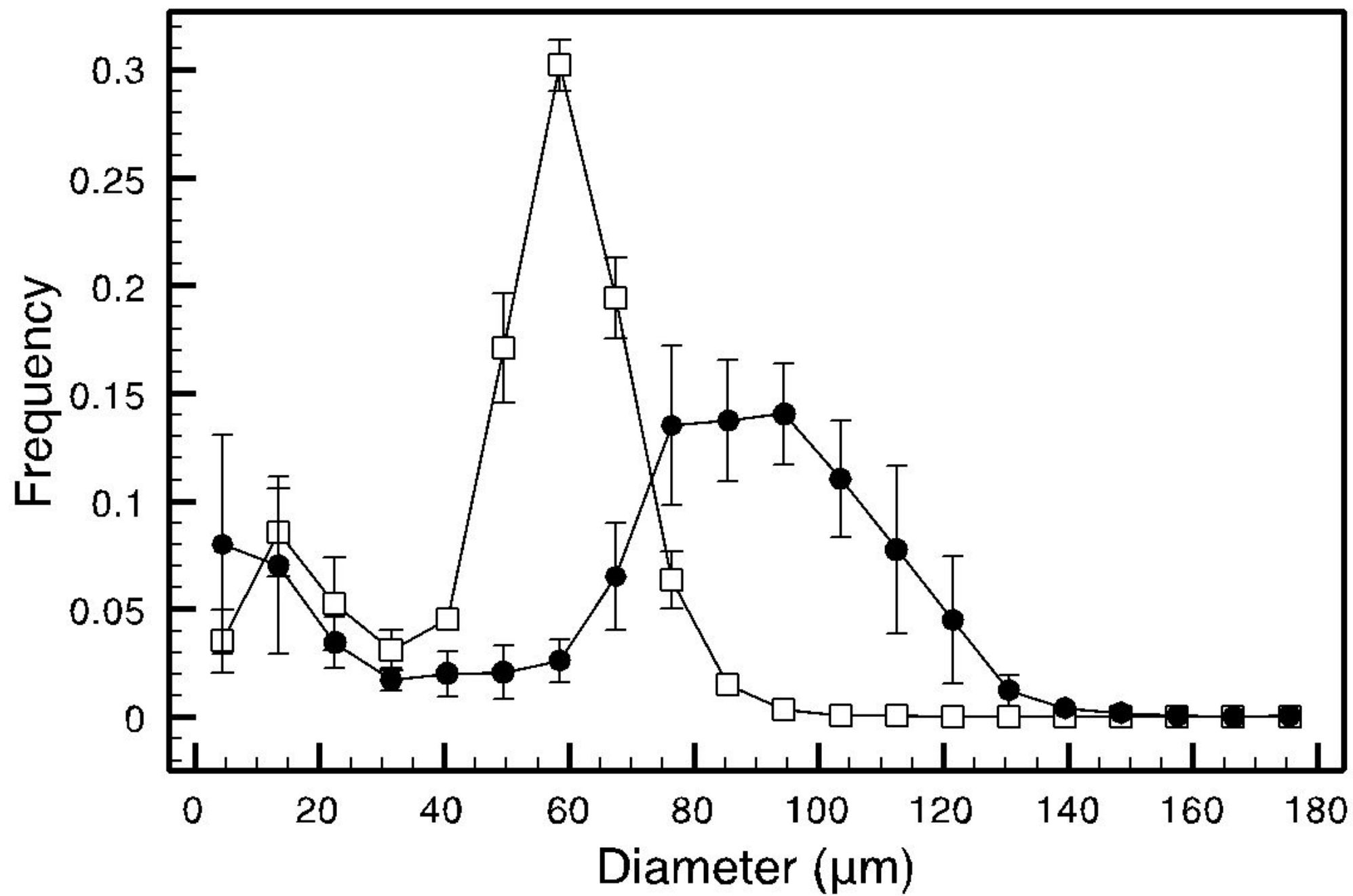


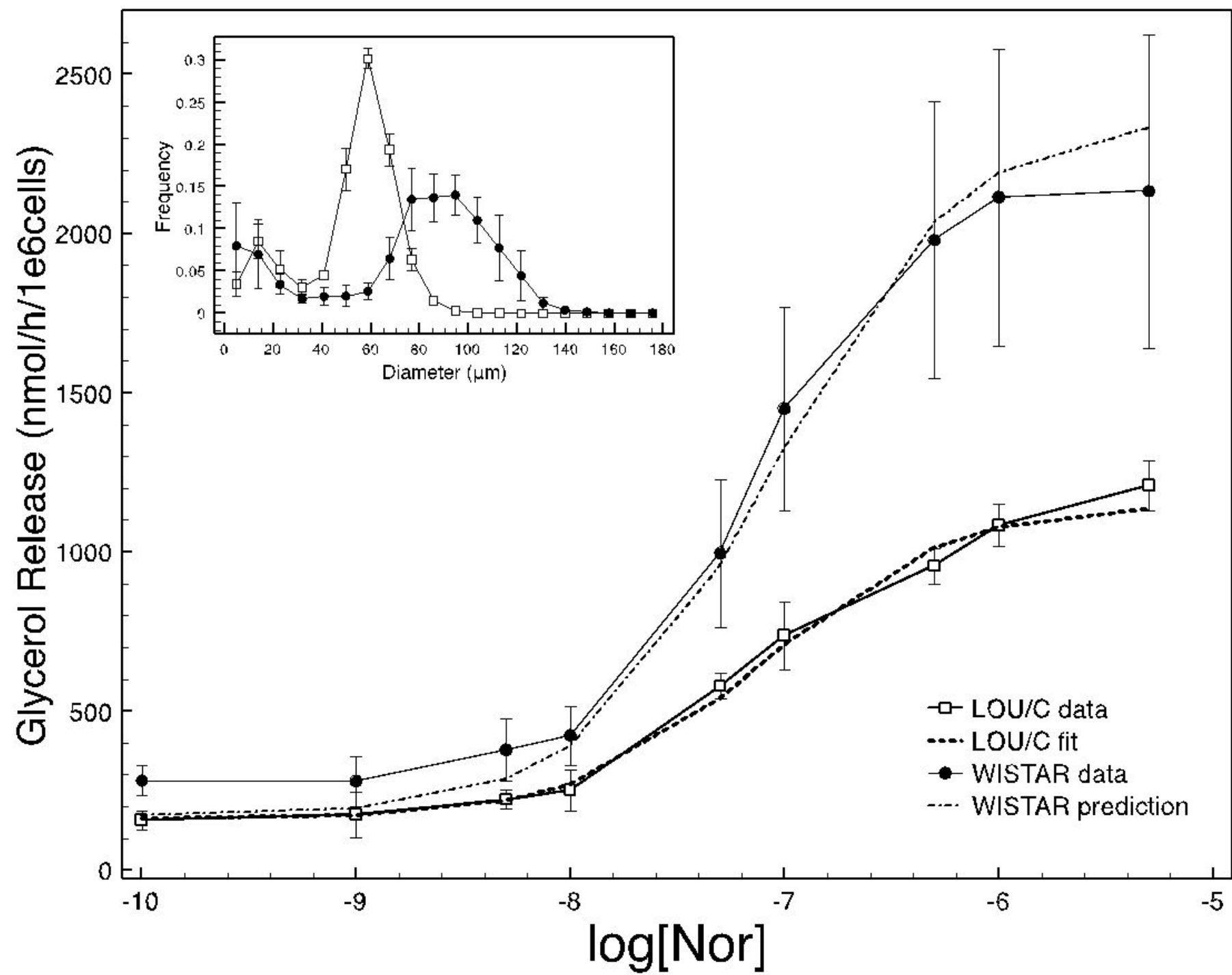


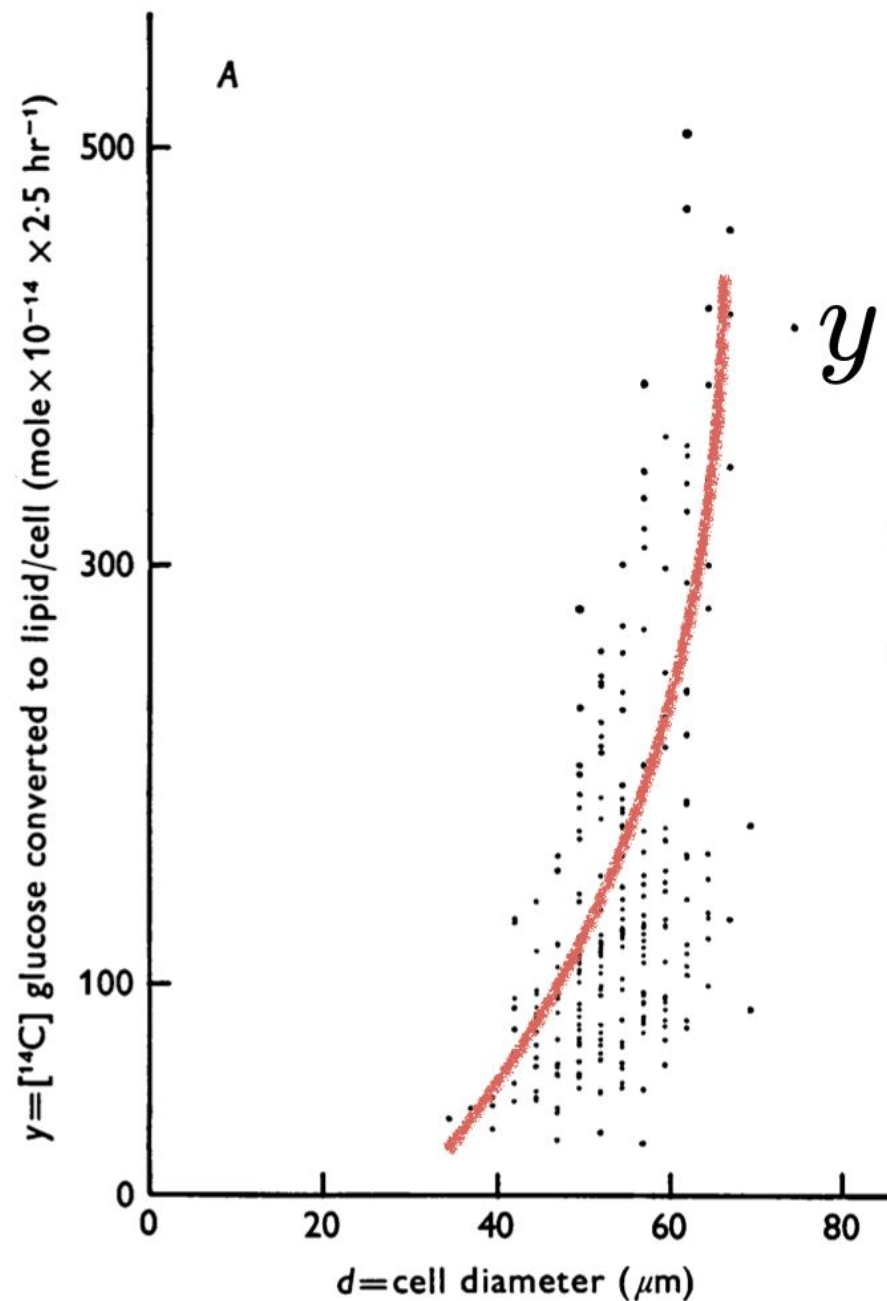
‘anti-obese’ LOU/C

‘obese’ Wistar









Sample no.	No. of cells (n)	Exponent ($\beta'i$)
1	252	2.61
2	100	1.98
3	101	1.54
4	101	1.19
5	68	1.80
6	130	1.90
7	97	1.60
8a	118	1.46
8b	102	1.79
8c	105	2.00
8d	107	2.70
8e	100	2.67

~ 2

LIPOGENESIS AND INSULIN SENSITIVITY OF
SINGLE FAT CELLS*

By J. GLIEMANN AND J. VINTEN
WITH AN APPENDIX BY AA. VØLUND

dynamics of adipose cells after fasting

- fasting induced lipolysis
- 20 rats (10 control 10 food restricted)
- restriction of 60%
- around 90 days

Mathematical predictions

- distribution shape invariant
- distribution shifted to the left

$$\frac{dV}{dt} = -\gamma r^2$$

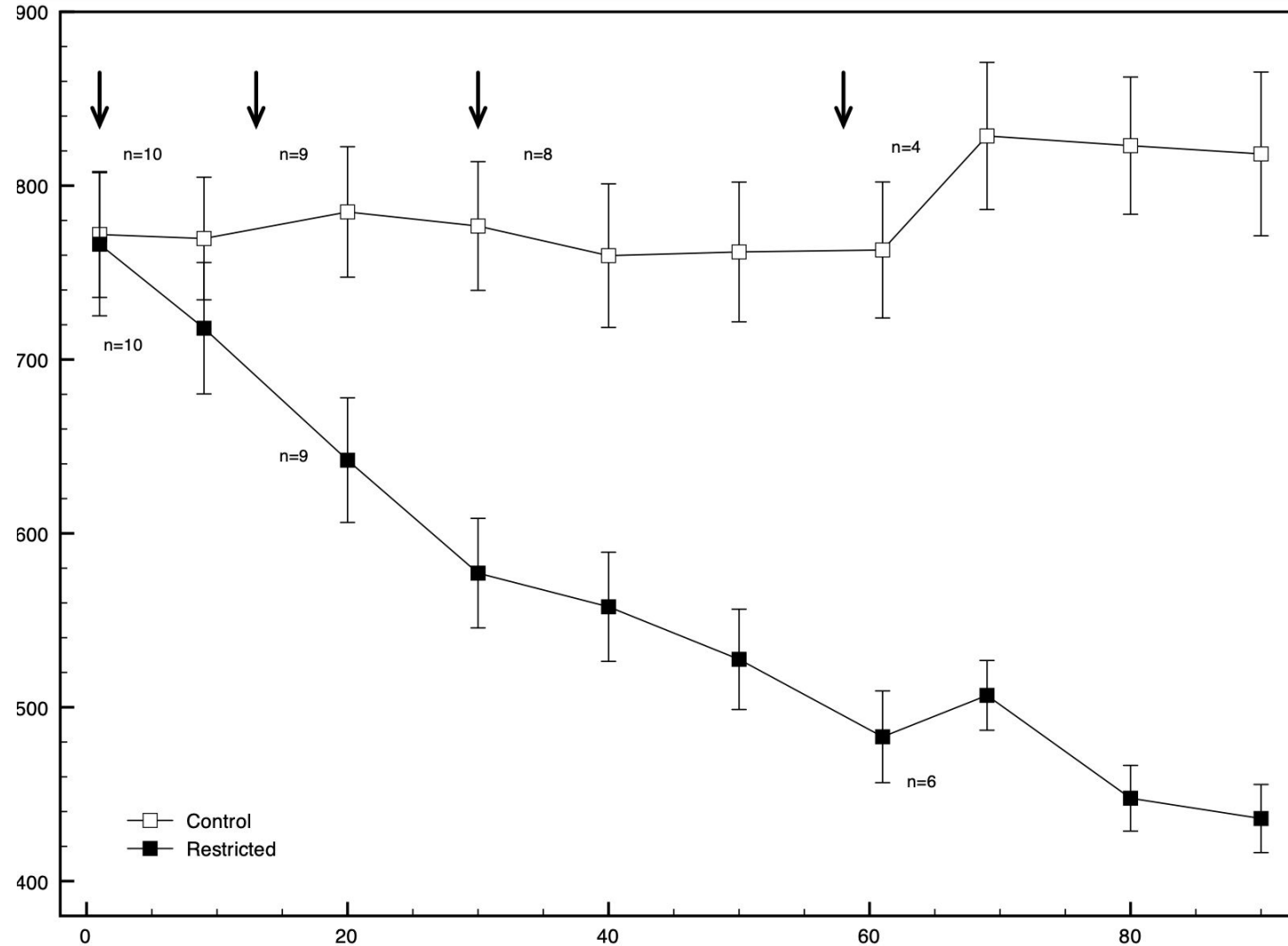
$$V = \frac{4\pi}{3} r^3$$

$$\frac{r}{dt} = -\frac{\gamma}{4\pi}$$

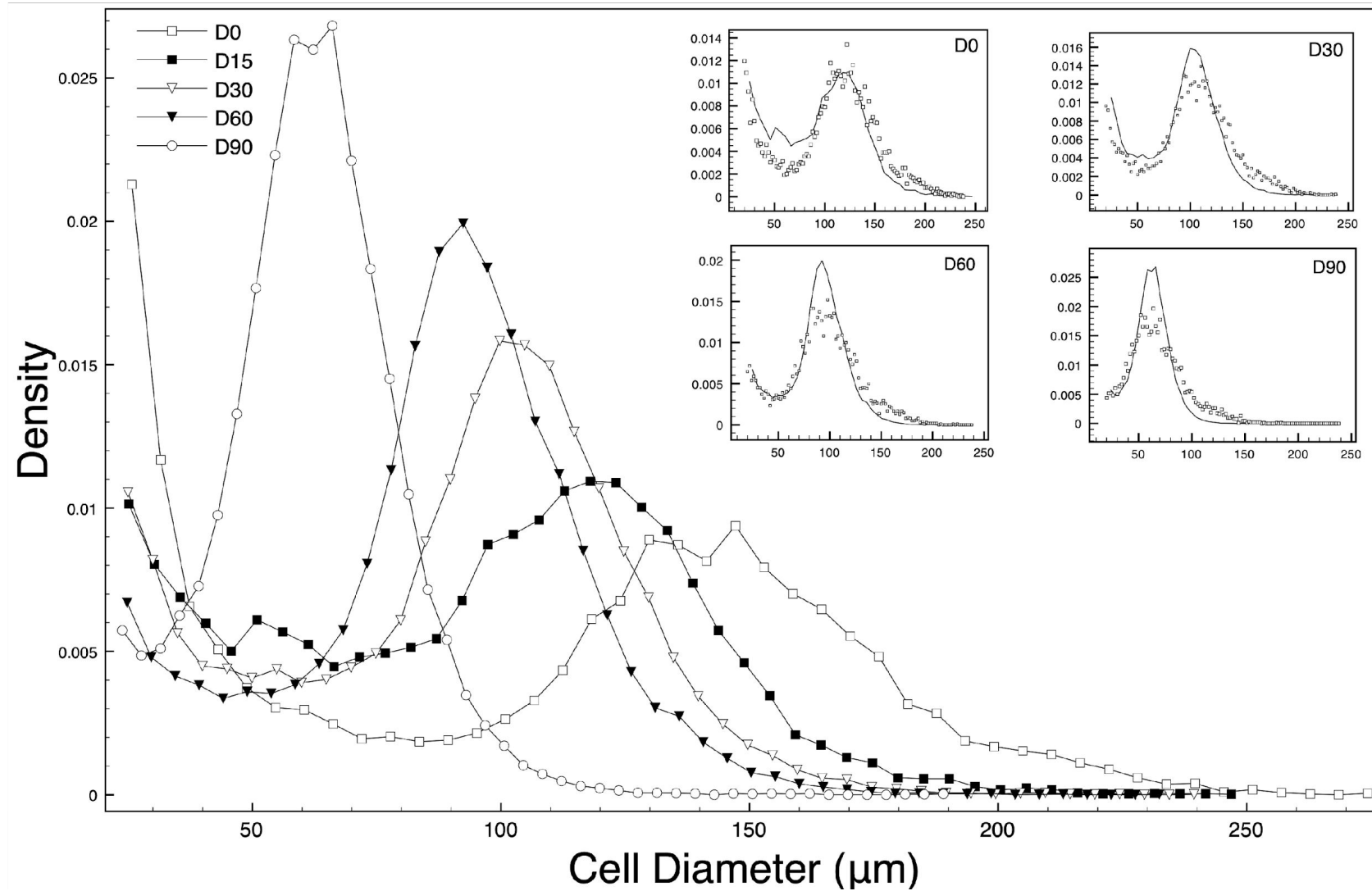
$$\frac{\partial \rho(r, t)}{\partial t} - \frac{\gamma}{4\pi} \frac{\partial \rho(r, t)}{\partial r} = 0$$

$$\rho(r, t) = \rho_0\left(r - \frac{\gamma}{4\pi} t\right)$$

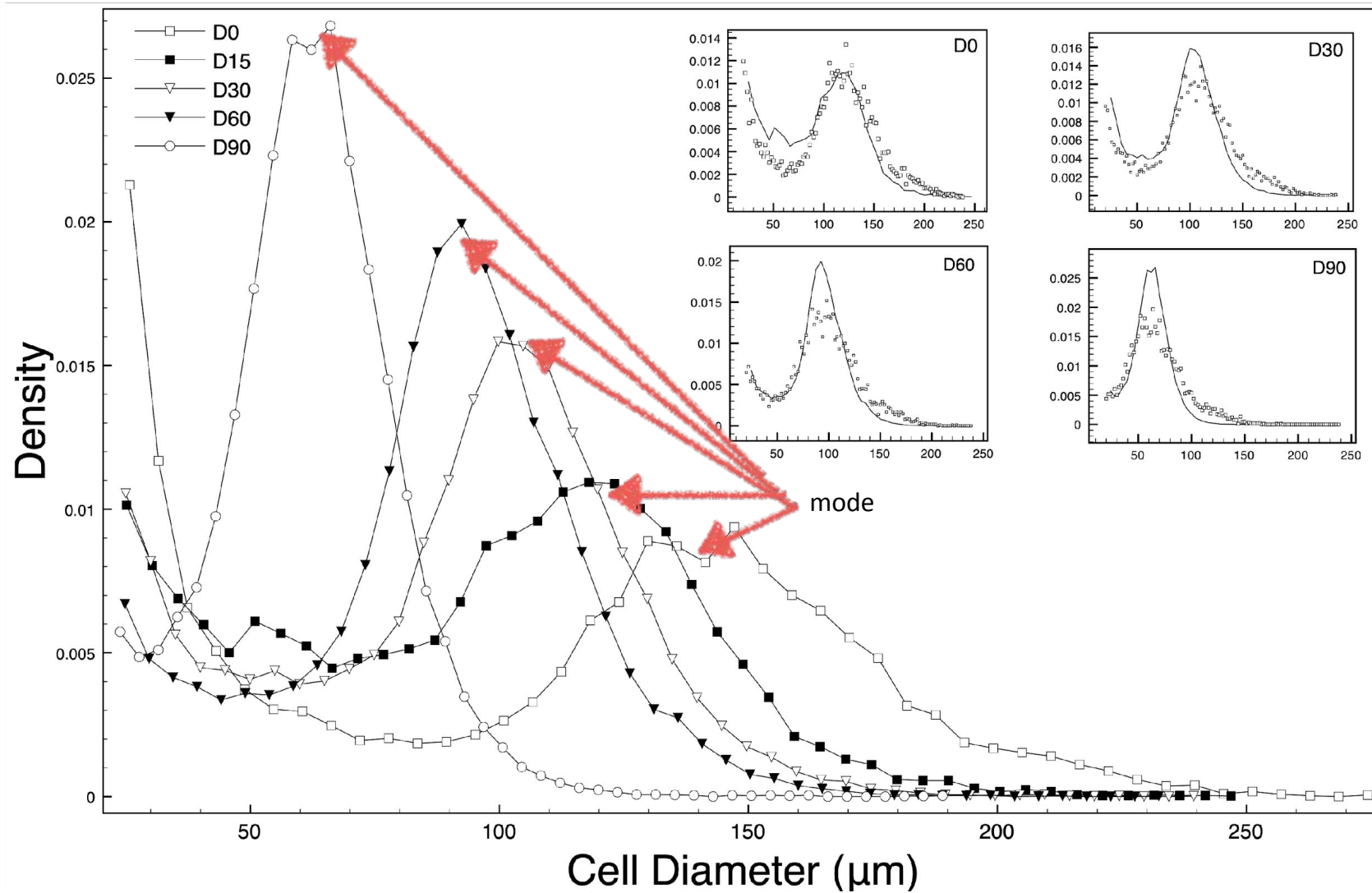
weight evolution (and bionssies dates)



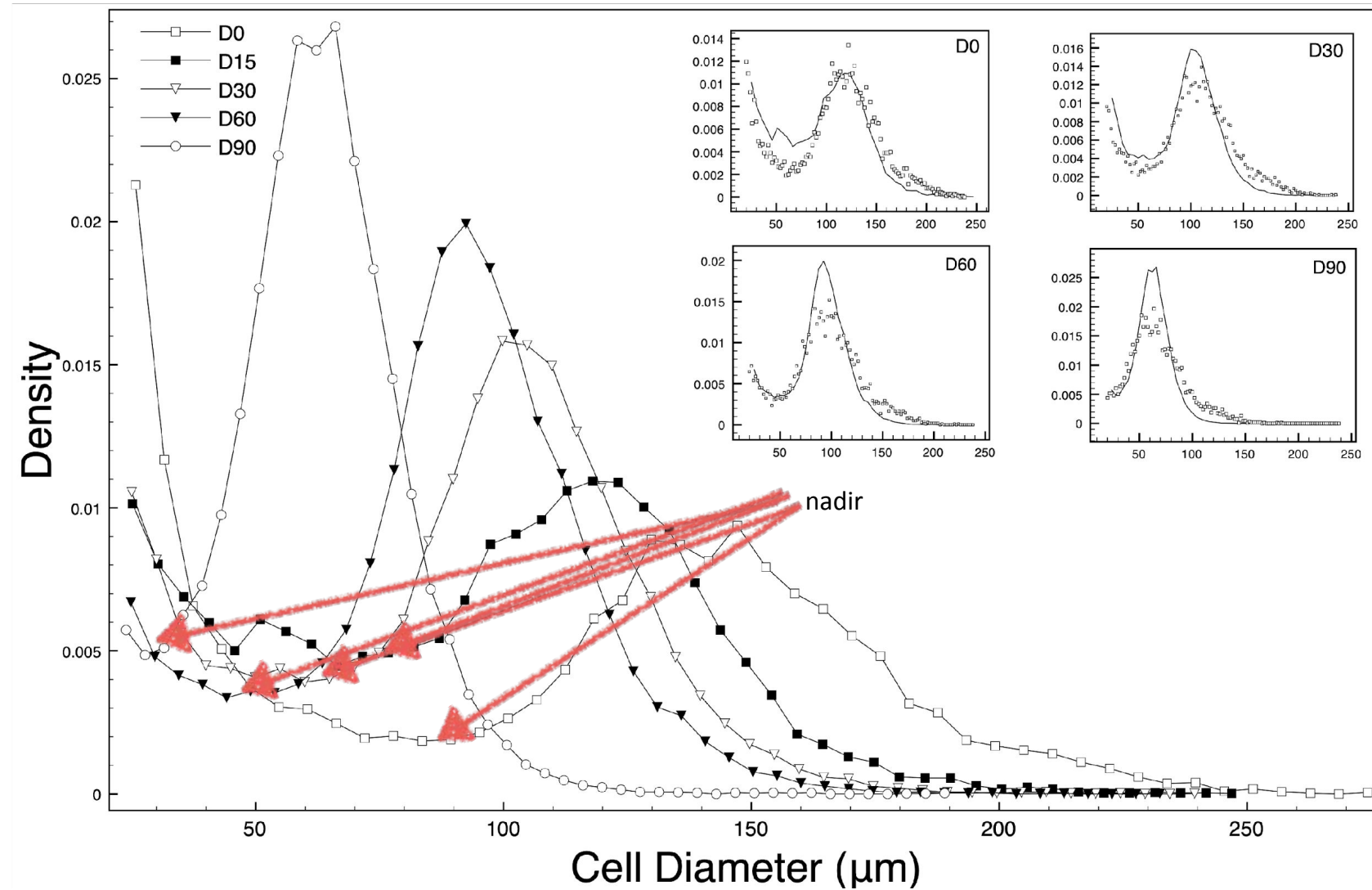
Individual evolution



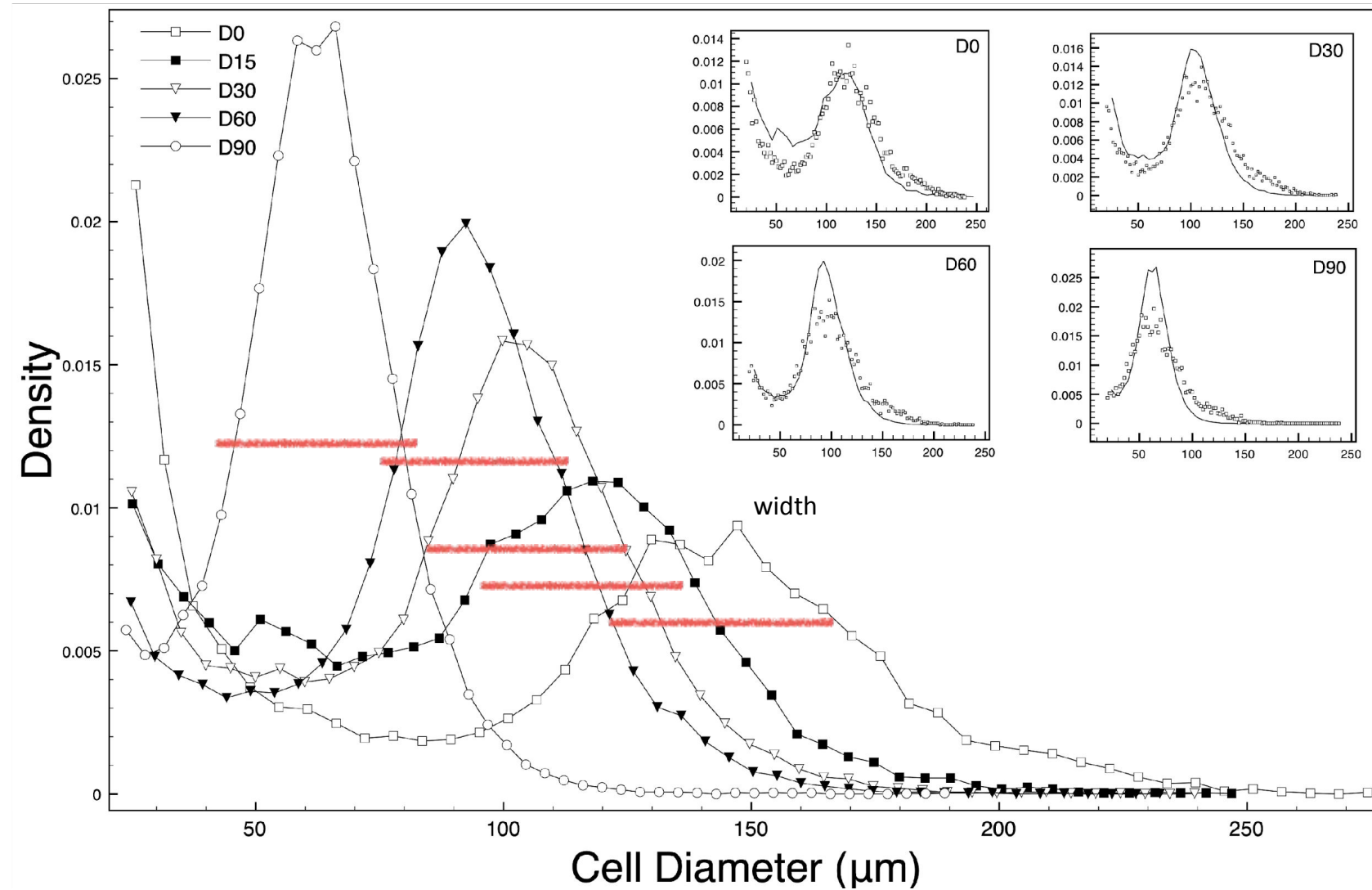
Individual evolution



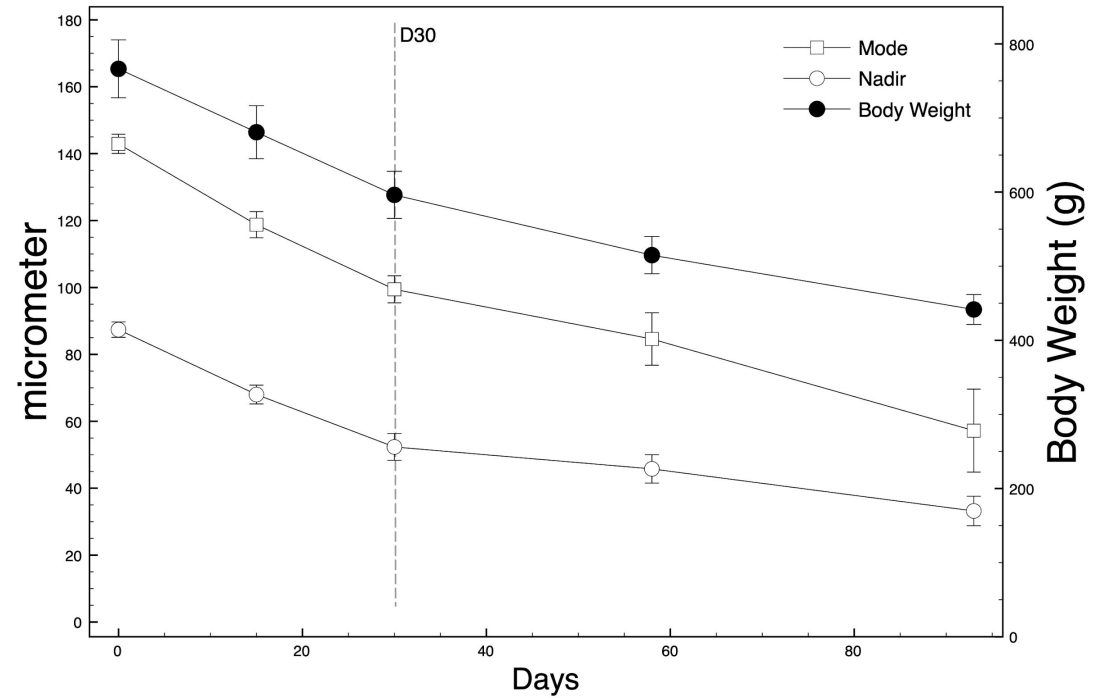
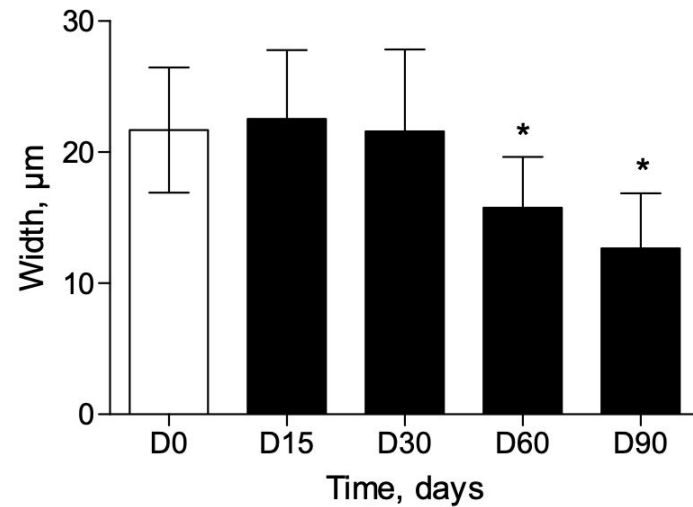
Individual evolution



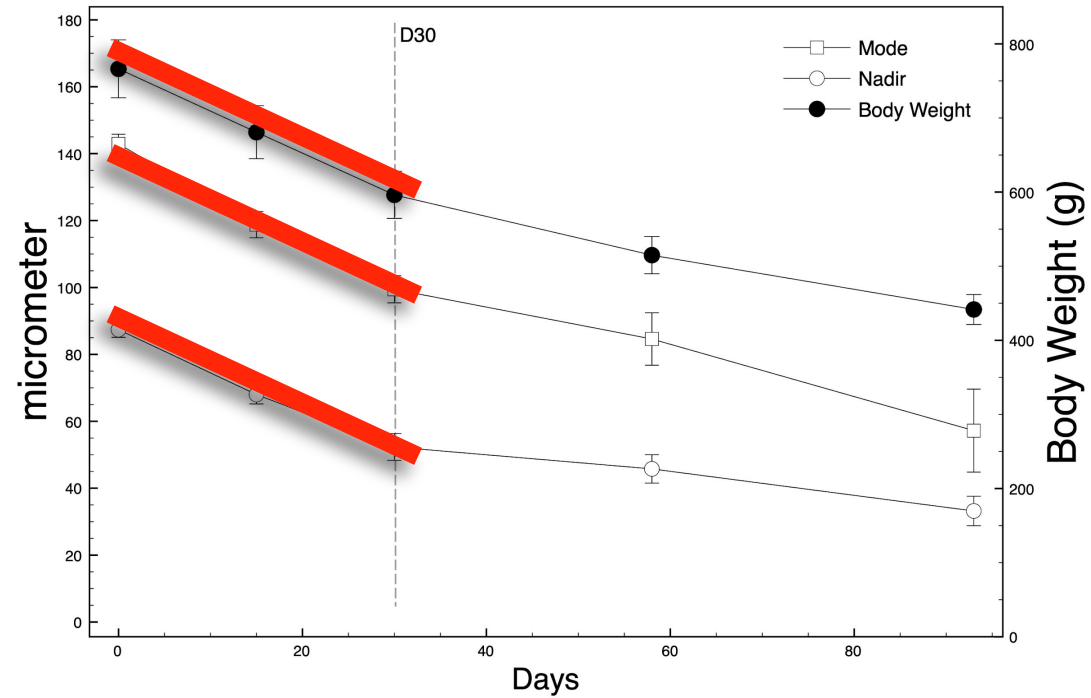
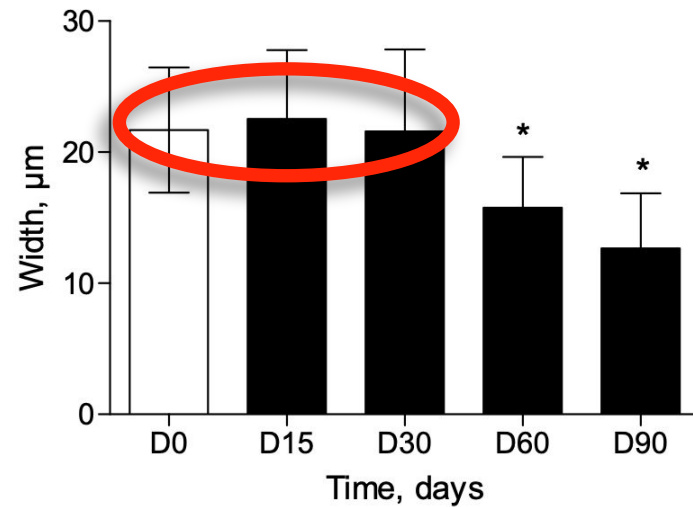
Individual evolution



distribution parameters evolution

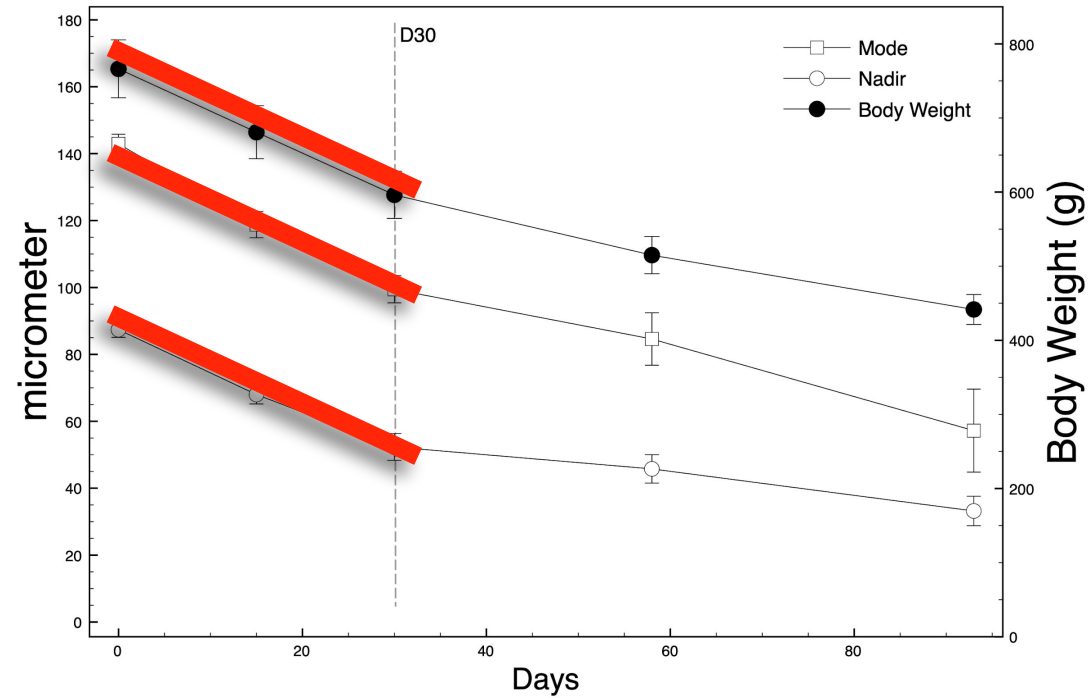
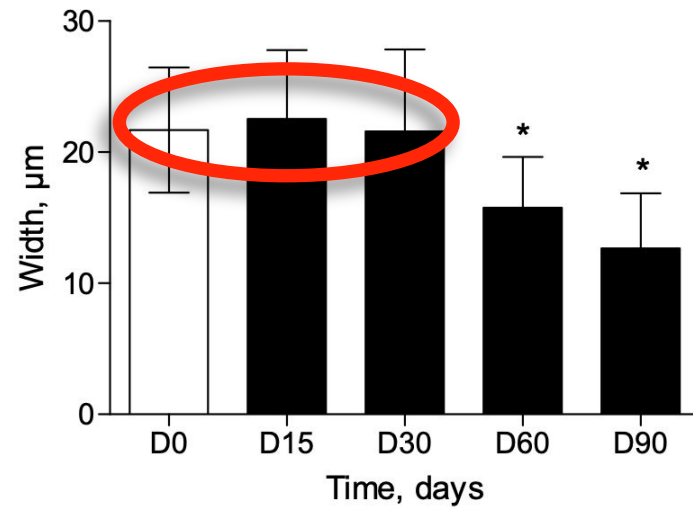


distribution parameters evolution



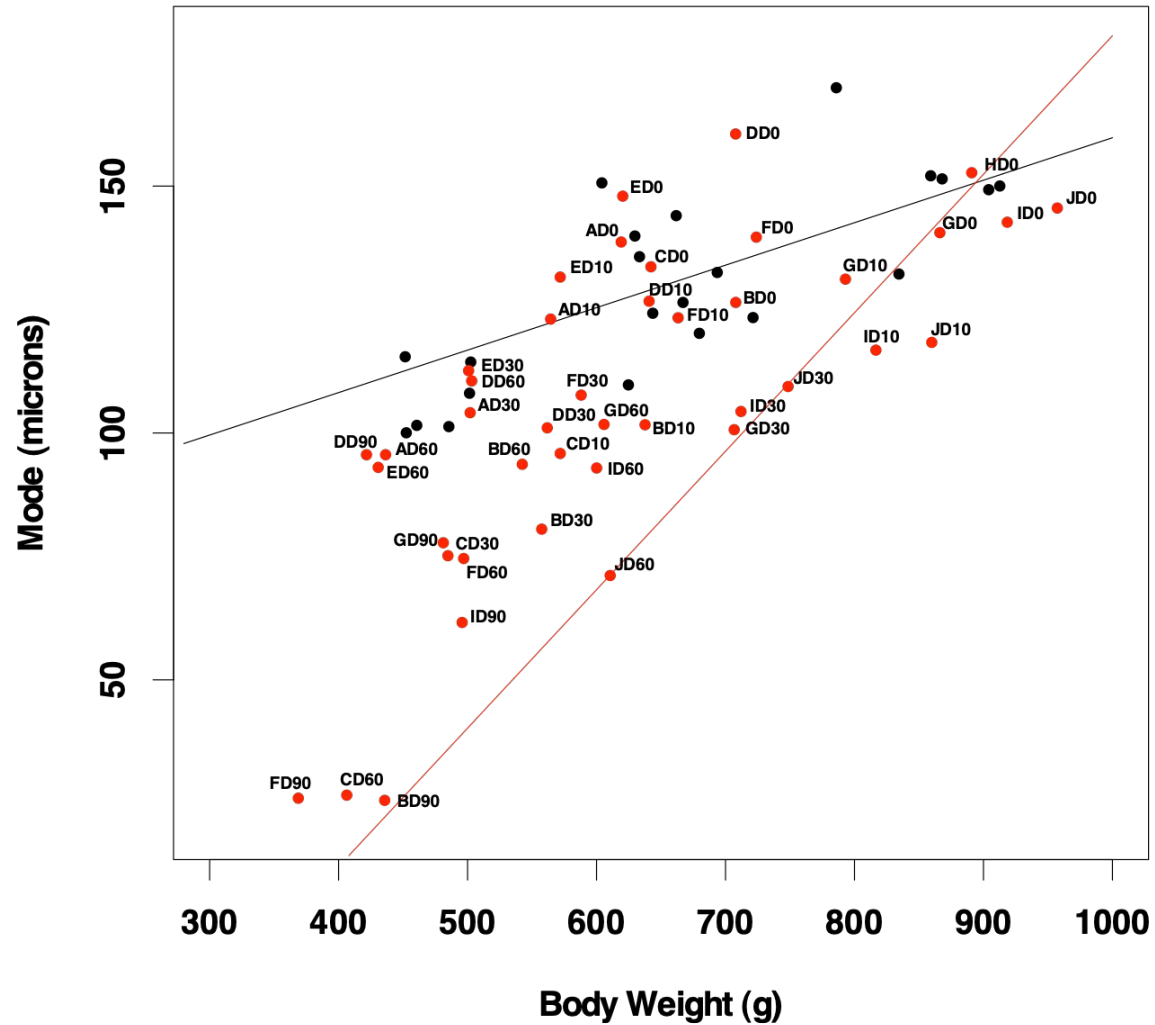
shape preserved

distribution parameters evolution

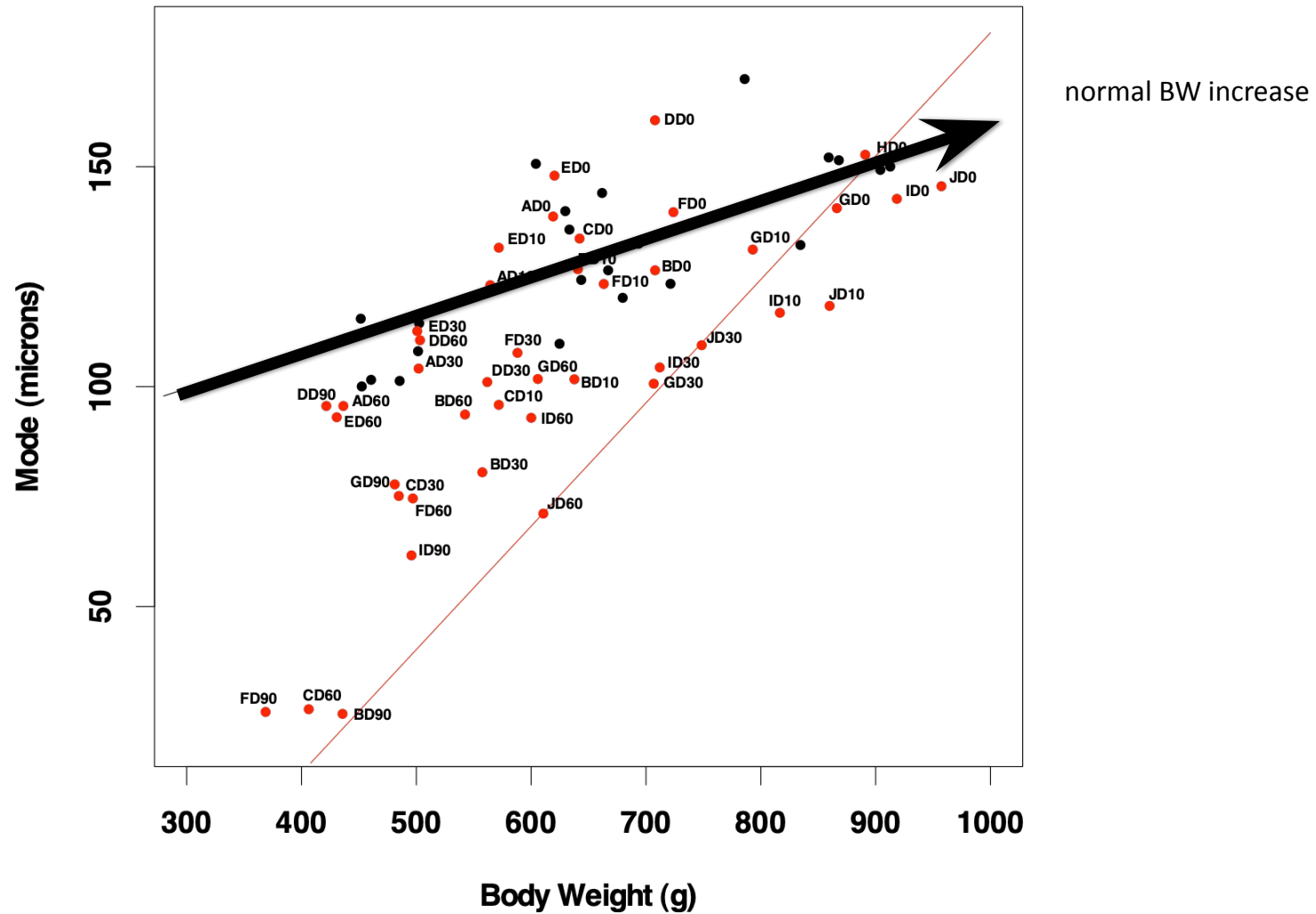


Whole body verification
of surface-based fluxes

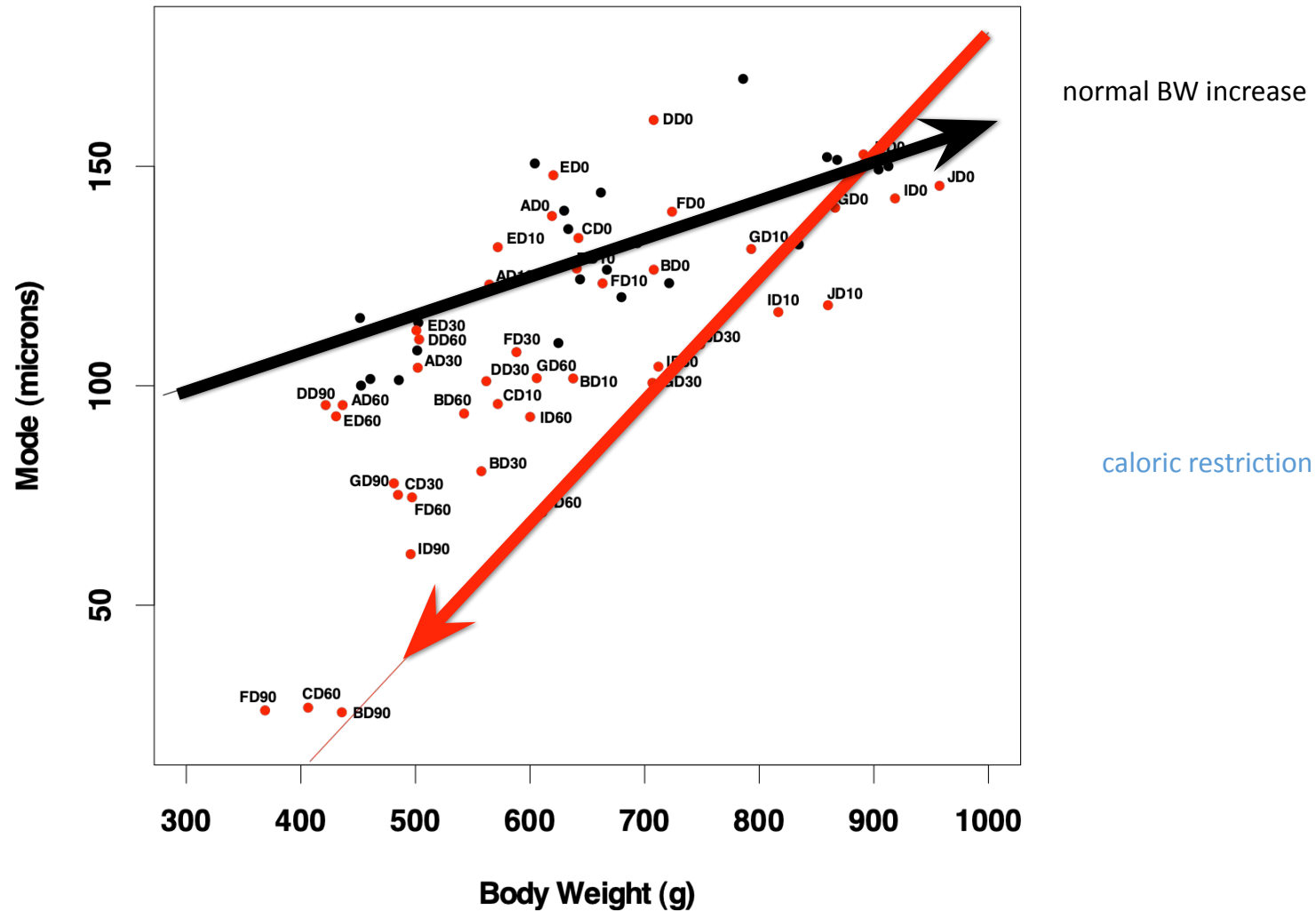
dynamics with history



dynamics with history



dynamics with history



$$\frac{dl}{dt} = \underbrace{ar^2 \frac{L}{L + \kappa_l} \frac{\kappa_r^n}{\kappa_r^n + r^n}}_{\text{lipogenesis}} - \underbrace{\left(B + br^2 \right) \frac{l}{k_t + l}}_{\text{lipolysis}}$$

radius lipogenesis



$$\frac{dl}{dt} = \underbrace{ar^2 \frac{L}{L + \kappa_l} \frac{\kappa_r^n}{\kappa_r^n + r^n}}_{\text{lipogenesis}} -$$

lipogenesis

radius lipolysis



$$\underbrace{\left(B + br^2 \right) \frac{l}{k_t + l}}_{\text{lipolysis}}$$

lipolysis

maximum size extension

radius lipogenesis

radius lipolysis

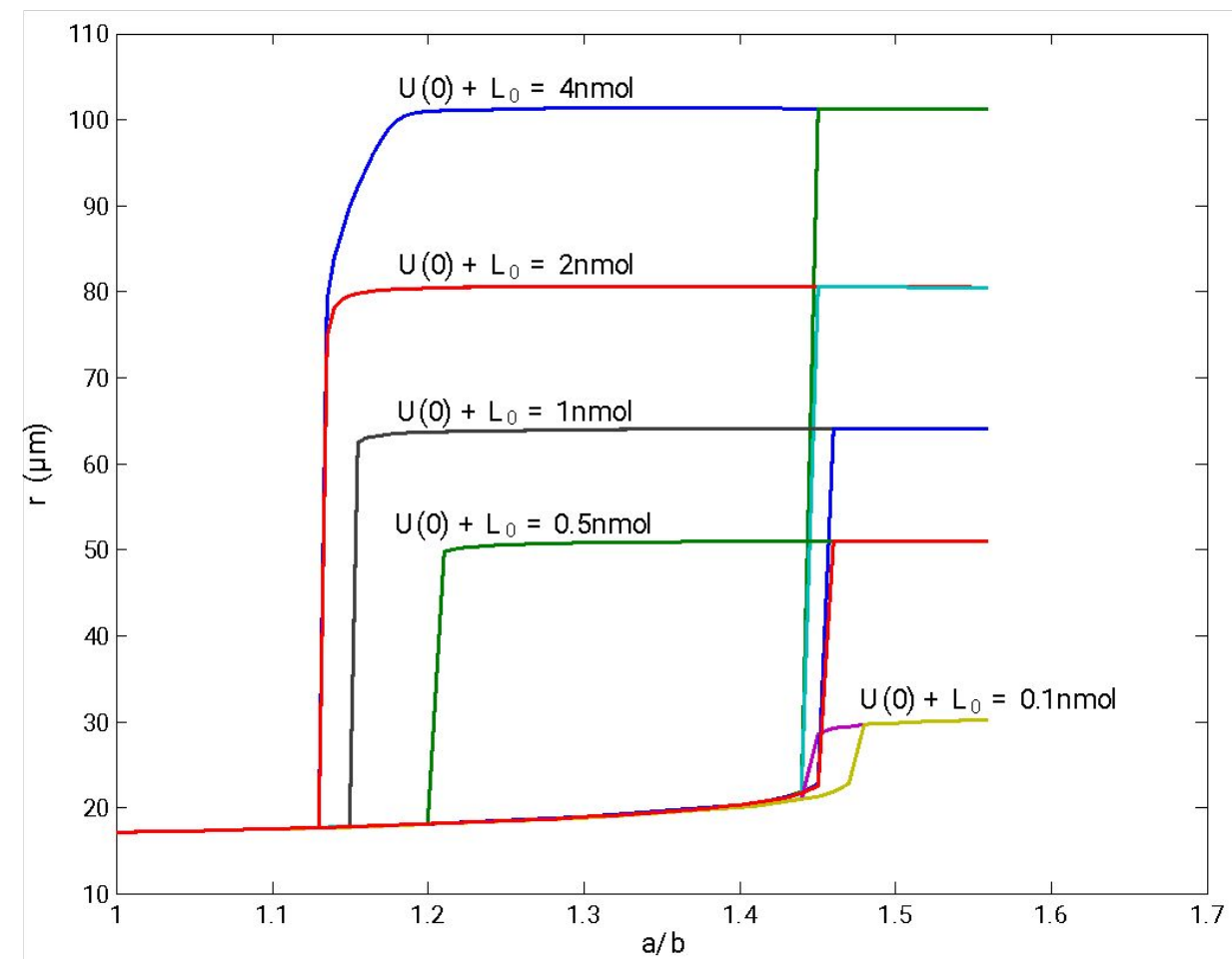
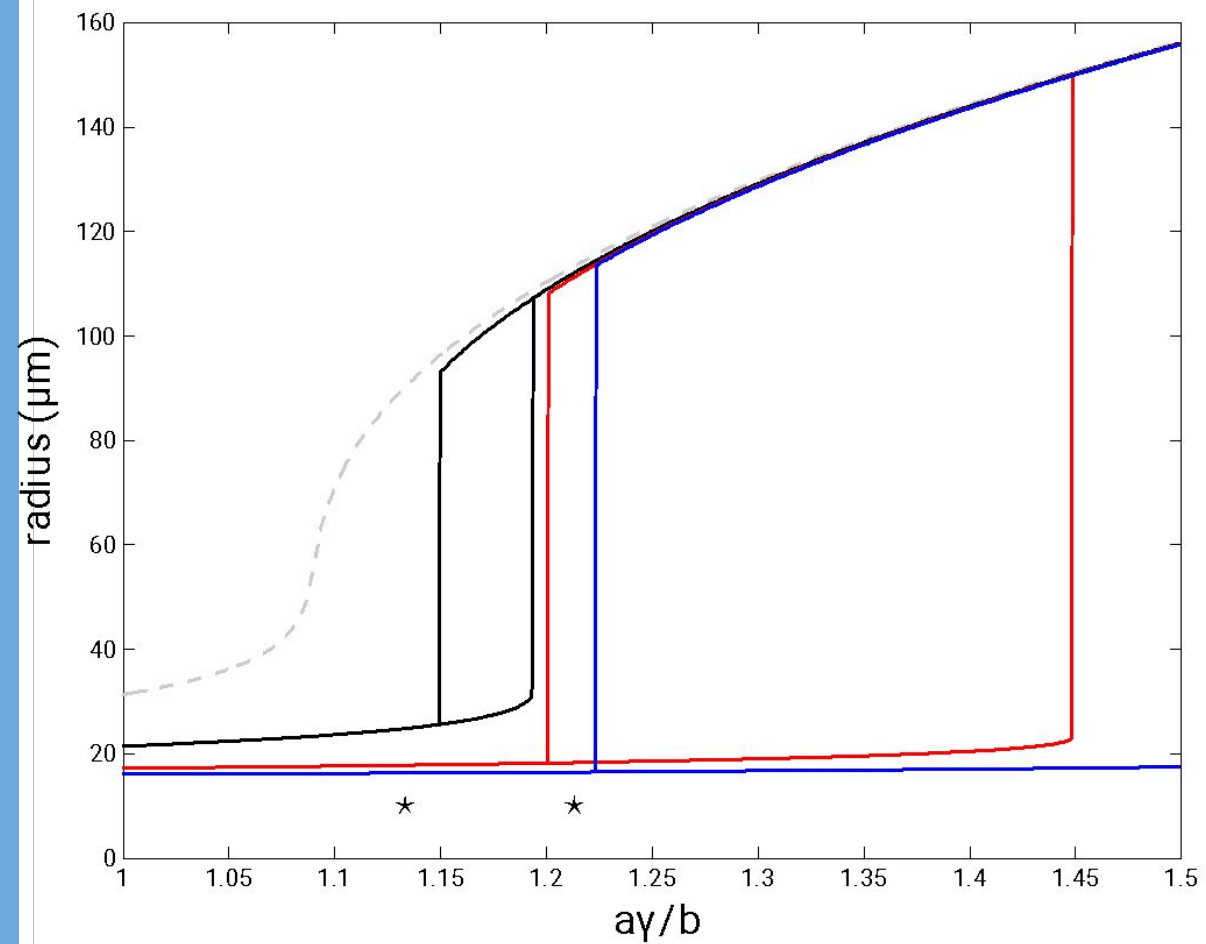
$$\frac{dl}{dt} = \underbrace{ar^2 \frac{L}{L + \kappa_l} \frac{\kappa_r^n}{\kappa_r^n + r^n}}_{\text{lipogenesis}} - \underbrace{(B + br^2) \frac{l}{k_t + l}}_{\text{lipolysis}}$$

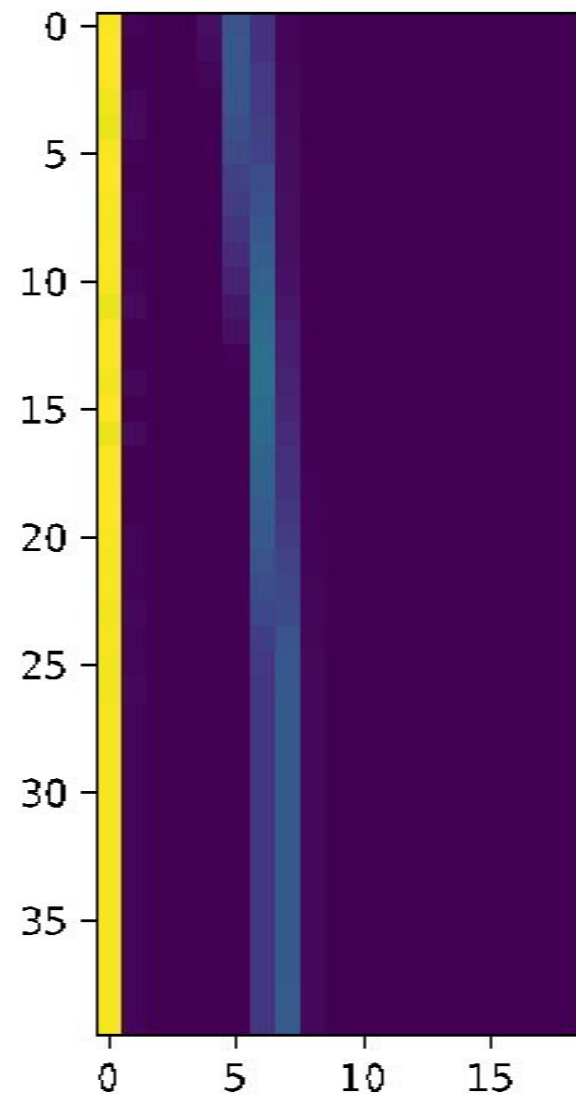
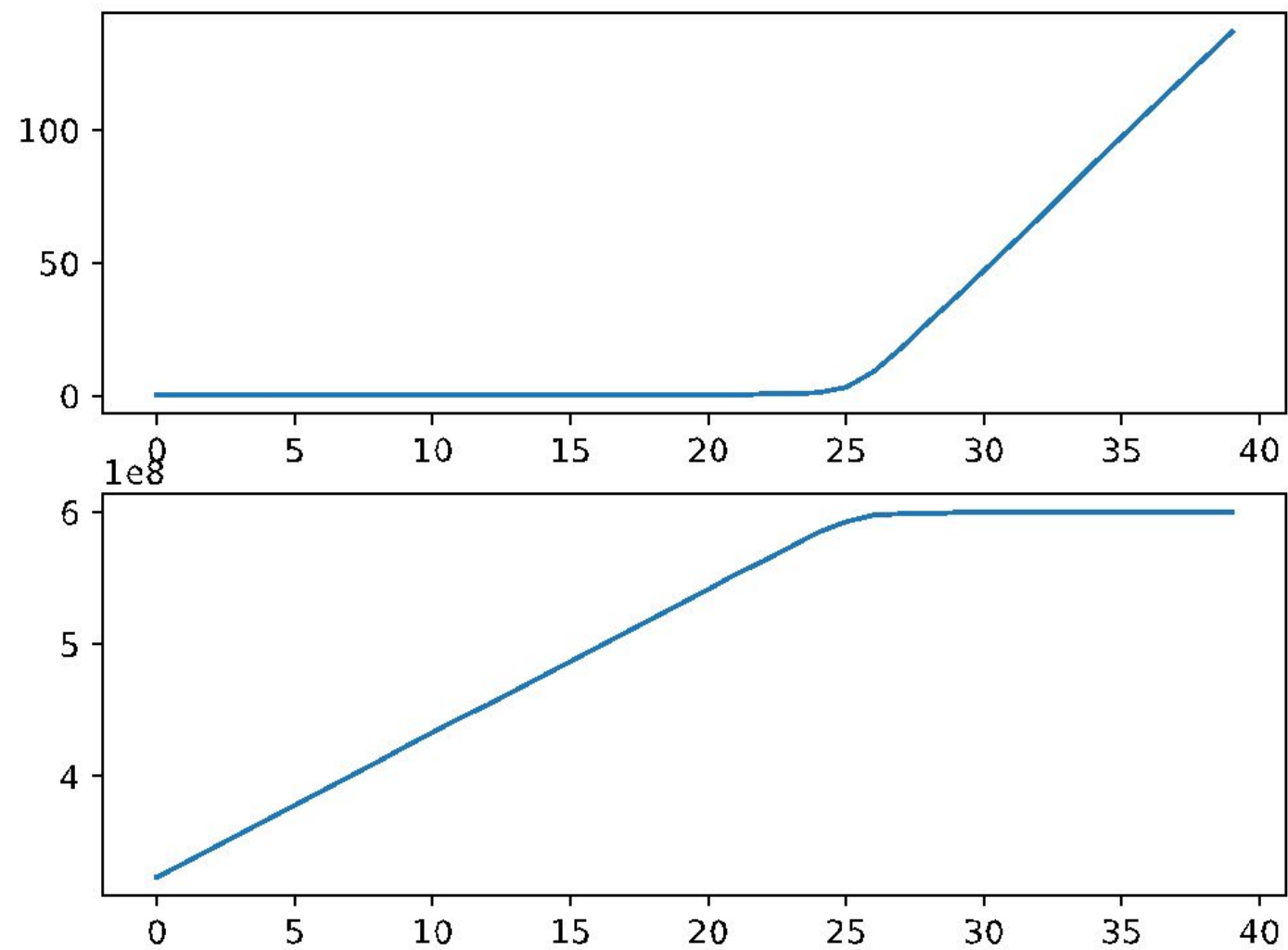
lipogenesis

lipolysis

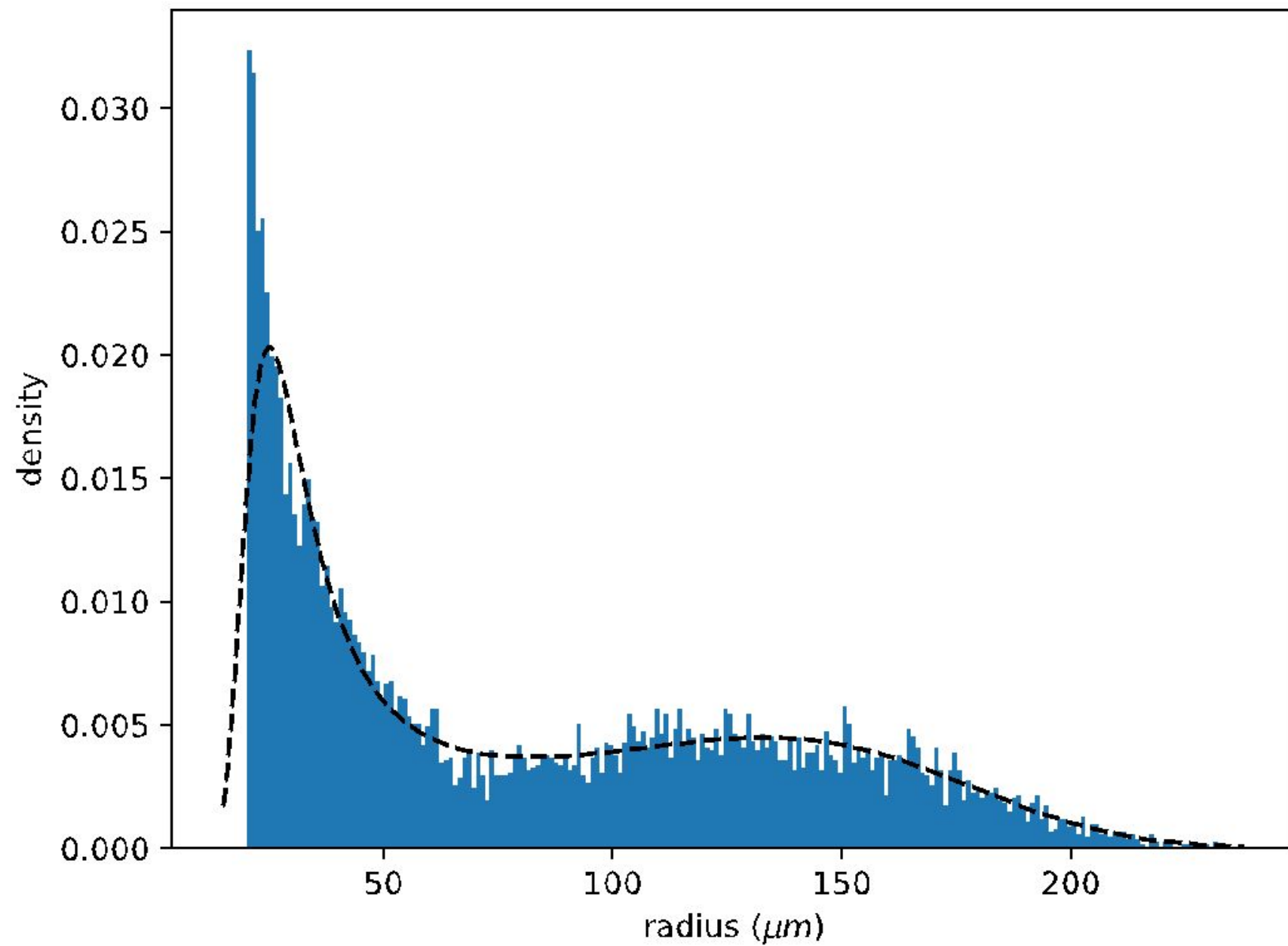
‘outside’ lipid content

‘inside’ lipid content





animal number3



Comrade SOULA is warning you

in 2025

You can

do MATHS

and IT





2014 Collaborative Research in
Computational Neuroscience (CRCNS)
PI Meeting

